Autonomous Navigation of Distributed Spacecraft using Intersatellite Laser Communications

by

Pratik K. Dave

B.S., University of Maryland (2009)

S.M., Massachusetts Institute of Technology (2014)

Submitted to the Department of Aeronautics and Astronautics
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Author .....
Department of Aeronautics and Astronautics
January 30, 2020

Certified by .....
Kerri L. Cahoy
Associate Professor of Aeronautics and Astronautics
Thesis Supervisor

Certified by .....
Richard Linares
Assistant Professor of Aeronautics and Astronautics

Certified by .....
Timothy M. Yarnall
Assistant Group Leader, MIT Lincoln Laboratory

Accepted by .....
Sertac Karaman
Associate Professor of Aeronautics and Astronautics
Chair, Graduate Program Committee

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Abstract

Autonomous navigation refers to satellites performing on-board, real-time navigation without external input. As satellite systems evolve into more distributed architectures, autonomous navigation can help mitigate challenges in ground operations, such as determining and disseminating orbit solutions. Several autonomous navigation methods have been previously studied, using some combination of on-board sensors that can measure relative range or bearing to known bodies, such as horizon and star sensors (Hicks and Wiesel, 1992) or magnetometers and sun sensors (Psiaki, 1999), however these methods are typically limited to low Earth orbit (LEO) altitudes or other specific orbit cases.

Another autonomous navigation method uses intersatellite data, or direct observations of the relative position vector from one satellite to another, to estimate the orbital positions of both spacecraft simultaneously. The seminal study of the intersatellite method assumes the use of radio time-of-flight measurements of intersatellite range, and a visual tracking camera system for measuring the inertial bearing from one satellite to another (Markley, 1984). Due to the limited range constraints of passively illuminated visual tracking systems, many of the previous studies restrict the separation between satellites to less than 1,000 kilometers (e.g., Psiaki, 2011), or simply drop the use of measuring intersatellite bearing and rely solely on obtaining a large distribution of intersatellite range measurements for state estimation (e.g., Xu et al., 2014). These assumptions have limited the assessment of the performance capability of autonomous navigation using intersatellite measurements for more general mission applications.

In this thesis, we investigate the performance of using laser communication (lasercom) crosslinks in order to achieve all necessary intersatellite measurements for autonomous navigation. Lasercom systems are capable of measuring both range and bearing to a receiving terminal with greater precision than traditional methods, and can do so over larger separations between satellites. We develop a simulation framework to model the measurements of intersatellite range and bearing using lasercom crosslinks in distributed satellite systems, with consideration of varying orbital op-

erating environments, constellation size and distribution, and network topologies. We implement two estimation algorithms: an extended Kalman filter (EKF) used with Monte Carlo sampling for performance analyses, and a Cramér-Rao lower-bound (CRLB) computation for uncertainty analyses.

We evaluate several case studies modeled off of existing and planned constellation missions in order to demonstrate the new capabilities of performing intersatellite navigation with lasercom links in both near-Earth and deep-space orbital applications. Performance targets are generated from the current state-of-the-art navigation capabilities demonstrated by Global Navigation Satellite Systems (GNSS) in Earth-orbit, and by radiometric tracking and orbit estimation using the Deep Space Network (DSN) in deep-space orbits.

For Earth-orbiting applications, we simulate a relay satellite system in geosynchronous orbit (GEO) inspired by current optical communications missions in development by both ESA and NASA, and Walker constellations in LEO inspired by the upcoming mega-constellations for global broadband internet service, such as those proposed by SpaceX and Telesat. In both case studies, we demonstrate improved navigation performance over the current state-of-the-art in GNSS receiver technologies by using intersatellite measurements from lasercom crosslinks. Monte Carlo simulations show median total position errors better than 3 meters in LEO, 12 meters in GEO, and 45 meters in high-altitude or highly-eccentric orbits (HEO), showing promise as an alternative navigation method to GNSS in Earth-orbiting environments.

We also simulate current and future Mars-orbiting missions as examples of deep-space applications. In one case study, we create an ad-hoc constellation comprised of low-altitude Mars exploration orbiters modeled off of current Mars-orbiting missions. In a second case study, we focus on a high-altitude constellation proposed for dedicated Earth-to-Mars networked communications. Again, in both case studies, we demonstrate improved navigation performance over the current state-of-the-art in DSN radiometric orbit solutions by using intersatellite measurements from lasercom crosslinks. Monte Carlo simulations show stable median total position errors better than 25 meters for Mars-orbit, which demonstrates a notable improvement both spatially and temporally versus DSN orbit estimation, mitigating the large cost and time constraints associated with DSN tracking.

These results demonstrate the promise of using lasercom intersatellite links for autonomous navigation, offering enhanced performance over current state-of-the-art capabilities, and a greater applicability to missions both near Earth and beyond.

Thesis Supervisor: Kerri L. Cahoy
Title: Associate Professor of Aeronautics and Astronautics

Thesis Committee Member: Richard Linares
Title: Assistant Professor of Aeronautics and Astronautics

Thesis Committee Member: Timothy M. Yarnall
Title: Assistant Group Leader, MIT Lincoln Laboratory

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Nomenclature

Symbols

α	Right-ascension angle
δ	Declination angle
λ	Longitude
μ	Standard gravitational parameter
ν	True anomaly
Ω	Right-ascension of the ascending node
ω	Argument of periapsis
ϕ	Azimuth angle
ρ	Range, or distance
σ	Uncertainty/noise
θ	Elevation angle
a	Semi-major axis
e	Eccentricity
h	Height, or altitude
i	Inclination

J_2	Dynamic oblateness, the gravity model spherical harmonic coefficient of degree 2 and order 0
M	Mean anomaly
n	Mean motion
R	Equatorial radius
T	Duration of time
t	Time

Satellite Missions

ARTEMIS	Advanced Relay and Technology Mission Satellite (ESA)
EDRS-A	First EDRS payload (ESA), on Eutelsat-9B
EDRS-C	Second EDRS payload (ESA), on dedicated satellite of same name
LADEE	Lunar Atmosphere and Dust Environment Explorer (NASA)
LCRD	Laser Communications Relay Demonstration (NASA), on STPSat-6
LLCD	Lunar Laser Communications Demonstration (NASA)
MAVEN	Mars Atmosphere and Volatile Evolution orbiter (NASA)
MEX	Mars Express orbiter (ESA)
MMS	Magnetospheric Multiscale (NASA)
MO	2001 Mars Odyssey orbiter (NASA)
MOM	Mars Orbiter Mission (ISRO), also called "Mangalyaan" which translates to "Mars-craft" in English
MRO	Mars Reconnaissance Orbiter (NASA)

NFIRE	Near Field Infrared Experiment satellite (DoD)
OICETS	Optical Inter-orbit Communications Engineering Test Satellite (JAXA), also called "Kirari" which translates to "glitter" or "twinkle" in English
STPSat-6	Space Test Program Satellite-6 (DoD)
TGO	ExoMars Trace Gas Orbiter (ESA/Roscosmos)
TSX	TerraSAR-X satellite (DLR)

Agencies & Organizations

AGI	Analytical Graphics, Inc. (U.S. company)
CNES	Centre National D'Etudes Spatiales (France), which translates to "National Center for Space Studies" in English
CSpOC	Combined Space Operations Center (DoD), formerly JSpOC
DLR	Deutsches Zentrum für Luft- und Raumfahrt (Germany), which translates to "German Center for Aviation and Space Flight" in English
DoD	Department of Defense (U.S.A.)
ESA	European Space Agency (22 member states)
ISRO	Indian Space Research Organisation (India)
JAXA	Japan Aerospace Exploration Agency (Japan)
JPL	Jet Propulsion Laboratory (NASA)
NASA	National Aeronautics and Space Administration (U.S.A.)
NGA	National Geospatial-Intelligence Agency (DoD)
NORAD	North American Aerospace Defense Command (U.S.A./Canada)

Roscosmos	State Corporation for Space Activities (Russia)

Official Programs, Products, or Technologies

DIODE	DORIS Immediate Orbit Determination (CNES)
DORIS	Doppler Orbitography and Radiopositioning Integrated by Satellite (CNES)
DSAC	Deep Space Atomic Clock (JPL)
DSN	Deep Space Network (NASA)
DSOC	Deep Space Optical Communications (NASA)
EDRS	European Data Relay System (ESA)
EGM2008	2008 Earth Gravitational Model (NGA)
GNSS	Global Navigation Satellite System
HORIZONS	On-line solar system data and ephemeris computation service (JPL)
HPOP	High-Precision Orbit Propagator (AGI)
IGS	International GNSS Service
MRO110C	2012 Mars gravity model with MRO data, to degree & order 110 (JPL)
STK	Systems Took Kit (AGI)

Other Abbreviations

ADC	Attitude determination and control
C&DH	Command and data-handling
CRLB	Cramér-Rao lower bound

DDOR	Delta differential one-way ranging
DF	Dual-frequency
DOR	Differential one-way ranging
DTE	Direct-to-Earth communication
EHS	Earth horizon sensor
EKF	Extended Kalman filter
FSOC	Free-space optical communication
GEO	Geosynchronous orbit
GSO	Geostationary orbit
HEO	Highly-elliptical orbit
ISL	Intersatellite links
KF	Kalman filter
Lasercom	Laser communication
LEO	Low Earth orbit
MEO	Medium Earth orbit
OD	Orbit determination
ODE	Ordinary differential equation
OpNav	Optical navigation
PF	Particle filter
PNT	Positioning, navigation, and timing
POD	Precise orbit determination

RF	Radio frequency
RMSE	Root-mean-square error
RTN	Radial, tangential/transverse, normal reference frame
SAR	Synthetic aperture radar
SF	Single-frequency
SLR	Satellite laser ranging
SRIF	Square-root information filter
SS	Sun sensor
SSO	Sun-synchronous orbit
ST	Star tracker
SWaP	Size, weight, and power
TLE	Two-line element set
TT&C	Tracking, telemetry, and command
UHF	Ultra-high frequency
UKF	Unscented Kalman filter
UTC	Primary world time standard, known as “Coordinated Universal Time" in English and “Temps Universel Coordonné” in French
VLBI	Very-long baseline interferometry
XNAV	X-ray pulsar-based navigation and timing

Chapter 1

Introduction

1.1 Motivation

The satellite industry is undergoing a shift towards smaller and more affordable spacecraft that can take advantage of reduced cost to launch [15]. This trend is ushering in new distributed satellite architectures to address critical mission areas. For example, Planet Labs Inc. has achieved unprecedented Earth imaging data with their 175+ small-satellite constellation providing global coverage and rapid revisits [16]. Beyond Earth, NASA JPL successfully demonstrated the first interplanetary CubeSats as technology demonstrators of a "carry-your-own-relay" concept, launching two MarCO spacecraft alongside the InSight Mars lander spacecraft specifically for the purpose of relaying data during InSight's critical landing sequence back to Earth [17].

Distributed architectures, however, come with a fair share of challenges. Though satellite development and manufacturing should benefit from the economy of scale, the same cannot yet be said for ground operations. Typical satellite operations for monitoring vehicle health and status, data processing, orbit determination, task planning and scheduling, and fault response are handled by teams of people per satellite. The brute-force method of handling the operations of many satellites would be building more ground stations and training more staff, thus resulting in greater cost and higher risk/complexity across a global system.

Another way to tackle concerns in operating large numbers of satellites is to de-

velop higher levels of autonomy within the satellite system. Tasks typically performed by an operator and/or ground station could be shared with or fully performed by the spacecraft. Examples include planning and scheduling tasks to meet multiple objectives (e.g., maximizing data return while minimizing latency) [18], on-board data-processing to identify issues [19] or deliver data products more quickly [20], and open-loop instrument operation to obtain higher-quality data (e.g., changing sensing configurations based on geolocation or features identified in the sensor data) [8].

In order to achieve higher levels of autonomous operations, satellites should be able to precisely estimate their position and velocity states in real-time. Many address orbit determination and navigation using Global Navigation Satellite Systems (GNSS). GNSS-based navigation has become more widely used as receivers and antennas have become more volume- and cost-effective, as they have shown the ability to provide sub-meter positioning accuracy for low Earth orbit (LEO) objects. However, achieving those small errors is highly dependent on regular or continuous receipt of ultra-rapid ephemerides and clock products from the International GNSS Service (IGS) [7]. GNSS-equipped LEO satellites more commonly achieve real-time navigation solutions on the order of 5 meters [21]. Missions in higher altitude orbits see further degraded performance as the primary service volume currently ends at 3,000 km in altitude [22].

Certain missions and applications require an alternative and enhanced autonomous navigation solution to GNSS. Military satellite applications may require redundancy in order to continue operations in potential GNSS-denied environments. Missions operating at high altitudes or orbiting other bodies (e.g., the moon or Mars) are not able to rely on GNSS-like navigation without additional relay/navigation satellite infrastructure.

The purpose of this thesis is to demonstrate an autonomous navigation method leveraging the distribution and potential connectivity of satellites in a constellation architecture to provide an alternative and enhanced navigation solution to GNSS for Earth-orbit and DSN for deep-space orbital missions. Section 1.2 provides an overview of the topics of satellite navigation methods, intersatellite links, and laser

communications in satellite systems. Section 1.3 summarizes the research objectives and contributions contained in this thesis.

1.2 Context

1.2.1 Satellite Navigation

The field of satellite navigation can be divided into two categories: (1) traditional methods of precise orbit determination (POD) using ground-sensors, and (2) autonomous methods using on-board sensors and navigational references. Autonomous navigation can be further separated into fully-autonomous methods, which are completely independent of man-made or ground-based resources, and semi-autonomous methods, which may use some external resources. Figure 1-1 summarizes the common methods in each of these categories. The following sections go into further detail on satellite navigation methods, and their relevance to this thesis.

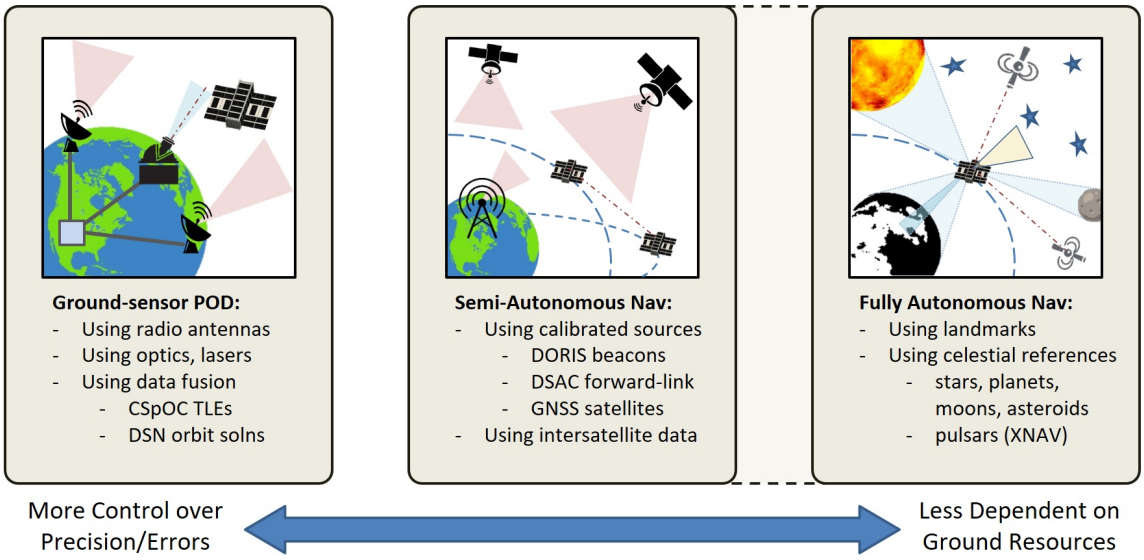


Figure 1-1: Overview of satellite navigation methods, categorized into three areas: ground-based precise orbit determination, semi-autonomous navigation, and fully autonomous navigation.

Traditional Methods

Traditional methods of satellite orbit determination involve the use of ground-based sensors to track satellites and collect measurements, such as radars for range and range-rate, and telescopes for bearing angles [1, 3, 4]. This data is then processed in a filtering algorithm to estimate the six-element state of the satellite, in the form of either classical Keplerian elements such as (a, e, i, Ω, ω, ν) or 3-dimensional position and velocity (x, y, z, $\dot{x}$, $\dot{y}$, $\dot{z}$).

The U.S. Combined Space Operations Center (CSpOC) tracks all satellites using a global network of radar and optical sensors, and publishes estimated state results in the form of two-line element sets (TLEs) for general public use. While this is useful state information for propagation forward or backward in time, performance is limited and degrades over time since the TLE epoch [23]. CSpOC's published TLE data is also prone to errors if/when satellites are in close proximity (e.g., when multiple satellites are inserted into the same orbit around the same time) [24].

Autonomous Methods

To mitigate reliance on TLEs or building additional networks of ground sensors, several methods of autonomous navigation have been proposed and used over the last few decades. Autonomous navigation refers to satellites performing on-board, real-time navigation without external input. Such methods allow for reduced cost and risk in ground operations by eliminating the computational and logistical load of regularly tracking the satellites, estimating their individual states, and transmitting the solutions to the spacecraft [25, 26].

Fully autonomous navigation is performed without any external input, only using the sensors and instruments on-board the spacecraft. Multiple techniques exist using some combination of sensors, such as cameras and infrared sensors (which can estimate state with respect to known reference points or landmarks) [13], magnetometers (which can estimate state with respect to a known magnetic field) [12], and X-ray detectors (which can estimate state with respect to known pulsars) [10]. Semi-

autonomous navigation is also performed using sensors and instruments on-board the spacecraft, but with some external input. This includes any observations made with respect to another man-made object, such as another satellite, a ground station, or a beacon. All satellite navigation using a Global Navigation Satellite System (GNSS) receiver is considered semi-autonomous, since it uses signals from a man-made source [27].

Figure 1-2 illustrates orders of magnitude of position errors demonstrated using several different techniques of precise orbit determination and autonomous navigation. Only two satellite payloads have demonstrated sub-meter real-time navigation accuracy: GNSS receivers and DORIS/DIODE. DORIS/DIODE is a positioning method developed by the French space agency, CNES, that relies on ground beacons transmitting to the DORIS receiver onboard the satellite [5]. Orbit determination software specially developed to process the DORIS data (DIODE) is used to generate navigation solutions on the order of centimeters, however this has only been demonstrated for low-altitude satellites in circular orbits [28].

Since the objective of this thesis is to provide an alternative navigation method to GNSS, other methods are investigated. The fully autonomous methods shown in Figure 1-2 are only capable of providing solutions on the order of 10s-100s of meters, and utilize sensors that are heavily dependent on orbit conditions and well-characterized references. For instance, magnetometers and landmark-based sensors will likely lose sensing observability at higher altitudes; plus magnetic field models and well-characterized landmarks cannot be readily assumed for other central-bodies.

The semi-autonomous navigation method using intersatellite data shows promise for use in distributed constellations. Intersatellite data refers to measurements of relative range, range-rate, or angles between two spacecraft. Previous work has found the intersatellite autonomous navigation method to be accurate on the order of 10s of meters [9, 11], which is better than the fully autonomous methods and resembles the performance of single-frequency GNSS receivers, though results have not yet been experimentally demonstrated on-orbit. Still, the promise of the intersatellite method to this thesis is realized in its independence of the orbit configuration of either satellite,

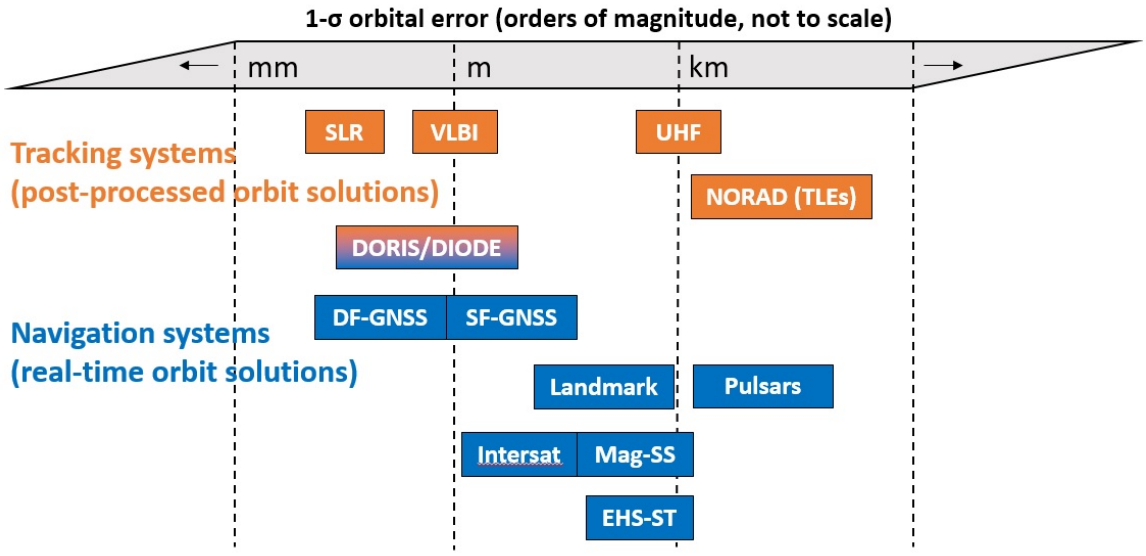


Figure 1-2: Typical 1-sigma orbit estimation error achieved using ground-based tracking and autonomous navigation methods. SLR: Satellite laser ranging, VLBI: Very-long baseline interferometry, UHF: Ultra-high frequency radio ranging, NORAD (TLEs): Two-line element sets published by the North American Aerospace Defense Command, DORIS/DIODE: Doppler Orbitography and Radiopositioning Integrated by Satellite ground-beacon system and on-board OD software, DF-/SF-GNSS: Dual-/Single-frequency GNSS receivers, Landmark: Autonomous navigation using landmarks, Pulsars: X-ray pulsar-based navigation, Intersat: Autonomous navigation using intersatellite measurement data, Mag-SS: Autonomous navigation using magnetometer and sun-sensors, EHS-ST: Autonomous navigation using Earth-horizon sensors and star-tracker [1–13].

and thus its ability to leverage greater separations between satellites, such as in a constellation. Assuming intersatellite range and angle measurements are available, this method is viable for satellites in distributed orbits, and could even perform better when used in a constellation.

Current State-of-the-Art

For near-Earth orbiting satellites, Global Navigation Satellite Systems represent the current state-of-the-art for spacecraft navigation, primarily for its capability as an autonomous method, its availability in different orbital regime, and most importantly, its performance. Recent literature has shown that GNSS receivers are capable of

achieving solution errors on the order of 3-5 meters in low Earth orbit (LEO) [21]. A few missions have also been used to test GNSS receivers at altitudes greater than the constellation altitude, and have admirably shown solution errors on the order of 12-15 meters in geosynchronous orbit (GEO) [29], and 25-65 meters in highly eccentric orbit (HEO) [22].

For deep-space applications, the Deep Space Network of tracking stations represents the current state-of-the-art for spacecraft navigation, capable of navigation accuracy (for Mars-orbiters) down to 25 meters during periods of DSN downlink, and 100 meters during periods without downlink [30].

These navigation performance values are used to generate performance targets for the intersatellite navigation method using lasercom crosslinks. In each case, we set a threshold to which we will compare the results of different orbital scenarios in this thesis. These are 3 meters for LEO, 12 meters in GEO, 45 meters in HEO, and 25 meters in Mars-orbit.

1.2.2 Crosslink Communications

Intersatellite measurements may easily be made if the constellation is designed for crosslink communications, also known as intersatellite links (ISLs). One benefit of implementing crosslinks in a satellite system is increasing ground access for spacecraft tracking, telemetry, and command (TT&C) operations without needing to add more ground stations. For communications and Earth-observing missions, crosslinks can also serve to increase data-to-ground by mitigating downlink bottlenecks and reducing latency between terminals [18, 31]. For Positioning, Navigation, and Timing (PNT) missions, crosslinks are used to improve the quality of the PNT information, increase system reliability and robustness, and enhance availability to higher altitude orbits [32, 33]. In addition to intersatellite measurements, crosslinks also enable other new capabilities in satellite systems, such as distributed processing, and new methods of navigation or formation-control by communicating measurements or state estimates between vehicles [34]. A number of previous studies make the assumption that intersatellite links exist in future constellation systems, and can be leveraged for

autonomous navigation. A more detailed review of previous work can be found in Section 2.2. This thesis also assumes the existence of crosslinks for use in autonomous navigation.

Lasercom Links

Lasercom provides improved energy efficiency, data rates, latency, and security compared with traditional radio-frequency (RF) communications systems [35]. Muri and McNair (2012) performed a survey of all intersatellite link demonstrations until 2012 as well as those that were expected to launch through 2015 [31]. Since the time of Muri and McNair (2012), there has been an notable uptick in the number of launched and proposed missions to utilize optical ISLs, as shown in Table 1.1. Recent years have seen multiple European Data Relay Satellite (EDRS) systems launched into GEO for the purpose of quickly and efficiently relaying critical Earth-observation data from ESA's Copernicus/Sentinel satellites to ground [36]. The upcoming slate includes similar relay technology demonstrators from JAXA, NASA, and Airbus.

Table 1.1: Survey of lasercom crosslink missions. Note that the line-break in the table distinguishes past and present missions (above the line) from future missions (below). Future missions are given *expected dates* (in italics) based on the references cited.

Year [ref]	**Platforms**	**Environment**	**Mission**
2001 [31]	SPOT-4 – ARTEMIS	LEO–GEO	Technology Demo. (1st one-way link)
2005 [31]	OICETS – ARTEMIS	LEO–GEO	Technology Demo. (1st two-way link)
2008 [31]	TerraSAR-X – NFIRE	LEO–LEO	Technology Demo. (1st LEO-LEO link)
2014 [37]	Sentinel-1A – Alphasat	LEO–GEO	GEO Relay Technology Demo. [ESA]
2016-present [36]	Sentinel (x4) – EDRS (x2)	LEO–GEO	Operational GEO Relay System [ESA]
2019 [38]	AeroCube-7B – ISARA	LEO–LEO	CubeSat Pointing Demo. [Aerospace Corp.]
2020 [39]	ILLUMA-T (ISS) – LCRD	LEO–GEO	GEO Relay Technology Demo. [NASA]
2020 [40]	ALOS-4 – JDRS	LEO–GEO	GEO Relay Technology Demo. [JAXA]
2020 [41]	SpaceX "Starlink" Constellation	LEO–LEO (Walker)	Global Broadband Communications
2021 [42]	CLICK-B – CLICK-C	LEO–LEO	CubeSat Payload Demo. [MIT, UF]
2021 [43]	Cloud Constellation "SpaceBelt"	LEO–LEO (Ring)	Global Data Security
2022 [44]	TeleSat LEO Constellation	LEO–LEO (Walker)	Global Broadband Communications
2022 [45]	Laser Light Comms. "HALO"	MEO–MEO (Ring)	Global Network Communications Service
2023 [39]	Gen-2 GEO Relay	LEO–GEO(–GEO)	Operational GEO Relay System [NASA]

McCandless and Martin-Mur (2016) also investigated a future DSN-like network using optical communications systems and performed simulations to compare position accuracy with the current capability of DSN radiometric tracking for a future Mars lander and Mars orbiter. The authors concluded that optical tracking stations could feasibly be an improvement over the current radiometric capability [30].

This thesis builds on the notion of using lasercom systems for crosslinks in order to collect intersatellite data measurements for autonomous navigation.

1.3 Thesis Overview

In order to measure intersatellite range and bearing between satellites in a constellation, we propose using lasercom crosslinks, expanding on the concept studied by Yong et al. (1983) [46]. Figure 1-3 provides a depiction of our proposed method.

Time-of-flight ranging measurements can be captured using time-embedded signals over communications links [47]. Laser communications payloads operate at optical wavelengths with greater frequency bandwidth than RF systems. The higher bandwidth enables lasercom systems to measure range between two terminals with more precision. RF systems typically achieve meter-level ranging precision [48]. Lasercom systems can achieve centimeter-level precision; the current state-of-the-art performance in satellite lasercom ranging was demonstrated to about 1-cm precision in two-way links from Earth to lunar orbit during the Lunar Laser Communication Demonstration (LLCD) mission in 2015 [49]. However, this was a ground-to-satellite demonstration, not satellite-to-satellite.

When compared to RF systems, intersatellite lasercom systems have narrow beamwidths, typically less than 1 degree [42]. In order to point the narrow beam signal directly at the receiving terminal, lasercom demands highly accurate pointing systems, which can be leveraged to obtain the bearing between the two satellites. Assuming a lasercom crosslink can be established between two satellites, we propose deriving intersatellite bearing by determining the body attitude of the spacecraft using an on-board star tracker, using any angular offsets between boresights of the lasercom transceiver and

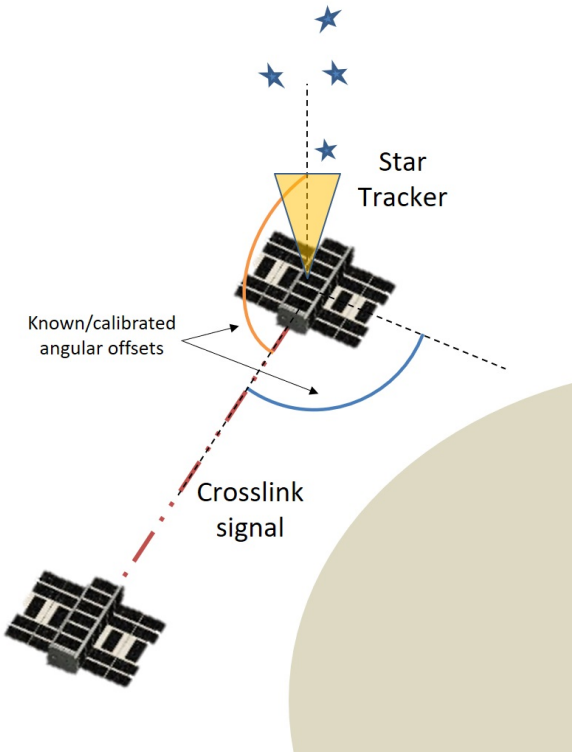


Figure 1-3: Diagram of proposed method of autonomous navigation using laser communication crosslinks between two or more satellites, co-orbital around any central-body with known gravity characteristics (mass and J2-term). This method assumes that the observing spacecraft have star-trackers for inertial attitude knowledge, and known angular offsets between the boresights of the star tracker and the lasercom terminal, and other on-board systems necessary to establish and maintain a crosslink signal with another satellite.

star tracker. If the lasercom payload is fixed to the spacecraft body, these offsets are fixed values that are either known from the design of the spacecraft, or defined using an on-board calibration process. If the lasercom payload is on a gimbal, then commanded offsets must also be included in order to fully capture angular offsets between the boresights of the payload and star tracker.

Long-range RF crosslinks typically cannot be used to similarly derive bearing to the receiving satellite due to the relatively large beamwidth of RF systems. For instance, GNSS satellite antenna half-beamwidths are on the order of 23-26 degrees [50]. In order to measure intersatellite bearing, additional sensors must also be used, such as telescopes or camera/beacon systems. Passive electro-optical sensors are

limited by observability constraints that effectively limit the separation between the two satellites in which bearing can be measured. For example, cameras are affected by apparent magnitude and proper illumination of the other spacecraft [48], and can only work for separations of a few hundreds of kilometers between satellites. Active sensors, such as beacon systems, can be used for longer ranges [51, 52]. Lasercom crosslinks can be considered a form of an active electro-optical system, capable of working over longer ranges [38, 42].

For this study, we assume that all satellites have at least one lasercom terminal, with parameters consistent with those offered by commercial companies [53] or currently being developed [42], along with the accompanying subsystems for power, attitude determination and control (ADC), and command & data handling (C&DH). These systems must operate together in order to establish and maintain a lasercom crosslink between two communicating spacecraft. We assume that crosslinks are established using methods such as performing a scanning maneuver or implementing a coarse beacon system [42].

1.3.1 Research Contributions

The three contributions in this thesis are:

- Creation of a simulation framework to estimate the performance of autonomous navigation methods using intersatellite measurements for varying orbital environments, satellite constellations, sensor network configurations, and measurement models. This includes a kinematic uncertainty approximation algorithm using Cramér Rao lower bound (CRLB) covariance estimates as a link selection heuristic when multiple links are available.

- Analysis of the navigation performance over relevant past, present, and future crosslinked satellite missions to demonstrate the method's applicability to both near Earth and deep space environments, including assessment of lasercom interconnectivity on constellation navigation.

- Demonstration of improved navigation performance using autonomous lasercom crosslinks over current spacecraft navigation techniques (GNSS for Earth orbiters, DSN for deep space) with median total position errors (from Monte Carlo simulations) better than 3 meters for LEO, 12 meters for GEO, 45 meters for HEO, and 25 meters for Mars orbiters.

1.3.2 Organization

Chapter 2 contains background in spacecraft navigation and a review of previous literature in the area of intersatellite navigation with the goal of identifying research gaps to be addressed by the work in this thesis. Chapter 3 describes the simulation approach, and Chapters 4 and 5 present the results of simulations in Earth-orbiting and Mars-orbiting applications, respectively. Chapter 6 summarizes the contributions and conclusions made in this thesis.

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Chapter 2

Background & Literature Review

This chapter provides a high-level review of traditional spacecraft navigation estimators and measurements, then performs a deeper analysis of previous work in autonomous navigation using intersatellite measurements in order to identify research gaps and highlight relevant insights towards achieving the objectives of this thesis.

2.1 Spacecraft Navigation Methodology

Spacecraft navigation estimation relies on using a time-series of measurements to estimate observables that are directly influenced by the gravitational forces existing in the space environment. This section describes commonly used estimation methods and measurements for spacecraft navigation.

2.1.1 Estimation Methods

State estimation can either be done by considering measurements all-at-once (a batch process) or as each measurement is made (a sequential process). Batch processes represent the traditional methods for determining post-processed orbit solutions. Sequential processes represent filtering methods that can be used for real-time navigation solutions.

Batch Least-Squares Processing

Least-squares estimation is a commonly used method for approximating the values of unknown parameters that best-fit the input data based on an expected model of behavior. The term "least-squares" refers to the objective of minimizing the sum of the square of the difference between an observed value and an expected value based on the model. This is captured by the following equation:

$$\hat{\mathbf{x}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{y} \tag{2.1}$$

where $\hat{\mathbf{x}}$ is the estimated state, and $\mathbf{H}$ is the measurement model that describes the relationship between the state vector, $\mathbf{x}$, and the measurements, $\mathbf{y}$. For a nonlinear, differentiable measurement model function $h(\mathbf{x})$ and measurement noise $\mathbf{v}$:

$$\mathbf{y} = h(\mathbf{x}) + \mathbf{v} \tag{2.2}$$

the following Jacobian can be calculated to linearize the model at each measurement time-step:

$$\mathbf{H} = \frac{\partial h}{\partial \mathbf{x}} \tag{2.3}$$

Sequential Kalman Filtering

The Kalman Filter (KF) is an optimal estimator for linear Gaussian systems. As a method of sequential processing, it is commonly used for real-time navigation solutions. The filter is divided into two phases, the prediction phase, which propagates the most recent state estimate and covariance to the current time-step, and the update phase, which uses any available measurements at the current time-step to update the predicted state estimate and covariance.

The dynamics model (or state-transition model) $\mathbf{F}_k$ is described by:

$$\mathbf{x}_k = \mathbf{F}_k\mathbf{x}_{k-1} + \mathbf{w}_k \tag{2.4}$$

where $\mathbf{x}_k$ are the state variables to be estimated at each filter step $k \in \{1, \cdots, N\}$,

and $\mathbf{w}_k$ is the process noise with normal distribution and covariance $\mathbf{Q}_k$.

The measurement model is described by:

$$\mathbf{y}_k = \mathbf{H}_k \mathbf{x}_k + \mathbf{v}_k \tag{2.5}$$

where $\mathbf{y}_k$ are the measurements observed at each filter step $k \in \{1, \cdots, N\}$, and $\mathbf{v}_k$ is the measurement noise with normal distribution and covariance $\mathbf{R}_k$.

The prediction phase is:

$$\hat{\mathbf{x}}_{k|k-1} = \mathbf{F}_k \hat{\mathbf{x}}_{k-1} \tag{2.6}$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}_k \mathbf{P}_{k-1} \mathbf{F}_k^T + \mathbf{Q}_k \tag{2.7}$$

where $\mathbf{x}_0$ and $\mathbf{P}_0$ reflect the *a priori* knowledge of the initial state and covariance values.

The update phase is:

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k[\mathbf{y}_k - \mathbf{H}_k \hat{\mathbf{x}}_{k|k-1}] \tag{2.8}$$

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k)\mathbf{P}_{k|k-1} \tag{2.9}$$

where $\mathbf{K}_k$ is the Kalman gain:

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R}_k)^{-1} \tag{2.10}$$

The Extended Kalman Filter (EKF) performs the same computations as the Kalman Filter except that the dynamics and measurement models may be nonlinear, differentiable functions, $f(\mathbf{x})$ and $h(\mathbf{x})$, respectively:

$$\mathbf{x}_k = f(\mathbf{x}_{k-1}) + \mathbf{w}_k \tag{2.11}$$

$$\mathbf{y}_k = h(\mathbf{x}_k) + \mathbf{v}_k \tag{2.12}$$

Jacobian matrices, $\mathbf{F}_k$ and $\mathbf{H}_k$, are computed matrices for use in the prediction and update phases:

$$\mathbf{F}_k = \left.\frac{\partial f}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_{k-1}} \tag{2.13}$$

$$\mathbf{H}_k = \left.\frac{\partial h}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_{k|k-1}} \tag{2.14}$$

The use of Jacobians linearizes the nonlinear functions around the most recent estimate, which is effective for low-order nonlinearity, but can be problematic for highly nonlinear functions for which Jacobian matrices cannot be easily derived (analytically or numerically). For these types of problems, the Unscented Kalman Filter (UKF) can be used. The UKF avoids linearization via Jacobians by computing mean and covariance estimates using a set of deterministic sample points (called sigma points) propagated through the nonlinear functions $f(\mathbf{x})$ and/or $h(\mathbf{x})$.

Kalman filters and its variants are commonly used for navigation problems, but are known to exhibit initialization and numerical stability issues. Applications that may be affected by such effects often implement a Square-Root Information Filter (SRIF) instead, which propagates the square-root of a state information matrix instead of a state-transition matrix that requires matrix inversions, leading to more robust numerical stability and the ability to handle high initial variance. Another common estimator is the Particle Filter (PF), which uses probability density functions to randomly generate sample points to model nonlinear dynamical systems. This is similar to the UKF, although PF methods tend to use a higher quantity of sample points. Thus PF methods have seen more use recently as computation and processing have become faster and cheaper. Only the EKF is used in this thesis, but other filters like SRIF and PF can be implemented in future work.

Cramér-Rao Lower Bound (CRLB)

The Cramér-Rao lower bound computes the lowest achievable covariance or expected errors assuming optimal filter performance. It is commonly used to evaluate state estimator precision and observability [48, 54–56]. Relevant to the context of this thesis, it can also be used to predict covariance for "efficient" estimators, those that

achieve the CRLB, such as the extended Kalman filter [54]. The CRLB is defined by the following equations:

$$\mathbf{P}_k \geq \mathbf{J}_k^{-1} \tag{2.15}$$

where

$$\mathbf{J}_k = (\bar{\mathbf{F}}_{k-1}\mathbf{J}_{k-1}^{-1}\bar{\mathbf{F}}_{k-1}^T + \mathbf{Q}_k)^{-1} + \bar{\mathbf{H}}_k^T\mathbf{R}_k^{-1}\bar{\mathbf{H}}_k \quad , \quad \mathbf{J}_0 = \mathbf{P}_0^{-1} \tag{2.16}$$

2.1.2 Measurement Types

This section presents background framework on common measurements made for spacecraft navigation. For each measurement type, we describe how the measurements are derived, and list a couple example systems that collect that type of measurement.

Range

Range is the measurement of distance or relative position between two points:

$$\rho^{(ij)} = |\boldsymbol{\Delta}\mathbf{r}^{(ij)}| \tag{2.17}$$

where $\boldsymbol{\Delta}\mathbf{r}^{(ij)} = \mathbf{r}^{(j)} - \mathbf{r}^{(i)}$ is the relative position vector between points i and j. In terms of spacecraft navigation, point i describes the Cartesian position of the observer, and point j describes the Cartesian position of the observed satellite. The observer may be a ground-based or space-based sensor.

Range is typically derived by measuring the one-way or two-way time-of-flight, t, of an electromagnetic signal between the two points. Since the signal travels at the speed of light, c, the distance traveled by the signal can be determined by the kinematic equation $d = c \times t$, assuming any sources of error are calibrated out, and any ambiguities are resolved using estimation algorithms. In two-way methods, $\rho = d/2$. This technique is commonly used by radar and laser sensors, such as those in Deep Space Network and Satellite Laser Ranging tracking stations.

Range can also be derived via signal intensity. One example of this technique is in optical astrometry, in which the size of an observed target on its sensor plane is mapped to a specific range value via a well-characterized and calibrated function of size versus range. This is commonly used for optical autonomous navigation (OpNav).

Range-Rate

Range-rate is the rate-of-change in the distance or relative position between two points:

$$\dot{\rho}^{(ij)} = \frac{\mathbf{\Delta r}^{(ij)} \cdot \mathbf{\Delta \dot{r}}^{(ij)}}{|\mathbf{\Delta r}^{(ij)}|} \tag{2.18}$$

where $\mathbf{\Delta \dot{r}}^{(ij)} = \mathbf{\dot{r}}^{(j)} - \mathbf{\dot{r}}^{(i)}$ is the relative velocity vector between points i and j. In terms of spacecraft navigation, point i describes the Cartesian position of the observer, and point j describes the Cartesian position of the observed satellite. The observer may be a ground-based or space-based sensor.

Range-rate is typically obtained by measuring the Doppler frequency shift, f_d, of an electromagnetic signal received or returned by the observed target. This relationship is described by the equation:

$$\dot{\rho} = \frac{\lambda f_d}{2} \tag{2.19}$$

where λ is the wavelength of the transmit signal. This technique is commonly used by radar sensors like DSN antennas, and beacon systems like the DORIS receiver.

Bearing (Angles)

A bearing measurement is the angle between a reference direction and the position of an object, in this case the observed satellite target. For spacecraft navigation, two bearing angles (ϕ and θ) are typically used to fully describe the relative direction to a target, either Azimuth and Elevation, or Right Ascension and Declination (which are equivalent angles in different reference frames):

$$\phi^{(ij)} = \arctan\left(\frac{\Delta y^{(ij)}}{\Delta x^{(ij)}}\right), \quad \text{from 0 to } 2\pi \tag{2.20}$$

$$\theta^{(ij)} = -\arcsin\left(\frac{\Delta z^{(ij)}}{|\mathbf{\Delta r}^{(ij)}|}\right) \tag{2.21}$$

where

$$\Delta x^{(ij)} = x^{(j)} - x^{(i)} \quad , \quad \Delta y^{(ij)} = y^{(j)} - y^{(i)} \quad , \quad \Delta z^{(ij)} = z^{(j)} - z^{(i)} \tag{2.22}$$

Bearing angles can be derived from measuring mechanical displacement (i.e., the angular displacement of a telescope mount or a gimbaled platform in mechanically-pointed sensors), measuring relative signal delays at known positions (e.g., the method used in very-long baseline interferometry, or VLBI), or via astrometry (i.e., determining the attitude of a sensor image relative to the directions of known, fixed objects in the image like stars).

2.2 Autonomous Navigation with Intersatellite Data

2.2.1 Seminal Research

The first study of using intersatellite range and inertially-referenced bearing measurements for determining the orbits of two satellites without any *a priori* information was conducted by F.L. Markley and published in 1984. The author first performed an observability analysis to determine the feasibility of such a method. Using a simple spherical Earth gravity model, Markley concluded that a few certain orbit cases are unobservable using this method: when both spacecraft have equal semi-major axis a, equal eccentricity e, equal "phasing" (i.e., mean-anomaly, ν), and are either coplanar, or "oriented so that the two spacecraft cross the line of intersection of the two planes simultaneously" when non-coplanar [9].

These findings were furthered in a study by M.L. Psiaki in 1999, using a higher-fidelity Earth gravitational model with J_2 secular perturbations. This study con-

cluded that only one orbit case is absolutely unobservable: the same case as described earlier (when both spacecraft have equal a, equal e, and equal ν), but specifically when the satellites are coplanar at the inclination i of 0° . This means that inclined and non-coplanar orbit cases are semi-observable due to the non-spherical Earth, and that errors would grow as the two-spacecraft system more closely resembles the absolute unobservable case. This conclusion largely affects satellites in GEO observing each other, as they are all in low-inclination circular orbits. Missions in other orbits can be affected too, though satellites in LEO and MEO tend to be inclined. We would also like to note that the absolute unobservable case can still be mitigated by observing satellites at other altitudes, such as between GEO and LEO. This is of particular interest for study in this thesis.

Psiaki also established some baseline performance references by reporting position uncertainty as a function of altering intersatellite geometry between two Earth-orbiting satellites. This analysis showed observability errors caused by small separations between satellites (on the order of 100 km), lack of measurement variability in specific state elements, proximity to the absolute unobservable case, orbital altitude, and high angle measurement uncertainty [11]. We intend to report on similar effects but in an expanded range of potential orbital geometries afforded by the longer separations possible using lasercom links.

In this thesis, while the source of the measurement is different from those used in the aforementioned observability analyses (RF systems vs. lasercom systems), the measurement types (range and bearing) have not changed; therefore an observability analysis is not repeated in this work. However, we are interested in evaluating the effect of estimation observability on navigation performance. As mentioned Section 2.1.1, one way to evaluate the effect of observability on error performance is to compute the CRLB. Thus, CRLB computations are an important part of our analyses.

2.2.2 Other Relevant Studies & Analyses

Table 2.1 shows a summary of previous work in autonomous navigation using intersatellite measurements. In reviewing this area of research, we report a few differenti-

ating characteristics of each study that are of particular interest to this thesis. These include the quantity and distribution of satellites and the uncertainty values of the measurements, all as modeled and used in each study's respective simulations. In the final column, we report the magnitude of the resulting navigation performance of each study.

Of the 17 studies reviewed, five studies simulate constellations larger than 4 satellites, five studies simulate intersatellite geometry with separations on the order of tens of thousands of kilometers, six studies model ranging precision below 1 meter, and six studies model angular precision below 5 arcsec. However, only four studies possessed two or more of these characteristics, while only one possessed all, Gao et al. (2014).

Psiaki (1999) modeled precise range and bearing measurements for largely separated satellites, but only for two-satellite Earth-orbiting systems [11]. Zhao et al. (2011) simulated a 30-satellite constellation comprised of distributed satellites in both LEO and GEO using a neutral value of ranging precision, but did not model any bearing measurements (possibly due to the large separations between satellites) [61]. Li et al. (2017) simulates a 6-satellite constellation with very precise bearing measurements, but a strangely high uncertainty in ranging [68]. This study also does not provide any details regarding the orbital configuration of the constellation, and therefore the actual modeled distribution of satellites is unknown.

Gao et al. (2014) was the only study to possess all of the characteristics of interest in this thesis [63]. In this study, the authors propose a navigation constellation comprised of 12 GNSS satellites in MEO and 4 satellites in Earth-Moon Lagrangian point orbits (LPO). Three cases are considered: (1) only GNSS satellites with only ranging measurements, (2) only GNSS satellites with range and angle measurements, and (3) GNSS + LPO satellites with only ranging measurements. The authors conclude that while range and angle measurements (case 2) show improved precision and stability over case 1, ranging with satellites in Lagrangian points demonstrates even better performance. Position errors were reported on the order to tens of meters. Case 2 is of most immediate relevance to this thesis, however no details are given regarding

Table 2.1: Review of previous studies in autonomous navigation using intersatellite measurements. Values in bold indicate those that fit the objective criteria of this thesis.

Year [ref]	**Lead Author**	**# of Sats**	**Range Prec. (m)**	**Bearing Prec. (arcsec)**	**Magnitude of ρ between satellites (km)**	**Magnitude of 1-σ pos. errors (m)**
1984 [9]	Markley	2	2	**2**	10^1-10^3	10^1
1987 [57]	Herklotz	**8**	266	—	$\mathbf{10^4}$	10^2
1999 [11]	Psiaki	2	**0.1**	**0.2-2**	10^1-$\mathbf{10^4}$	10^0-10^2
2004 [58]	Yim	2	—	**3.6**-360	10^2-10^3	10^2-10^4
2005 [59]	Hill	2	1	—	$\mathbf{10^4}$	10^1
2007 [60]	Psiaki	2	1.2	5	10^2	10^1
2011 [61]	Zhao	**30**	**0.75**	—	10^2-$\mathbf{10^4}$	10^1
2013 [48]	Xiong	4	3	5	10^2-$\mathbf{10^4}$	10^1-10^2
2014 [62]	Xu	**24**	3	—	10^3	10^1-10^2
2014 [63]	Gao	**16**	**0.3**	**1**	10^2-$\mathbf{10^4}$	10^1
2016 [64]	Xiong	3	—	**0.7-2.5**	10^2	10^0-10^3
2016 [65]	Wang	2	**0.1**	(0.1 m)*	10^1	10^1-10^2
2016 [66]	Davis	2	**0.2**	—	10^2-10^3	N/A **
2016 [67]	Ou	2	**0.1**	—	10^1	10^1
2017 [68]	Li	**6**	6	**0.3**	N/A ***	10^1
2018 [69]	Ou	2-3	1	5	10^1	10^1
2018 [70]	Ou	2-3	1	(1 m)*	10^1	10^0-10^1

* These studies model intersatellite measurements as relative position vectors in the inertial reference frame, and not as separate range and bearing measurements; therefore, the relative position vector precision is shown instead of angular precision.

** This paper reports on a proposed experiment, but does not provide any expected or simulated results. No follow-on papers were found at the time of this literature review.

*** This study does not provide information on the orbital geometry of the 6-satellite constellation, therefore the separation between satellites is unknown.

how the measurements are made in order to achieve such precise uncertainty values. It is possible that this analysis was simply a thought-exercise in order to demonstrate the enhanced performance of the third case. Case 3 is notable in itself as it highlights another possible orbital scenario that takes advantage of the circular restricted three-body problem (CRTBP) dynamical system, though it does not model any bearing measurements.

2.2.3 Research Gaps

Based on our review of literature in the research area of autonomous navigation using intersatellite measurements, we identify a couple of research gaps that we address in this thesis. One gap is the development of simulation frameworks for evaluating navigation error performance using intersatellite measurements with consideration for variable orbital scenarios, noise/uncertainty models, and network architectures. Another is satellite navigation simulations of distributed constellation architectures using measurement and link-access models consistent with lasercom crosslink systems. Addressing these gaps is part of the research objectives and contributions of this thesis.

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Chapter 3

Simulation Approach

This chapter introduces the simulation approach we use in this thesis to predict and evaluate the performance of the proposed navigation method using intersatellite lasercom measurements. The simulation environment developed for this study is written in Matlab with input data generated using AGI's Systems Tool Kit (STK) software. An overview of the full simulation process is shown in Figure 3-1, and an overview of the case studies considered in this thesis can be found in Table 3.5 (at the end of the chapter). Each of the steps of the simulation framework is described in detail in the following sections.

Throughout this chapter, we use a two-satellite GEO-LEO configuration to serve as an illustrative example for the simulation approach. This configuration is modeled off of the satellites ARTEMIS (in GEO) and OICETS (in LEO), which were used in the first bi-directional lasercom crosslink in 2005 [31]. Figure 3-2 shows the primary end-product of our simulation approach for this example scenario, which plots total position error of the satellites in the scenario over the full simulation period, as affected by geometric and network constraints.

3.1 Orbit Generation & Propagation

The first step in our approach is to select and define the simulation scenario to be studied. The constellation is configured in STK, which is used to propagate and

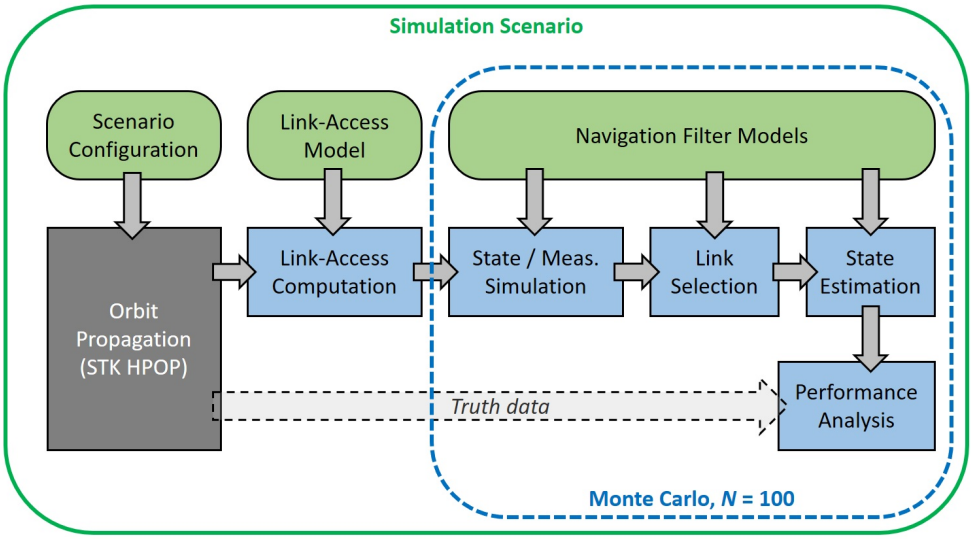


Figure 3-1: Block diagram of simulation approach: green ovals indicate inputs, blue boxes indicate written functions, and the gray box denotes external software or functions.

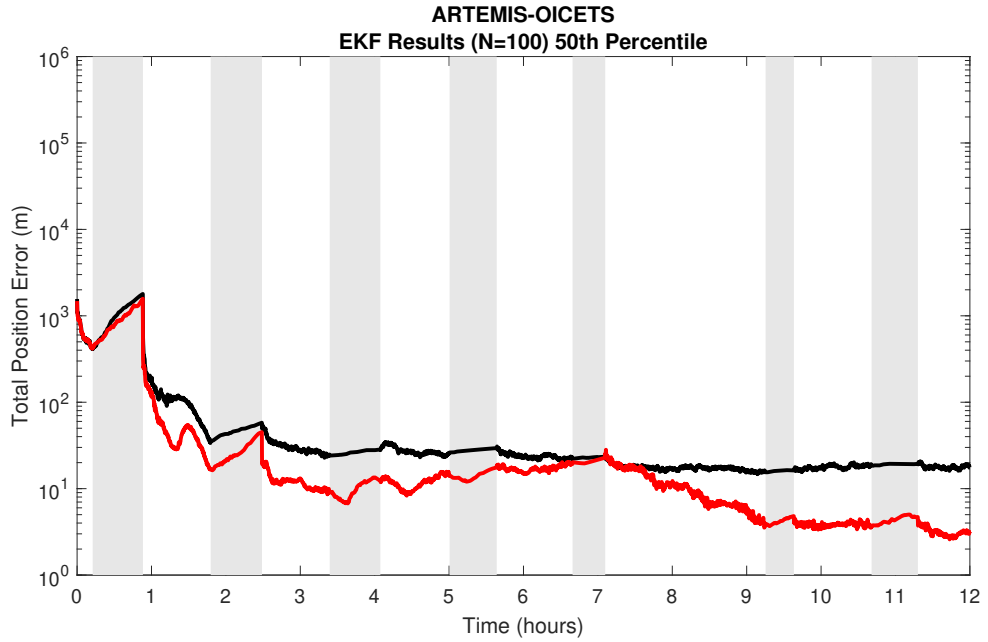


Figure 3-2: Results for the GEO-LEO example case: 50th percentile navigation error over 100 Monte Carlo samples of simulated lasercom crosslink measurements fed through an EKF intersatellite navigation filter. Results for ARTEMIS (GEO) are in black, while results for OICETS (LEO) are in red. Gray vertical bars are used to denote data-gap periods due to inaccessible crosslink geometry.

output orbital state vectors for all satellites in the constellation to be processed by the next steps in the simulation.

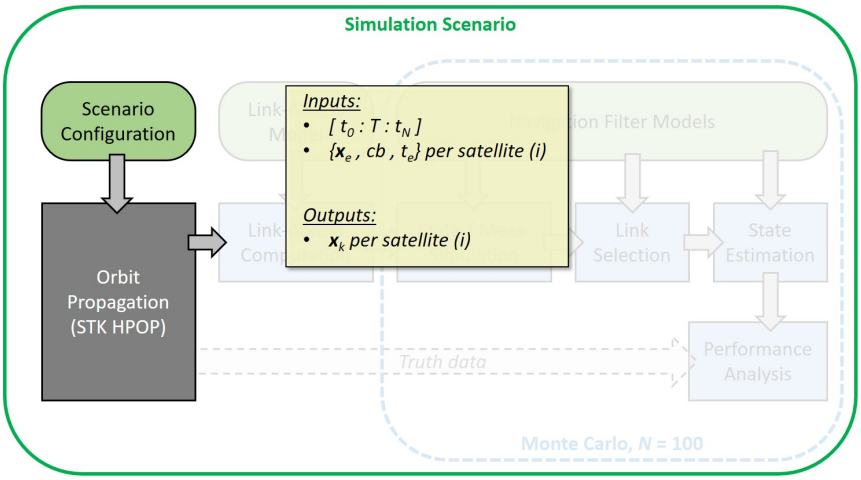


Figure 3-3: Summary of inputs and outputs for the "orbit generation and propagation" step in the simulation approach.

3.1.1 Input: Scenario Configuration

A simulation scenario is comprised of three elements: time parameters to define the simulation period, a constellation configuration to define the intersatellite geometry of the simulated satellites, and network topology parameters to define how the constellation is connected. The simulation period and intersatellite geometry are defined and used in this step to generate and propagate orbits, while the network topology is defined and used in the next step to compute link availability. Figure 3-4 and the following sections provide more detail on the steps to configuring the orbital scenario.

Time Parameters

The first set of time-based parameters establish the time period of the simulation scenario. These are defined as t_0 and t_N, which are the start and end times, respectively, of the simulation period. The time-period and duration of each case study is different, depending on the satellites involved. For instance, our example case for ARTEMIS-OICETS is simulated over a 12-hour time period between t_0 = 16 Mar

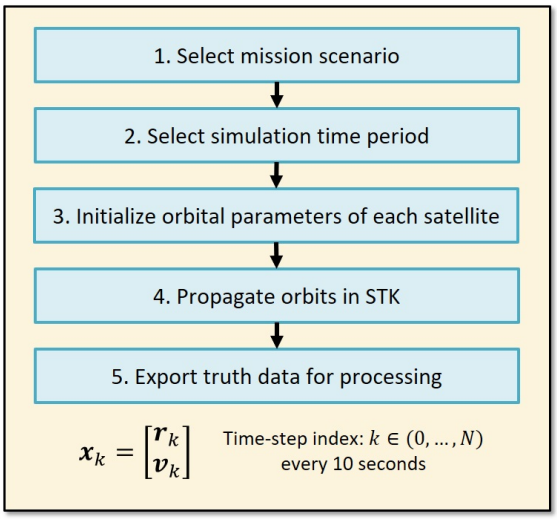


Figure 3-4: Summary of actions taken to configure a simulation scenario.

2015 16:00:00 UTC and t_N = 17 Mar 2015 04:00:00 UTC, while most other scenarios in this thesis are run for 24-hour studies.

A time-step parameter, T, is also defined as the duration between each step k in the navigation filter. Measurements are collected and fed into the navigation filter at every time-step. For all simulation shown in this thesis, we assume a constant time-step of $T = 10$ seconds. A sensitivity analysis of the simulation approach on this input parameter is shown later in Section 3.5.2.

Constellation Configuration

Next, each satellite in the constellation scenario must be initialized. The satellites are indexed by $i \in \{1, \cdots, N_s\}$, where N_s is the total number of satellites ($N_s \geq 2$). One major benefit of the proposed concept of intersatellite navigation is that the spacecraft involved can be operating from a wide set of orbits. The satellites can be in the same or similar orbits, like those typical of LEO constellations, or they can be in very different orbits at very different altitudes. For instance, in our example scenario of ARTEMIS-OICETS, one satellite is stationed in GEO while the other is in LEO. Satellites can also be orbiting other central bodies (e.g., Mars), or neutral-gravity points between central bodies (e.g., halo orbits), or entirely separate bodies

(e.g., Earth and Mars). Therefore, each satellite's initial state, $\mathbf{x}_e^{(i)}$, must be defined (in either Keplerian or Cartesian elements) relative to its central body, $cb^{(i)}$, which has an equatorial radius of R. Generally, the epoch of the initial state $t_e^{(i)}$ is the start time of the simulation period, though it can be defined for a different time, in which case that time must also be specified.

For clarity, all of the studies in this thesis are co-orbital about a single central body (e.g., all Earth-orbiting, or all Mars-orbiting). Though not a focus in this thesis, multiple central body scenarios are possible (e.g., a deep-space crosslink between relay nodes separately orbiting Earth and Mars, or Earth and the Moon), and can be implemented and studied as future work.

Table 3.1 lists the constellation configuration parameters as used for our example scenario between ARTEMIS (in GEO) and OICETS (in LEO). ARTEMIS is no longer in a stable geostationary orbit, as the spacecraft was decommissioned and disposed into a graveyard orbit beyond GEO in late 2017 [71]. Just prior to that, the spacecraft was maneuvered in 2016 from its original operating longitude at 21.5° E to 123° E to serve a commercial mission for Indonesia [72]. In order to present a more true-to-life simulation of the historic GEO-LEO lasercom demonstration, we chose to initialize their orbital states using historical TLE data from a time period prior to these maneuvers.

Table 3.1: Orbital parameters for GEO-LEO example scenario.

Orbital Element	**ARTEMIS (GEO)**	**OICETS (LEO)**
n (rev/day)	1.0027	14.9869
e	3.42e-4	1.58e-3
i (deg)	11.70	98.09
ω (deg)	313.01	312.12
Ω (deg)	42.13	109.62
M (deg)	230.48	99.95
t_e	2015/03/16	2015/03/16
(UTC)	02:03:25.9	01:28:42.75

3.1.2 Output: "Truth" State Data for All Satellites

The scenario configuration information is used to generate orbits and propagate state vector data, $\mathbf{x}_k^{(i)}$, which captures a time-series of position and velocity in Cartesian coordinates for all satellites within the simulation scenario, to be used as "truth" data for the remaining steps in the simulation. For this task, we use the High-Precision Orbit Propagator (HPOP), licensed under the STK software package. HPOP was chosen as it allows us to select the fidelity of the underlying models and computations as needed. Table 3.2 shows the settings that were used for all of the simulations in this thesis. The following sections go into more detail regarding how settings were chosen for the HPOP force models and the coordinate frame of the output state vector data.

Table 3.2: STK settings as used to producing "truth" data for all simulations.

Parameter	**Setting**
Propagator	HPOP
Step Size	1 sec
Earth Gravity File	EGM2008.grv
Mars Gravity File	MRO110C.grv
Maximum Degree	2
Maximum Order	0
Secular Variations	No
Solid Tides	None
Use Ocean Tides	No
Third Body Gravity	None
Use Drag	No
Use SRP	No
Include Albedo	No
Include Thermal	No
Include Relativistic Accelerations	No
Integration Method	RKF 7(8)
Integrator Settings	Defaults
Report Style	Cartesian Position & Velocity
Report Units	km & km/s
Report Coord System	True of Epoch

Force Models

A number of perturbing forces can affect a given satellite's orbit to varying degrees. Satellites in GEO and other high-altitude orbits are more affected by third-body perturbations and solar radiation pressure, while satellites in LEO are more affected by gravitational perturbations and atmospheric drag. A satellite's navigation filter is generally only as good as how well its dynamics model matches the forces affecting that satellite's orbit. While a sufficient navigation filter would account for just the primary perturbing forces, a highly accurate filter would need to account for any smaller forces as well. Therefore, depending on the application or what is being studied, different force models can be used to represent the orbital dynamics.

The research scope of this thesis is to evaluate the performance of intersatellite navigation measurements from lasercom crosslinks. Given that the navigation filter based on intersatellite measurements gains observability in estimating the absolute states of the spacecraft based on how their relative position vector is affected by their individual orbital dynamics, a higher fidelity of perturbing forces included in the dynamics model can lead to greater performance. However, high-fidelity dynamics models also require a higher knowledge of certain time-varying parameters, which can introduce additional sources of error, and also require a higher order of computation and processing in the filter.

In order to focus on the capabilities of the lasercom measurements, and not on how accurately the dynamics model reflects the force models of the propagator, we sought to strike a balance between including any relevant force models and limiting additional sources of filtering error, and deemed it sufficient to only include the perturbing forces that are necessary and can be easily implemented in the dynamics model. As such, we chose to limit the perturbing force models used in the filter dynamics model to only gravitational effects based on the J2-term of the non-spherical gravity model coefficients.

Atmospheric drag and solar radiation pressure are both highly dependent on the geometric characteristics of a spacecraft's physical design and its orientation with

respect to the source of the perturbing force. While the parameters related to the geometric design of each spacecraft would be constant, the attitude of each spacecraft's body, and the magnitude and direction of the forces acting upon the spacecraft are all time-variant. Third-body perturbations are easier to model than atmospheric drag or solar radiation pressure, but were not implemented for the simulations in this thesis, and are slated for future work.

For clarity, though STK includes a built-in analytic J2 propagator, this was not used in this study due to potential for introducing unknown "black-box" sources of error. HPOP, a fully numerical propagator, is preferred, as it can be transparently configured as needed, and also provides the opportunity for future iterations of this work to readily implement and activate additional force models, such as third-body effects.

Coordinate Frames

Including non-spherical gravitational perturbations in the filter's dynamics model leads to a need to properly select the coordinate frame in which the "truth" state vector data is generated. Gravity model coefficients are based on a specific definition of the central body's Z-axis, and each coordinate frame defines their axes to different references.

In order to mitigate the need for time-variant coordinate transformations, we sought to perform all computation in a coordinate frame that is non-rotating (inertial) and shares the same Z-axis as the Fixed coordinate system, the frame in which the gravity model is defined. After inspecting and testing each of the coordinate frames natively offered in STK[1], we determined that the "True of Epoch" coordinate system, with the epoch defined as the start time of each simulation scenario, was best-suited for our simulation approach.

[1]See summary of the different coordinate frames that are offered in STK at <http://help.agi.com/stk/index.htm#stk/referenceFramesCBdescriptions.htm>.

3.2 Link-Access Availability Computation

The second step in our simulation approach is to select and define the link-access model of the constellation in the simulation scenario. This model is used to generate a list of every possible satellite-pairing that can establish a crosslink in the constellation, and compute link-access availability for each pairing over the full duration of the simulation.

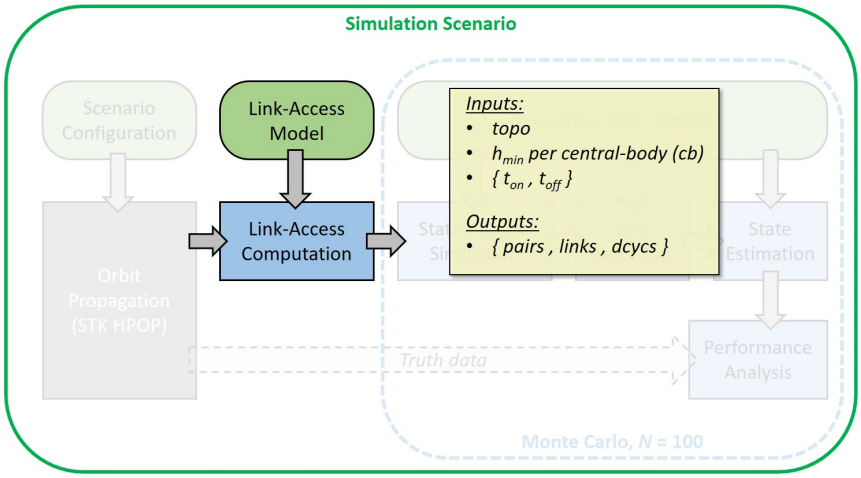


Figure 3-5: Summary of inputs and outputs for the "link-access availability computation" step in the simulation approach.

3.2.1 Input: Link-Access Model

The link-access model describes any restrictions when a link between any two satellites in the chosen constellation is not available. This can include constraints based on the network, geometry, or time. In this thesis, we model one constraint for each of these categories, as shown in Figure 3-6, and described in the following sections.

Network Topology

The network topology input variable, *topo*, is implemented as a set of subsets of satellite indices used to define which other satellites are potential link partners for that particular satellite:

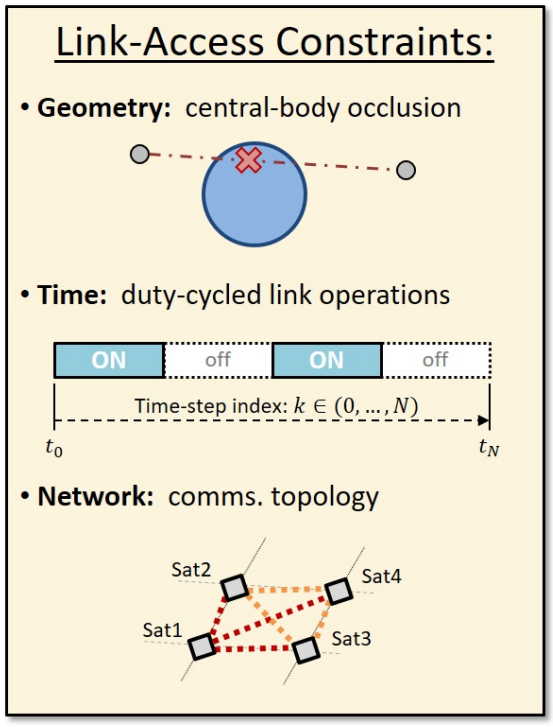


Figure 3-6: Summary of link-access constraints developed for use in this thesis.

$$topo = \left\{ sats^{(i)} \right\}, \quad i \in \{1, \cdots, N_s\} \tag{3.1}$$

where $sats^{(i)} \subseteq \{1, \cdots, N_s\}$ is the set of potential partners for the satellite of that row/index i.

This variable is used to model potential network constraints based on the connectivity of each satellite in the constellation. A constellation with the highest interconnectivity employs what we call an "all-to-all" type of network topology, which is implemented in the *topo* variable as:

$$topo_{\text{ALL}} = \left\{ sats^{(i)}_{\text{ALL}} \right\}, \quad i \in \{1, \cdots, N_s\} \tag{3.2}$$

where $sats^{(i)}_{\text{ALL}} = \{1, \cdots, N_s\}$, which effectively means that every satellite is capable of establishing a crosslink with every other satellite. Note that this is strictly from a network perspective, and does not consider any other constraints like those based on

geometry or other system design parameters.

Different topology combinations can be created for each constellation. The number of combinations is dependent on the size of the constellation. For instance, in two-satellite systems like that of our ARTEMIS-OICETS example scenario, there is only one possible combination of network topology, defined as:

$$topo_{\mathrm{2SAT}} = \begin{Bmatrix} \{1,2\} \\ \\ \{1,2\} \end{Bmatrix} = \begin{Bmatrix} \{2\} \\ \\ \{1\} \end{Bmatrix} \tag{3.3}$$

If we were to introduce a third satellite into this constellation, there would now be four possible combinations of network topology (not counting those that exclude a satellite since that would effectively reduce down to a two-satellite system):

$$topo_{3SAT} = \begin{Bmatrix} \{2,3\} & ; & \{2,3\} & ; & \{2\} & ; & \{3\} \\ \{1,3\} & & \{1\} & & \{1,3\} & & \{3\} \\ \{1,2\} & & \{1\} & & \{2\} & & \{1,2\} \end{Bmatrix} \tag{3.4}$$

The first combination is an "all-to-all" type of topology for a three-satellite system, where each satellite is capable of linking with every other satellite. The other three combinations are what we consider different variants of a "some-to-some" network topology. In particular, with only three satellites, the three "some-to-some" combinations are equivalent to a "hub-node" type of model, where particular satellites (the "hubs") are able to establish links with all of the other satellites, but the other satellites (the "nodes") can only establish links with hubs and not other nodes.

As one can imagine, were the constellation to have more than three satellites, the number of possible network topology combinations would grow exponentially. Examining all possible combinations for any given constellation is not within the scope of this thesis, however this can be studied in future work. Instead, we define a small set of topologies to study for any constellations larger than two satellites, based on what we expect to be the mission of that given scenario. For example, in relay missions, we expect that a few satellites will serve as hubs in the constellation, and

the rest will serve as nodes that can only establish links with hubs. For each of the larger satellite constellation scenarios studied in this thesis, topology definitions will be clearly stated.

Minimum Link Altitude

The minimum link altitude input parameter, h_{min}, is used to model a simple geometric constraint for the lowest altitude at which a crosslink can be established. Therefore, the link-access availability computation will consider the grazing altitude of all potential crosslinks, and only allow those that are above the h_{min} set for that scenario. At a minimum, this value could be 0, effectively using the central-body alone as a link-occluding geometric constraint, based on its average radius R. In order to account for atmospheric density and its effect in occluding crosslink communications, we use the Karman line as the minimum link altitude for both Earth and Mars, set to 100 and 80 kilometers, respectively [73].

Duty-cycle Timing

A simple duty-cycle model is used to emulate potential time constraints due to other mission operations and priorities restricting the time or ability for crosslink communications. This could be due to limited resources (e.g., power) or due to mission operating modes (e.g., science vs. communications, or downlink vs. crosslink). Two variables are used to define the duty-cycle timing, T_{on} and T_{off}, which are the durations for how long crosslink communications mode is "on" and "off", respectively. During the "on" mode, crosslinks are available and intersatellite measurements are captured. During the "off" mode, the satellite may be idle and charging or performing any other task besides crosslink communications. One duty-cycle is defined as one T_{on} duration followed by one T_{off} duration, followed by the next duty-cycle, and continuously repeating for the duration of the simulation.

Other potential parameters

Other potential constraints were considered but not implemented, and can be explored in future work. These include constraints based on relative range, relative velocity, or link duration, which can be used to model different design or operations configurations of the lasercom system. Consideration can also be given to geometric keep-out zones with respect to background objects (e.g., sun, planets, moons), or implementing a time-delay model to account for the time needed to establish links based on the state uncertainty of link-access partners.

3.2.2 Output: Satellite Pairs, Available Links, & Duty Cycles

Generating Satellite-Pairings

The first main output variable is a set of all possible satellite-pairings in the simulated constellation, coded as the $N_p \times 2$ matrix *pairs*, where N_p is the total number of satellite-pairings. This is generated by expanding each entry in the network topology model, *topo*, and trimming any self-to-self entries and repeat connections, such that:

$$pairs = \{i, j\}, \quad i \in \{1, \cdots, N_s\}, \quad j \in \{sats^{(i)}\}, \quad i \neq j \tag{3.5}$$

where i is the observing (or receiving) satellite, and j is a potential link partner.

To provide an example of a repeat connection, an ARTEMIS-OICETS pairing is the same as an OICETS-ARTEMIS pairing in a two-way lasercom crosslink topology. Note that this would not necessarily be true for a different scenario. For instance, in a broadcast RF crosslink topology, the order of the pairing would now matter, since either satellite involved could operate in either transmit-only, receive-only, or both transmit-and-receive modes.

Computing Link-Availability

The second output variable is an $N_p \times N$ matrix *links* encoding Boolean data for the availability of link-access between satellite pairings over the full simulation period.

Link-access availability is evaluated for each satellite-pairing $p \in \{1, N_p\}$ at each time step $k \in \{1, N\}$ using the propagated truth data from STK, with consideration of the network, geometry, and time constraints supplied by the link-access model.

Gaps in link-access availability are coded as 0, while periods of available links are coded as $l_k^{(p)} > 0$, as the incremental link-access number for that satellite-pairing p. N_l is the total number of available links for the full constellation (between all satellite-pairings) over the full simulation period, and is computed as:

$$N_l = \sum_{p=1}^{N_p} \max_{k \in [N]} l_k^{(p)} \tag{3.6}$$

Simulating Duty-Cycles

The third output variable is another matrix encoding Boolean data, this time for the on/off mode schedule of duty-cycles over the full simulation period, *dcycs*. As mentioned earlier, one duty-cycle is comprised of one T_{on} duration of time immediately followed by one T_{off} duration of time, continuously repeating from the start time of the simulation, t_0, to the end time, t_N. The *dcycs* matrix is implemented similar to *links*, where "off" durations are coded as 0, while "on" durations are coded as $d_k > 0$, as the incremental duty-cycle number over the course of the simulation period.

N_d is the total number of duty-cycles over the full simulation period, and is computed as:

$$N_d = \max_{k \in [N]} d_k \tag{3.7}$$

To simplify the model, and mitigate the complexity of mismatched duty-cycles within a constellation, all satellites in a simulation scenario are assumed to be synchronized to follow the same duty-cycle schedule. This means that the full constellation goes into a "crosslink" mode during the "on" durations of the *dcycs* schedule, and collects zero intersatellite measurements during the "off" durations. Note that for a particular link to be active and collecting intersatellite measurements at given time, that row/column entry must be greater than zero in both *links* and *dcycs*.

Note that while separate duty-cycles for individual satellites are not implemented for this study, the developed framework allows for easy implementation as needed in future work.

3.3 State & Measurement Simulation

The next step in our simulation approach is to define and implement the navigation filter models in order to generate the following products: an initial state estimate and uncertainty, state predictions based on that uncertain initial estimate, and simulated measurements.

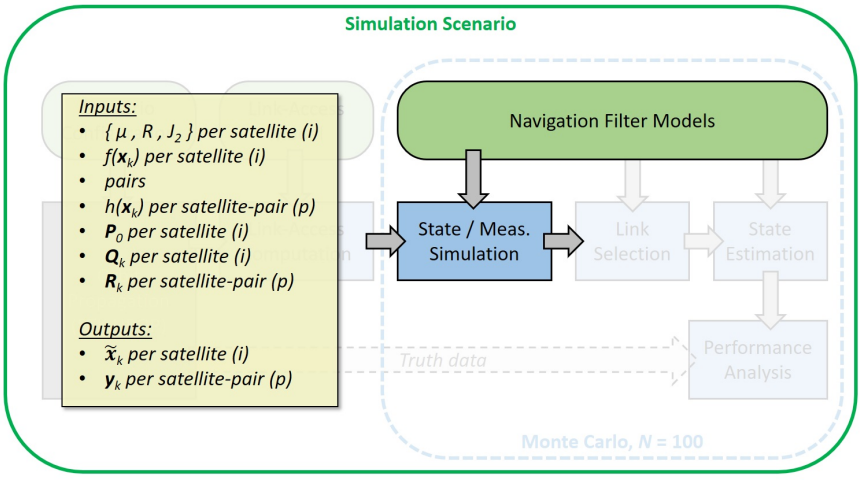


Figure 3-7: Summary of inputs and outputs for the "state and measurement simulation" step in the simulation approach.

3.3.1 Input: Navigation Filter Models

This step initializes the models of spacecraft dynamics, intersatellite measurements, and simulated uncertainty for use in the navigation filter.

State Model

The full state vector for estimation includes the six-element Cartesian position and velocity states (in meters and m/s, respectively) of each satellite in the scenario:

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}^{(1)} \\ \vdots \\ \mathbf{x}^{(N_s)} \end{bmatrix} \quad , \quad \mathbf{x}^{(i)} = \begin{bmatrix} \mathbf{r}^{(i)} \\ \dot{\mathbf{r}}^{(i)} \end{bmatrix} \quad , \quad \mathbf{r}^{(i)} = \begin{bmatrix} x^{(i)} \\ y^{(i)} \\ z^{(i)} \end{bmatrix} \quad , \quad \dot{\mathbf{r}}^{(i)} = \begin{bmatrix} \dot{x}^{(i)} \\ \dot{y}^{(i)} \\ \dot{z}^{(i)} \end{bmatrix} \tag{3.8}$$

where $i \in 1, \ldots, N_s$ is the satellite index.

Initial State Uncertainty Model

The initial uncertainty in the position (σ_{0r}) and velocity (σ_{0v}) of each spacecraft is assumed to be 1000 meters and 1 m/s, respectively, in each axis:

$$\mathbf{P}_0 = \text{diag}(\begin{bmatrix} \mathbf{p}_0^{(1)} \\ \vdots \\ \mathbf{p}_0^{(N_s)} \end{bmatrix}) \quad , \quad \mathbf{p}_0^{(i)} = \begin{bmatrix} \sigma_{0r}^2 \\ \sigma_{0r}^2 \\ \sigma_{0r}^2 \\ \sigma_{0v}^2 \\ \sigma_{0v}^2 \\ \sigma_{0v}^2 \end{bmatrix} = \begin{bmatrix} (1000)^2 \\ (1000)^2 \\ (1000)^2 \\ (1.0)^2 \\ (1.0)^2 \\ (1.0)^2 \end{bmatrix} \tag{3.9}$$

These values follow those used in literature [48]. A sensitivity analysis of the simulation approach on this input parameter is shown later in Section 3.5.2.

Dynamics Model

The dynamics model for satellite motion with J2 gravitational perturbations is described by the following nonlinear functions:

$$f(\mathbf{x}_k^{(i)}) = \dot{\mathbf{x}}_k^{(i)} = \begin{bmatrix} \dot{x}_k^{(i)} \\ \dot{y}_k^{(i)} \\ \dot{z}_k^{(i)} \\ \frac{-\mu x_k^{(i)}}{|\mathbf{r}_k^{(i)}|^3}\left\{1 + \frac{3}{2}J_2\left(\frac{R}{|\mathbf{r}_k^{(i)}|}\right)^2\left[1 - 5\left(\frac{z_k^{(i)}}{|\mathbf{r}_k^{(i)}|}\right)^2\right]\right\} \\ \frac{-\mu y_k^{(i)}}{|\mathbf{r}_k^{(i)}|^3}\left\{1 + \frac{3}{2}J_2\left(\frac{R}{|\mathbf{r}_k^{(i)}|}\right)^2\left[1 - 5\left(\frac{z_k^{(i)}}{|\mathbf{r}_k^{(i)}|}\right)^2\right]\right\} \\ \frac{-\mu z_k^{(i)}}{|\mathbf{r}_k^{(i)}|^3}\left\{1 + \frac{3}{2}J_2\left(\frac{R}{|\mathbf{r}_k^{(i)}|}\right)^2\left[3 - 5\left(\frac{z_k^{(i)}}{|\mathbf{r}_k^{(i)}|}\right)^2\right]\right\} \end{bmatrix} \tag{3.10}$$

where $k \in 0, \ldots, N$ is the time-step index. Known constants are used to characterize the central body that each satellite orbits [48]. This includes the mass-based gravitational constant (μ), equatorial radius (R), and J2-term of gravity model coefficients (J_2), which are taken directly from the gravity model files used to propagate the "truth" data in STK HPOP. The values used in this thesis for Earth and Mars are shown in Table 3.3. Note that additional forces can be implemented for future work, as described in literature [74, 75].

Table 3.3: Central-body constants used in dynamics model, taken directly from STK gravity model files.

Parameter	**Earth (EGM2008)**	**Mars (MRO110C)**
μ (m^3/s^2)	3.986004415e14	0.4282837564e14
R (m)	6.3781363e6	3.396000e6
J_2	1.082626174e-3	1.956609159e-3

This nonlinear dynamics function is used in the state-transition matrix $\mathbf{\Phi}_k$ such that:

$$\mathbf{x}_k = \mathbf{\Phi}_k \mathbf{x}_{k-1} \tag{3.11}$$

$$\mathbf{\Phi}_k = \mathbf{I} + \mathbf{F}T + \frac{1}{2}(\mathbf{F}T)^2 + \cdots = \mathrm{e}^{\mathbf{F}T} \tag{3.12}$$

where F is the Jacobian of the dynamics model, and T is a duration of the time-step between $k-1$ and k. This representation of the dynamics model can be used

in conjunction with an ordinary differential equation (ODE) solver to propagate a known state $\mathbf{x}_{k-1}$ forwards or backwards in time. In our simulation process, we use the Matlab *ode45()* solver for this purpose.

Process Noise Model

The process noise is assumed to be 2e-5 meters in position and 1e-4 meters/sec in velocity to account for potential unmodeled terms in the dynamics model, and to help with numerical stability. These values were chosen to be consistent with literature [48].

$$\mathbf{Q}_k = \text{diag}(\begin{bmatrix} \mathbf{q}_k^{(1)} \\ \vdots \\ \mathbf{q}_k^{(N_s)} \end{bmatrix}) \quad , \quad \mathbf{q}_k^{(i)} = \begin{bmatrix} \sigma_r^2 \\ \sigma_r^2 \\ \sigma_r^2 \\ \sigma_v^2 \\ \sigma_v^2 \\ \sigma_v^2 \end{bmatrix} = \begin{bmatrix} (2 \times 10^{-5})^2 \\ (2 \times 10^{-5})^2 \\ (2 \times 10^{-5})^2 \\ (1 \times 10^{-4})^2 \\ (1 \times 10^{-4})^2 \\ (1 \times 10^{-4})^2 \end{bmatrix} \tag{3.13}$$

A sensitivity analysis of the simulation approach on this input parameter is shown later in Section 3.5.2.

Measurement Model

The relative position vector between two satellites is estimated from measurements of its magnitude and directional components. The magnitude component is fully captured by measuring intersatellite range. The direction component is captured by measuring two intersatellite bearing angles in inertial space from the perspective of the "observer" satellite to the other.

Thus the measurement model is represented by the following nonlinear functions:

$$h(\mathbf{x}_k^{(i)}, \mathbf{x}_k^{(j)}) = \begin{bmatrix} \rho_k^{(p)} \\ \phi_k^{(p)} \\ \theta_k^{(p)} \end{bmatrix} = \begin{bmatrix} |\mathbf{\Delta r}_k^{(p)}| \\ \arctan\left(\frac{\Delta y_k^{(p)}}{\Delta x_k^{(p)}}\right) \\ -\arcsin\left(\frac{\Delta z_k^{(p)}}{|\mathbf{\Delta r}_k^{(p)}|}\right) \end{bmatrix} \tag{3.14}$$

where the satellite-pairing index $p \in \{1, N_p\}$ maps to the satellite indices i and j according to:

$$pairs^{(p)} = \{i, j\}, \quad \{i, j\} \in \{1, \cdots, N_s\}, \quad i \neq j \tag{3.15}$$

Measurement Noise Model

We assume a conservative value of 10-cm uncertainty to represent ranging using lasercom systems [76]. The precision of the bearing measurement is typically limited by the uncertainty of the star tracker. We assume 2-arcsec bearing uncertainty based on the performance of satellite star trackers currently in use and development [77, 78].

$$\mathbf{R}_k = \text{diag}(\begin{bmatrix} \mathbf{r}_k^{(1)} \\ \vdots \\ \mathbf{r}_k^{(N_p)} \end{bmatrix}) \quad , \quad \mathbf{r}_k^{(p)} = \begin{bmatrix} \sigma_\rho^2 \\ \sigma_\phi^2 \\ \sigma_\theta^2 \end{bmatrix} = \begin{bmatrix} (10)^2 \\ (2)^2 \\ (2)^2 \end{bmatrix} \tag{3.16}$$

A sensitivity analysis of the simulation approach on range and angular measurement noise is shown later in Section 3.5.2.

3.3.2 Output: Predicted States & Simulated Measurements

Initial State Estimate

The "truth" state vector data of each satellite at the start-time of the simulation, $\mathbf{x}_0^{(i)}$, is supplied zero-bias Gaussian white noise, based on the initial state uncertainty, σ_0, to generate an uncertain initial state estimate for each satellite as follows:

$$\tilde{\mathbf{x}}_0 = \mathbf{x}_0 + \sigma_0 \quad , \quad E\{\sigma_0 \sigma_0^T\} = \mathbf{P}_0 \tag{3.17}$$

$$\sigma_0 = [\sigma_{0r} \ , \ \sigma_{0v}]^T \tag{3.18}$$

A priori State Prediction

The dynamics model, $f(\mathbf{x}_k^{(i)})$, is then used to propagate state vectors for all satellites for the full simulation period from those uncertain initial estimates, $\tilde{\mathbf{x}}_k(i)$, using the Matlab *ode45* solver over all time-steps, T, between the start and end times of the simulation scenario, t_0 and t_N, respectively. This output product is not used by the navigation filter (as that computes a new prediction based on the latest estimate at each time-step) but by the link-selection algorithm described in the next section.

Measurement Simulation

Using the "truth" state vector data $\mathbf{x}_k^{(i)}$ and the *pairs* list of satellite-pairings, true measurements are computed for each link at each time step using the measurement model $h(\mathbf{x}_k^{(i)}, \mathbf{x}_k^{(j)})$. All measurements are then supplied zero-bias Gaussian white noise, based on the measurement uncertainty $\mathbf{v}_k$, to generate simulated measurements to be used in the navigation filter. The measurement noise is applied as follows:

$$\mathbf{y}_k = h(\mathbf{x}_k, \mathbf{x}_k) + \mathbf{v}_k \quad , \quad E\{\mathbf{v}_k \mathbf{v}_k^T\} = \mathbf{R}_k \tag{3.19}$$

$$\mathbf{v}_k = [\sigma_\rho \ , \ \sigma_\phi \ , \ \sigma_\theta]^T \tag{3.20}$$

3.4 Link-Access Selection

Generally, a single satellite lasercom system may only establish and maintain one link at a given time, due to the narrow beamwidth of the signal requiring precise pointing to the receiving terminal. Additional links can be supported by a given satellite by including additional lasercom payloads in the spacecraft design, although this can become difficult since each payload would also require separate pointing and tracking

systems. This limitation in simultaneous links adds some complexity to the overall network communications architecture of a lasercom crosslink constellation, since a satellite may have more links available than it is able to support at any given time.

For comparison, the larger beamwidth of RF transmission allows for broadcast signals (which can be transmitted from a single source and received by many), and also low-gain antennas (which can receive signals from a wide range of directions). This simplifies the communications architecture, as it enables satellites to transmit to and/or receive from any other satellites within a given satellite's antenna pattern, and mitigates the need for planning and scheduling particular links.

In order to solve this problem for lasercom constellations, there must be some process by which links can be planned and scheduled for crosslink communications. Typically, this will be dictated by other priorities within the constellation's mission, in particular routing or distributing data. However, in this thesis, we seek to determine the best possible navigation performance achievable by crosslink networked satellites, and we therefore require a process of selecting links based on navigation-specific priorities.

Having predicted the future states of each satellite based on the uncertain initial estimates, $\tilde{\mathbf{x}}_k^{(i)}$, we can use this knowledge to predetermine a schedule by using some heuristic to help decide which links are preferred over others. This step in the simulation process provides one way to approach this problem using a heuristic of predicted uncertainty derived using the Cramér-Rao lower bound (CRLB).

3.4.1 Overview of Inputs and Outputs

3.4.2 Output: Selected Links based on CRLB

CRLB uses the same filtering structure and input models as a Kalman filter, as described in Section 2.1.1. Thus, the CRLB is able to provide a relevant and accurate prediction of the effect that a given intersatellite link will have on that spacecraft's state uncertainty. Therefore, we have chosen to use the CRLB computed uncertainty as the heuristic for deciding which links to select and schedule during the simulation

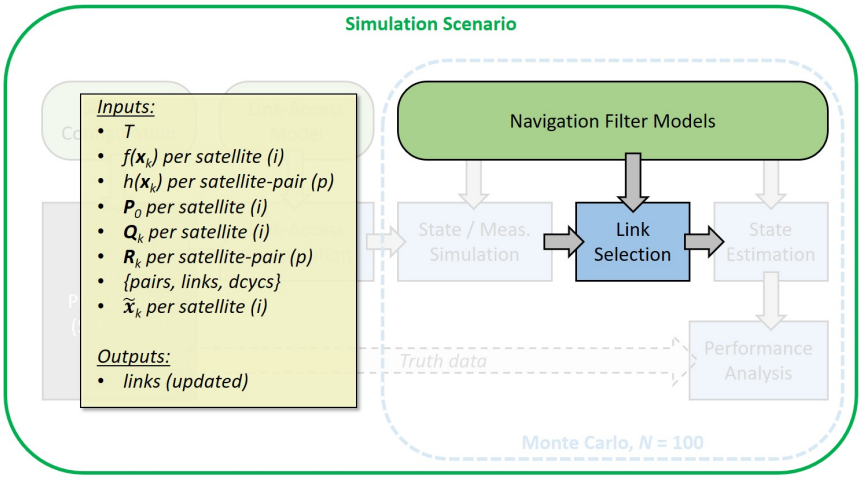


Figure 3-8: Summary of inputs and outputs for the "link-access selection" step in the simulation approach.

period.

However, this means that the predicted uncertainty of every possible link must be somehow compared. This is slightly problematic, because different links tend to start and end at different times, leading to an unfair comparison unless they are evaluated over the same duration of time. For a fair comparison over a common time period, we use our implementation of the duty-cycle, where the full constellation cycles between an "on" duration when links can be made, and an "off" duration when no links are available. Regardless of the duration of the on/off modes (or whether an "off" period even exists), the full duty-cycle duration is used as the time period over which links are directly compared. Therefore, a spacecraft may switch from one potential link partner to another at the start/end of every duty-cycle.

Still, how a link is scheduled still needs to be addressed. The brute-force method of addressing this would be to generate all possible schedules given all combinations of all possible link-selections that can be made over the full simulation period. One can imagine that this would require a large amount of computation, especially for large constellations and highly-connected network topologies. Instead of an exhaustive search for the best or optimal schedule, such as the brute force method, we have decided to implement a greedy approach instead. As such, decisions are made after

evaluating each possible link for a given duty-cycle, instead of at the end of the simulation period. This means that the link selected for a given duty cycle is completely independent of past or future decisions that can be made. Although this cannot guarantee the optimal solution will be found, it provides a faster method of making selections with less computation. And it is likely to be close to the optimal solution if uncertainty is minimized over each duty-cycle.

Based on this strategy, we use the following framework to evaluate and select links for a given satellite in the simulated constellation:

Pseudo-code:

for each duty cycle
    if {first duty cycle}
        initialize $\tilde{P}_0$ with P_0
    else
        initialize $\tilde{P}_0$ with $\tilde{P}_N$ from last duty cycle
    end
    for each link partner
        compute 2-sat CRLB for duration of current duty cycle
    end
    select partner with min CRLB at end of duty cycle
    (optional) configure rest of constellation based on selection
    compute constellation CRLB for duration of current duty cycle
end

Figure 3-9 provides a notional diagram for this process of selecting links based on CRLB.

The final product of this link-selection framework is an updated version of the *links* variable that maintains only the nonzero values of links that are scheduled, while all others are set to 0.

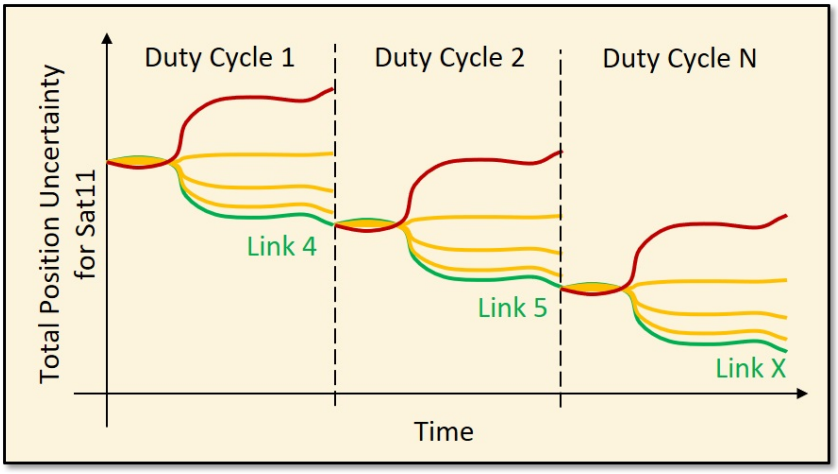


Figure 3-9: Notional diagram for the link-selection process developed for use in this thesis. Each simulation scenario is divided in time into duty-cycles, which are used as evaluation periods for link-selection. Each possible link-access partner is used in a two-satellite CRLB computation with a particular satellite (Sat11 in this case). The final CRLB value of each possible partner is compared, and the link that minimizes this value is selected. The full scenario is propagated forward using this selection, up until the start of the next duty-cycle or evaluation period. This process is repeated until the end of the simulation scenario.

Caveats

We acknowledge that the framework we have implemented for link-selection is not fully robust for use in all possible simulation scenarios, and have thus included the "optional" step shown in the pseudo-code above to allow for specific configurations for special cases, or simplifications for resolving any identified issues.

One example of a special case is that of a symmetric constellation, for which we assume that the best link for a given satellite is the best link for all satellites in similar constellation positions (or phase slots), and thus only run the link-selection algorithm for each unique constellation position. Another example is that of a relay constellation, for which we assume that all relay satellites maintain crosslinks with other relay satellites when available, and only select/switch between links with user nodes.

One of the issues that we have identified is that of conflict resolution. If the framework is run for all satellites in a given constellation, two satellites may prefer

the same link-partner for the same duty-cycle, or one satellite may prefer a particular link-partner that may prefer yet another link-partner, and so on. These situations are only confronted in communications architectures with limits on the number of simultaneous connections.

For any special cases and issues such as these, we implement a set of "network rules" into the "optional" step mentioned above in order to resolve special cases or potential conflicts. These "network rules" will be clearly identified for the scenarios in which they are used.

We also acknowledge that a heuristic based on CRLB computation for every possible link in each duty-cycle is not very efficient in terms of computations and processing time. Some of the simpler heuristics that could be used instead are those based on geometry (largest intersatellite range, largest change in intersatellite range), or based on time (longest total access time), or some combination of the two. However, we chose to use CRLB to provide a prediction of estimation uncertainty, with direct application of the same dynamics and measurement models and filtering structure as the EKF estimator. Future work should identify any potential algorithms that can be used to address any/all of these issues or provide more robustness or computational efficiency.

3.5 Navigation Performance Estimation & Analysis

The final step in our simulation approach is to compute the navigation estimates and uncertainty. For performance and error analysis, we use the Extended Kalman Filter (EKF) on a Monte Carlo sampling of simulations. In order to mitigate running every possible variation of every scenario through a Monte Carlo analysis, we first perform an uncertainty analysis to predict the navigation uncertainty of variant scenarios using the Cramér-Rao Lower Bound (CRLB). The following sections go into detail on both of these methods, and how their outputs are analyzed to compute performance results. An overview of the case studies considered in this thesis can be found in Table 3.5.

3.5.1 Overview of Inputs and Outputs

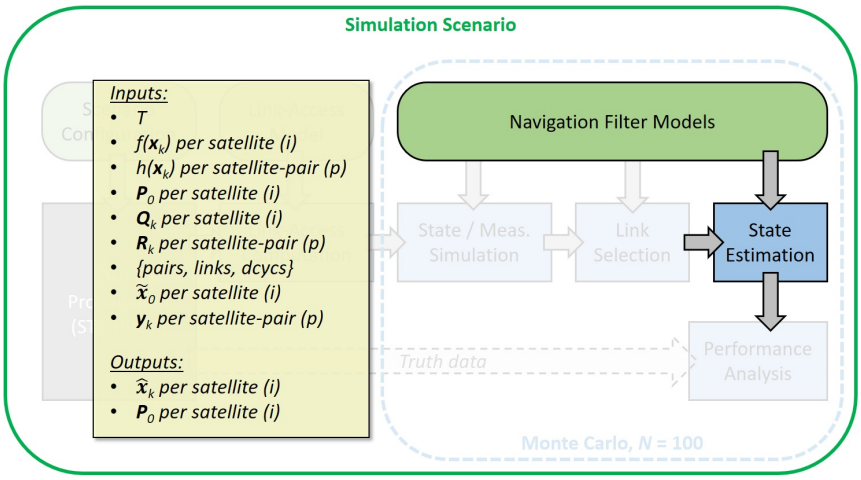


Figure 3-10: Summary of inputs and outputs for the "state estimation" function in the simulation approach.

Since both the dynamics and measurement models are nonlinear, we use an EKF estimation approach that requires calculation of Jacobian matrices, $\mathbf{F}_k$ and $\mathbf{H}_k$, at each time-step $k \in \{1, N\}$. However, as not every spacecraft in the constellation is actively used at every time-step (based on the links available and selected in *links*), certain elements of $\mathbf{H}_k$ are not needed for the update phase. Instead of resizing matrices to accommodate inactive links and measurements at every time-step, we chose to handle this situation by assigning any inactive links and measurements with an artificially high measurement noise value, in this case 1×10^{20}. This is virtually an "infinite" amount of noise compared to that of active measurements, and effectively negates any potential contribution from these inactivated measurements in the update phase of the navigation filter. Therefore, any satellites in the simulated constellation that are not actively used at a given time-step simply receives a new prediction or propagated state based on most recent update. Note that since the CRLB computation uses the same structure as the EKF, this approach applies to both methods.

3.5.2 EKF Output: Estimation Error & Uncertainty

For a given simulation scenario, the input models and simulated observations are fed into the EKF estimator. Each step is evaluated for potential measurement updates based on which links are available and selected in *links*, as described earlier. The main outputs from the EKF are the state estimates $\hat{\mathbf{x}}_k^{(i)}$ and resulting covariance $\mathbf{P}_k^{(i)}$ for each satellite.

Performance Analysis Metrics

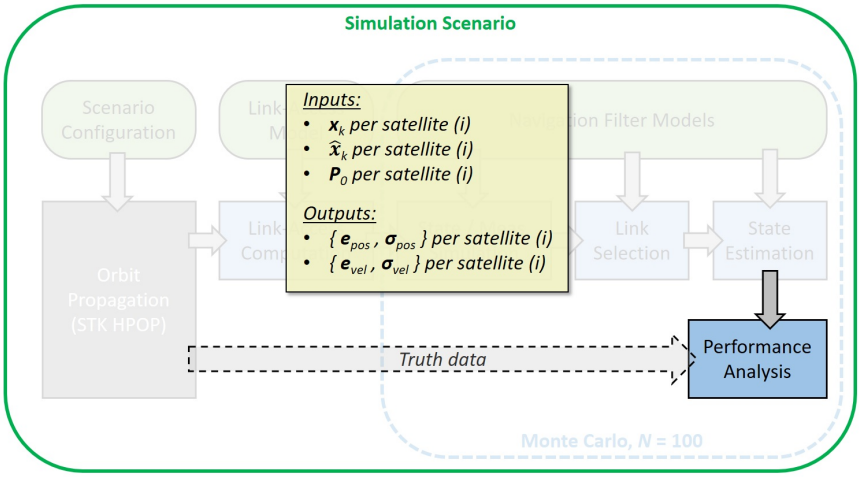


Figure 3-11: Summary of inputs and outputs for the "performance analysis" function in the simulation approach.

The main metric used for analysis is the "total position error", which is simply the L2 norm of the residual error between the estimated and true position vectors:

$$e_{pos} = |\hat{\mathbf{r}} - \mathbf{r}| \tag{3.21}$$

Similarly, the "total velocity error" is the L2 norm of the residual error between the estimated and true velocity vectors:

$$e_{vel} = |\hat{\dot{\mathbf{r}}} - \dot{\mathbf{r}}| \tag{3.22}$$

The variance, σ^2, of each element of the state vector is along the diagonal of this

covariance matrix. Therefore, we compute the total position and velocity 1-sigma uncertainties of each spacecraft as:

$$\sigma_{pos} = |\sqrt{\mathbf{p}_{rr}}| \quad , \quad \sigma_{vel} = |\sqrt{\mathbf{p}_{\dot{r}\dot{r}}}| \tag{3.23}$$

where

$$\mathbf{p}_{rr} = \text{diag}(\mathbf{P}_{rr}) \quad , \quad \mathbf{p}_{\dot{r}\dot{r}} = \text{diag}(\mathbf{P}_{\dot{r}\dot{r}}) \quad , \quad \begin{bmatrix} \mathbf{P}_{rr} \mathbf{P}_{r\dot{r}} \\ \mathbf{P}_{\dot{r}r} \mathbf{P}_{\dot{r}\dot{r}} \end{bmatrix} = \mathbf{P} \tag{3.24}$$

Monte Carlo Analysis

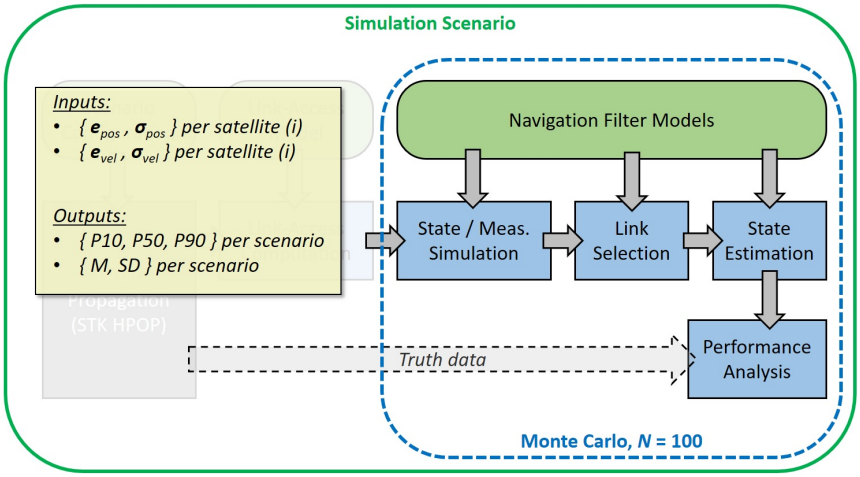


Figure 3-12: Summary of inputs and outputs for the "Monte Carlo analysis" step in the simulation approach.

The results from the estimator are then analyzed for performance metrics and statistics. We perform a Monte Carlo analysis (N=100) in order to sample noise / uncertainty variations and obtain statistical results. Results are typically displayed in figures as the 10th, 50th, and 90th percentiles of these Monte Carlo simulations.

Sensitivity to Input Parameters

Figures 3-13 through 3-17 show the results of a sensitivity analysis of our simulation approach to different values of input parameters in our example scenario between

ARTEMIS and OICETS. Table 3.4 shows the parameters and values over which we analyzed sensitivities.

Table 3.4: Input parameters and values tested for sensitivity in simulation approach. Note that the values selected for use in this thesis are shown in bold.

Input Parameter(s)	**Sensitivity Study Sets/Values**
T (s)	[5, **10** 30, 60, 180]
σ_{0r} (m) σ_{0v} (m/s)	$\begin{bmatrix} 10 \\ 0.01 \end{bmatrix}, \begin{bmatrix} 100 \\ 0.1 \end{bmatrix}, \begin{bmatrix} \mathbf{1000} \\ \mathbf{1.0} \end{bmatrix}, \begin{bmatrix} 10000 \\ 10 \end{bmatrix}, \begin{bmatrix} 100000 \\ 100 \end{bmatrix}$
σ_r (m)	[2e-9, 2e-7, **2**e-**5**, 2e-3, 2e-1]
σ_v (m/s)	[1e-6, 5e-6, 1e-5, 5e-5, **1**e-**4**, 5e-4, 1e-3, 5e-3, 1e-2, 5e-2, 1e-1]
σ_ρ (m) $\sigma_\phi, \sigma_\theta$ (arcsec)	$\begin{bmatrix} 0.01 \\ 1 \end{bmatrix}, \begin{bmatrix} \mathbf{0.1} \\ \mathbf{2} \end{bmatrix}, \begin{bmatrix} 1.0 \\ 4 \end{bmatrix}$

As expected, simulated navigation error tends to grow linearly with the time-step parameter, since increased duration between measurement updates results in longer periods of propagation from potentially old or erroneous estimates. We use $T = 10$ seconds for all simulations in order to balance minimizing time-based errors with maintaining a manageable step-size for filter computations. This sensitivity can be improved in future work by using a smoothing estimator.

The EKF is very sensitive to high initial state uncertainty errors, which is also expected. As described in Section 2.1.1, other filters can be used for scenarios with initial uncertainty values greater than 10 km and 10 m/s.

The navigation filter seems to be fairly insensitive to changes in process noise in the position elements of the state vector, though it is extremely sensitive to process noise in the velocity elements. We assume 1e-4 m/s of velocity process noise, chosen to be consistent with literature [48]. It would seem that this value was purposely selected in order to tune out numerical instabilities induced by the process noise.

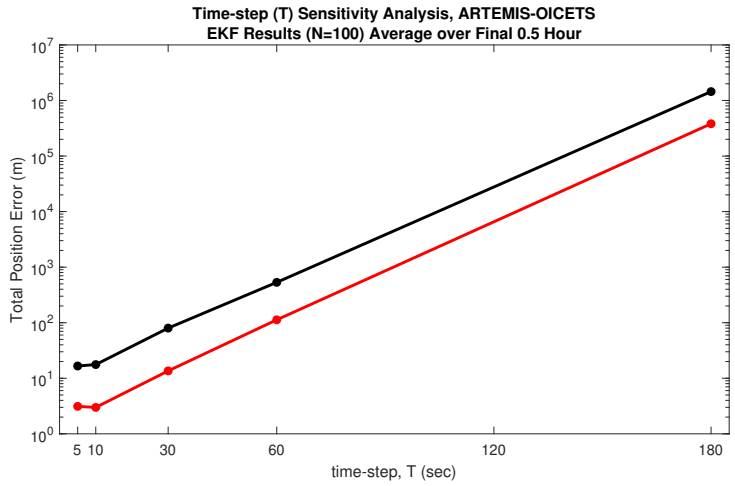


Figure 3-13: Summary of EKF results (50th percentile error averaged over final 0.5 hour) from Monte Carlo simulations (N=100) of simulation approach sensitivity to time-step, T, in the GEO-LEO example case: a single GEO-relay satellite, ARTEMIS (black), with a single LEO user, OICETS (red).

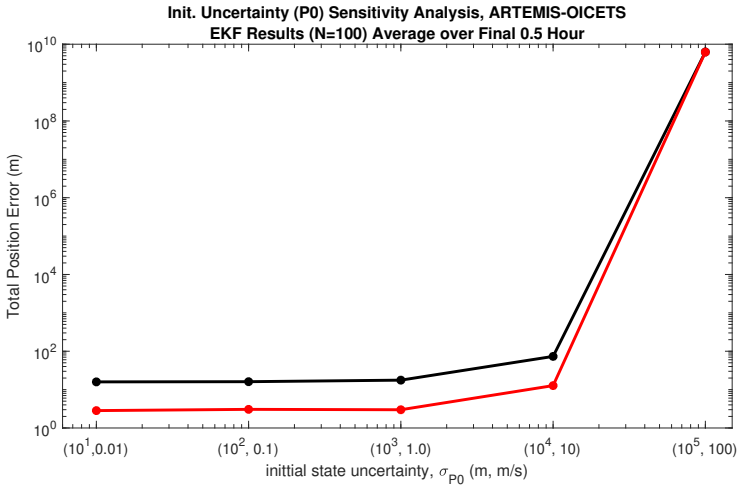


Figure 3-14: Summary of EKF results (50th percentile error averaged over final 0.5 hour) from Monte Carlo simulations (N=100) of simulation approach sensitivity to initial state uncertainty, $\mathbf{P}_0$, in the GEO-LEO example case: a single GEO-relay satellite, ARTEMIS (black), with a single LEO user, OICETS (red).

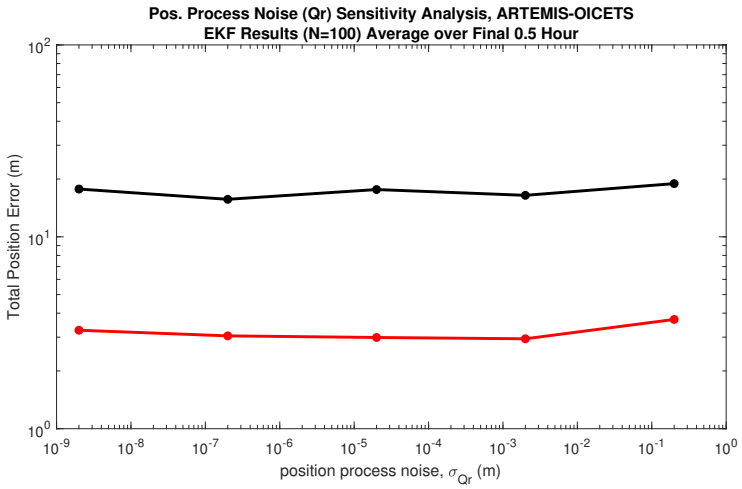


Figure 3-15: Summary of EKF results (50th percentile error averaged over final 0.5 hour) from Monte Carlo simulations (N=100) of simulation approach sensitivity to position-estimation process noise, $\mathbf{Q_r}$, in the GEO-LEO example case: a single GEO-relay satellite, ARTEMIS (black), with a single LEO user, OICETS (red).

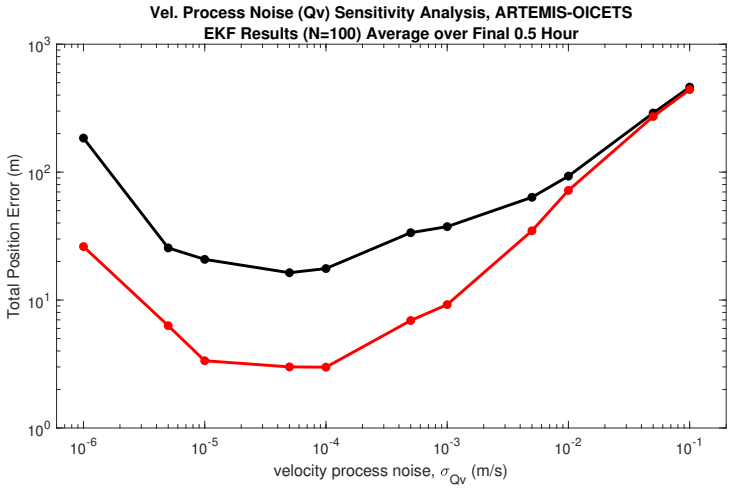


Figure 3-16: Summary of EKF results (50th percentile error averaged over final 0.5 hour) from Monte Carlo simulations (N=100) of simulation approach sensitivity to velocity-estimation process noise, $\mathbf{Q_v}$, in the GEO-LEO example case: a single GEO-relay satellite, ARTEMIS (black), with a single LEO user, OICETS (red).

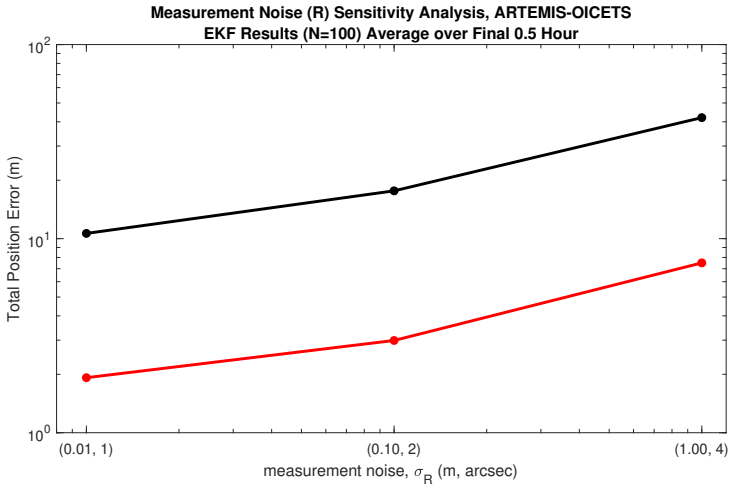


Figure 3-17: Summary of EKF results (50th percentile error averaged over final 0.5 hour) from Monte Carlo simulations (N=100) of simulation approach sensitivity to measurement noise, $\mathbf{R}$, in the GEO-LEO example case: a single GEO-relay satellite, ARTEMIS (black), with a single LEO user, OICETS (red).

3.5.3 CRLB Output: Predicted Uncertainty

For studies that explore potential trends based on wide ranges of parameter values or orbital configurations, we first perform an uncertainty analysis in order to mitigate running full Monte Carlo error analyses on every variation of every scenario. For a given uncertainty analysis study, we compute and compare the CRLB predicted uncertainty of each variant scenario in order to determine which scenario has the greater likelihood of performing better.

To achieve this, the input models and initial state estimate are fed into the CRLB computation. Each step is evaluated for potential measurement updates based on which links are available and processed through link-selection, as described earlier. The main output from the CRLB computation are the covariance estimates for each satellite in the simulated constellation at every time-step in the simulation period, $\hat{\mathbf{P}}_k^{(i)}$.

Similar to the estimated variance from the EKF, the predicted variance, $\hat{\sigma}^2$, of each element of the state vector is along the diagonal of the predicted covariance matrix. Therefore, we compute the total position and velocity 1-sigma uncertainties of each spacecraft as:

$$\hat{\sigma}_{pos} = |\sqrt{\hat{\mathbf{p}}_{rr}}| \quad , \quad \hat{\sigma}_{vel} = |\sqrt{\hat{\mathbf{p}}_{\dot{r}\dot{r}}}| \tag{3.25}$$

where

$$\hat{\mathbf{p}}_{rr} = \text{diag}(\hat{\mathbf{P}}_{rr}) \quad , \quad \hat{\mathbf{p}}_{\dot{r}\dot{r}} = \text{diag}(\hat{\mathbf{P}}_{\dot{r}\dot{r}}) \quad , \quad \begin{bmatrix} \hat{\mathbf{P}}_{rr}\hat{\mathbf{P}}_{r\dot{r}} \\ \hat{\mathbf{P}}_{\dot{r}r}\hat{\mathbf{P}}_{\dot{r}\dot{r}} \end{bmatrix} = \hat{\mathbf{P}} \tag{3.26}$$

In particular, we use the minimum uncertainty value achieved over a given simulation period as the primary metric for comparison between scenarios.

Table 3.5: Overview of case studies considered in this thesis.

Case Study	**Const. Size/Cfg.**	**Network Topology**	**Duty-cycled Operations**	**Time-step**	**Noise Models**	**Analyses**
OICETS–ARTEMIS (baseline example)	Fixed ($N_s : 2$)	Fixed (2-sat)	Varies	Varies	Varies	Sensitivity to Fixed Input Parameters
E1: GEO Relay Satellite System	Varies ($N_s : 2-5$)	Fixed (hub-node)	Off	Fixed	Fixed	Navigation Performance vs. Intersatellite Geometry
E2: LEO (Walker) Constellations	Varies ($N_s : 6-48$)	Fixed (all-all)	Off	Fixed	Fixed	Navigation Performance vs. Const. Configuration
M1: Ad-hoc Const. of Mars-orbiters	Varies ($N_s : 2-6$)	Varies	On (30:30 mins)	Fixed	Fixed	Navigation Performance for Mars-orbiter Missions
M2: Mars Comms. Constellation	Fixed ($N_s : 6$)	Varies	On (30:30 mins)	Fixed	Fixed	Navigation Performance vs. Network Architecture

Chapter 4

Earth-Orbiting Applications

This chapter describes the results and analysis from evaluating the performance of the proposed navigation method using intersatellite lasercom measurements for different Earth-orbiting applications. In the first case study, we explore the potential performance of intersatellite navigation using lasercom in a geostationary relay satellite system operating with users in different orbits. In the second case study, we analyze lasercom navigation performance for LEO constellations of varying size and distribution.

4.1 Case Study E1: GEO Relay Satellite Systems

In this case study, we evaluate the navigation performance for system architectures designed around optical relay satellites in geosynchronous orbit. This configuration has relevance to past, present, and future missions. The first demonstrations of lasercom crosslinks occurred between satellites in GEO and LEO [31]. More recently, the European Space Agency partnered with Airbus to launch two lasercom payloads into geosynchronous orbit (EDRS-A in 2016 and EDRS-C in 2019) in an effort to establish what it calls a "SpaceDataHighway" for its Copernicus program of Earth-observation satellites [36]. NASA is currently preparing its own GEO optical relay satellite, LCRD, estimated to launch in late 2020 as part of an initial phase towards optical communications for both near-Earth and deep-space missions [39]. It is clear

that GEO optical relay satellites are an important part of future space communications programs. Based on these current missions and future plans, we simulate a few potential navigation scenarios for GEO relay satellites serving users in both low and high-altitude orbits.

4.1.1 Setup of Scenarios

All scenarios are simulated over the 24-hour time period between t_0 = 16 Mar 2015 16:00:00 UTC and t_N = 17 Mar 2015 16:00:00 UTC, with a time-step of T = 10 seconds.

Different scenarios are defined using different constellation configurations, with at least one GEO relay satellite as the common spacecraft existing in all scenarios, along with some set of users in either near-Earth orbits, such as LEO, MEO, and GEO, or in a highly-elliptical orbit (HEO). Figure 4-1 provides a diagram of the orbital scenario.

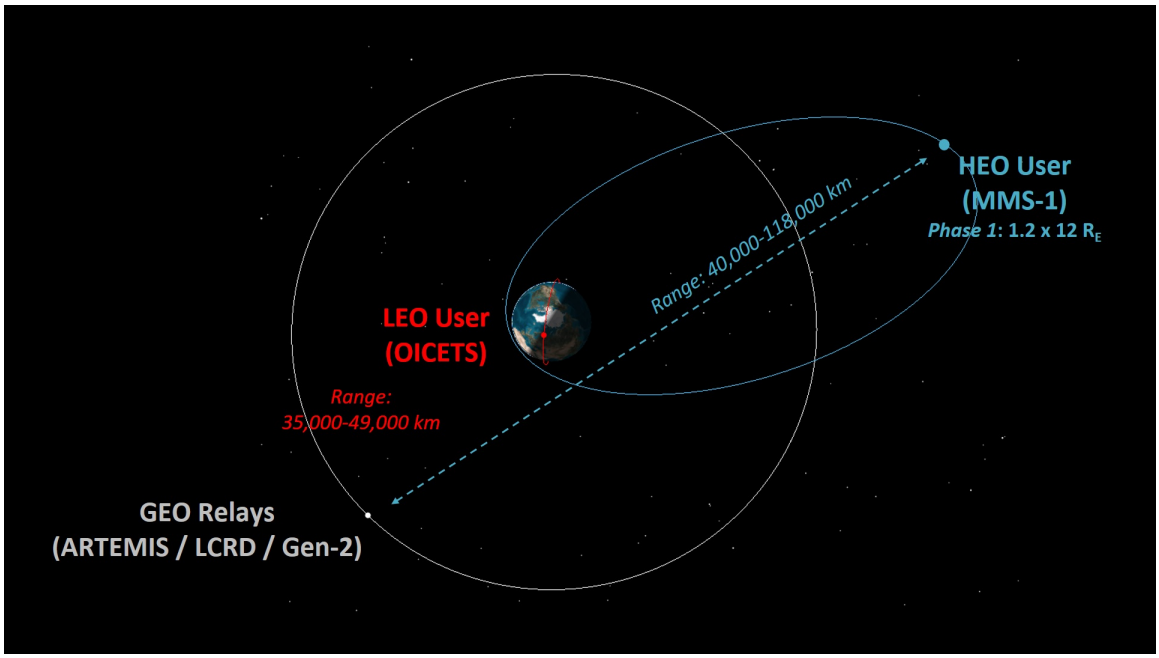


Figure 4-1: Diagram of orbital scenario for the GEO Relay case study, generated using AGI STK.

Initial states for spacecraft modeled off of existing satellites are provided in the

form of TLEs using the online database Space-Track.org [79]. The downloaded TLEs provide Keplerian element states at a given time of epoch, $t_e^{(i)}$, for the satellites ARTEMIS, OICETS, and MMS-1, which are used as representative examples for GEO, LEO, and HEO spacecraft, respectively. Table 4.1 shows the orbital elements and epoch for these spacecraft.

Table 4.1: Orbital parameters for GEO-Relay scenario.

Orbital Element	**ARTEMIS (GEO)**	**OICETS (LEO)**	**MMS-1 (HEO)**
n (rev/day)	1.0027	14.9869	1.0216
e	3.42e-4	1.58e-3	8.33e-1
i (deg)	11.70	98.09	28.81
ω (deg)	313.01	312.12	19.87
Ω (deg)	42.13	109.62	31.65
M (deg)	230.48	99.95	92.98
t_e	2015/03/16	2015/03/16	2015/03/16
(UTC)	02:03:25.9	01:28:42.75	08:46:03.73

We generated satellite orbits for use in parameter-sweep studies, which were initialized for the epoch $t_e = t_0$ with the orbital elements shown in Table 4.2.

Table 4.2: Orbital parameters for simulated near-Earth orbiters in parameter-sweep analysis, valid at time of epoch $t_e = t_0$.

Orbital Element(s)	**Parameter-Sweep Sets/Values**
a (km)	R_E+[400, 800, 1250, 2500, 4000, 8000, 20180, 35785]
e	0
i (deg)	[0, 18, 37, 52, 70, 86, 98]
ω, Ω, M (deg)	0

We also generated a set of fictional GEO-relay satellites at different longitudes that together are used to represent constellations of multiple relay spacecraft with ideal spatial coverage. The initial state parameters used to propagate orbits for these

satellites are shown in Table 4.3.

Table 4.3: Orbital parameters for simulated GEO-Relay satellites, modeled off of the orbital parameters of EDRS-C, valid at time of epoch $t_e = t_0$.

Orbital Element	**GEO0**	**GEO120**	**GEO180**	**GEO240**
n (rev/day)	1.0027	1.0027	1.0027	1.0027
e	4.90e-4	4.90e-4	4.90e-4	4.90e-4
i (deg)	0.0815	0.0815	0.0815	0.0815
ω (deg)	0	0	0	0
λ (deg)	0	120	180	240
ν (deg)	0	0	0	0

For all two-satellite cases, the network topology is obviously defined as $topo_{\text{2SAT}}$. For cases with multiple users, a "hub-node" model is implemented, where the GEO relay satellites act as "hub" satellites (with multiple simultaneous connections possible) and all user satellites serve as "nodes" (with only one link possible with any hub at any given time). For cases with multiple relays, the relay "hubs" maintain continuous connection with one another as available based on the link-access model constraints.

The minimum link altitude is defined for Earth at 100 km and kept constant for all cases. In order to show "best-case" performance results, all satellites are assumed to have no "off" mode duration ($T_{\text{off}} = 0$), and so link-availability is not affected by duty-cycles, only by the link-access model. However, T_{on} is still defined with a non-zero value in order to maintain evaluation periods for the link-selection framework. For LEO satellites, link-selection is evaluated every 15 minutes, while HEO satellites are evaluated every 180 minutes.

In the following sections, we gradually increase the complexity of the scenarios, starting with simple cases with one relay satellite and one user, followed by more complex cases of multiple relays and/or multiple users.

4.1.2 Single GEO-Relay with Single User

Baseline Example: ARTEMIS & OICETS (2005)

We first simulate and analyze the performance of a baseline example between satellites in GEO and LEO. For this, we use the same example scenario as used in Chapter 3, between ARTEMIS (in GEO) and OICETS (in LEO). The total position error for this baseline example is shown in Figure 4-2.

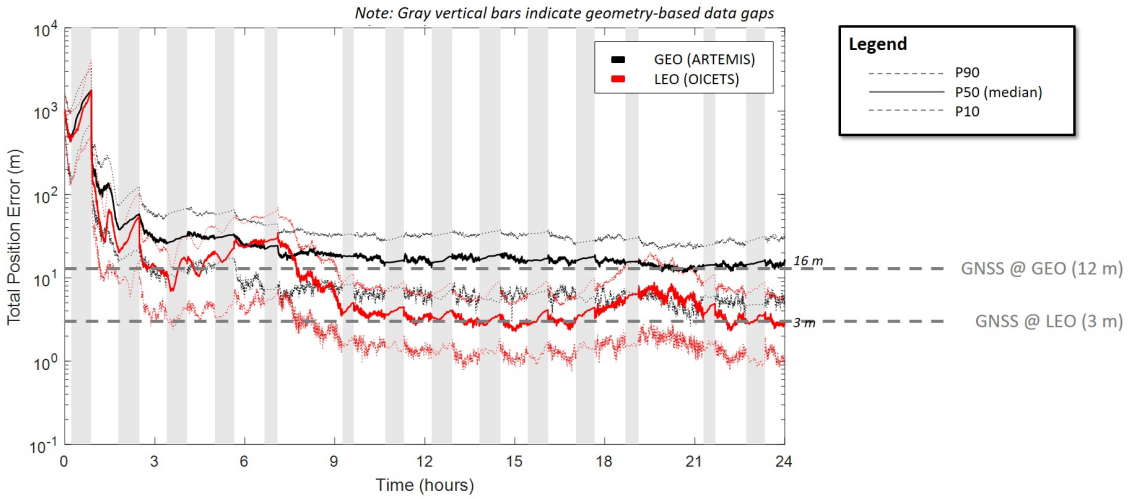


Figure 4-2: EKF results (10/50/90 percentiles) from Monte Carlo simulations (N=100) of baseline GEO-relay example: a single GEO-relay satellite, ARTEMIS (black), with a single LEO user, OICETS (red).

Gray vertical bars are used to indicate periods of geometry-based outages in the link between the two spacecraft. When the link is available, measurements are collected for the full availability period at a time-step of $T = 10$ seconds (a sample frequency of 0.1 Hz). After 24 hours of observation under these conditions, the navigation filter is able to achieve median total position errors of about 12 meters for GEO and 3 meters for LEO, which exactly meets the performance objective used in this thesis based on the average performance of GNSS.

Notice that there are two temporaneous periods of higher errors for the LEO spacecraft at around t = 7 hours and 19.5 hours. These are examples of the impact of intersatellite geometry on observability in the state estimation problem. When the

orbital plane of one satellite is nearly perpendicular to the relative position vector between the two satellites, this results in a semi-constant intersatellite range, causing increased uncertainty in the state estimation. This geometry occurs twice a day, nearly 12 hours apart, between a GEO satellite and a polar LEO satellite, which is consistent with the results shown in Figure 4-2.

This result serves as a baseline for navigation performance of a near-Earth user with a GEO relay satellite, to which we will compare the results of users with different intersatellite geometries, generated from other potential orbits based on the parameters in Table 4.2.

Parameter-Sweep Analysis of Other Circular Earth Orbits

We now perform a parameter-sweep analysis for users in other circular Earth orbits, using CRLB predicted uncertainty as a metric for comparing results.

The parameters that we alter to generate different orbit configurations are the altitude and inclination of the user satellite, while keeping the orbit of the GEO relay satellite fixed to that from the baseline scenario. See Table 4.2 for the full sets of altitude and inclinations used in this study. We selected a few typical altitudes for LEO missions, ranging from 400 km (the average altitude of the International Space Station) up to 2,500 km, along with a few MEO altitudes (including that in which GPS operates), and finally the altitude for GEO at 35,785 km. The set of inclinations selected for this parameter-sweep analysis is based on common inclinations for various missions, ranging from equatorial orbits at 0° up to sun-synchronous orbits at around 98°.

Figure 4-3 shows the variability of navigation uncertainty with respect to the user's orbit configuration, from the state estimation perspective of the GEO relay satellite. From these results, it is clear that there is a relationship with estimation uncertainty growing proportionally to the altitude of the user satellite. This is expected based on the observability of the intersatellite navigation filter. With respect to the GEO relay satellite, high altitude users will generate both lower spread of values and less variability in the relative position vector, which is the key observable in our navigation

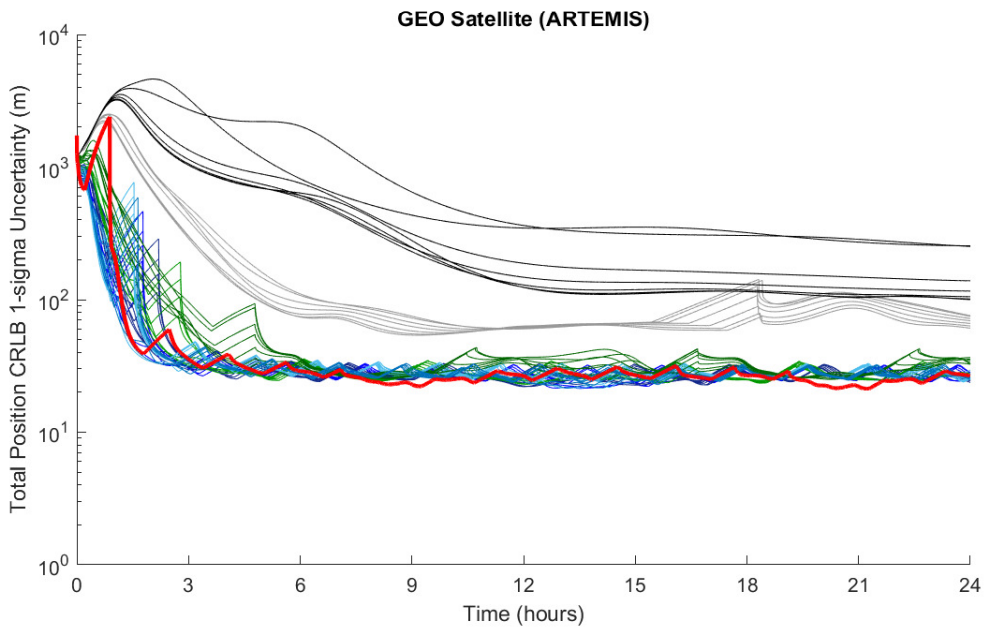


Figure 4-3: CRLB analysis results for GEO satellite (ARTEMIS) with other Earth-orbiting satellites in varying orbital altitudes and inclinations. LEO altitudes are shown in blue, lower MEO altitudes in green, GPS altitude in gray, and GEO altitude in black. Baseline GEO-LEO example of ARTEMIS & OICETS is shown in red. Lower altitude is better.

filter. This leads to higher uncertainty for higher orbits like those in GEO and higher MEO altitudes, and lower uncertainty for those at LEO and lower MEO altitudes. Interestingly, it is difficult to distinguish a difference between the LEO and MEO curves.

The variations of inclination are nearly inconsequential for users at lower altitudes. However, this is not the case for the geosynchronous altitude. Observability issues are expected at this altitude as it creates an intersatellite geometry that is nearly equal to the absolute unobservable case for navigation using intersatellite measurements (co-planar satellites at equal altitudes and phase). The impact of this unobservability effect can be seen in the two curves producing the highest uncertainty. These curves represent the results for $i = 0$ and 18 degrees, which are the closest inclination values to that of the GEO relay satellite at $i = 11.7$ deg. However, thanks to the slight offset in values, these cases maintain some partial observability, though with a high

uncertainty.

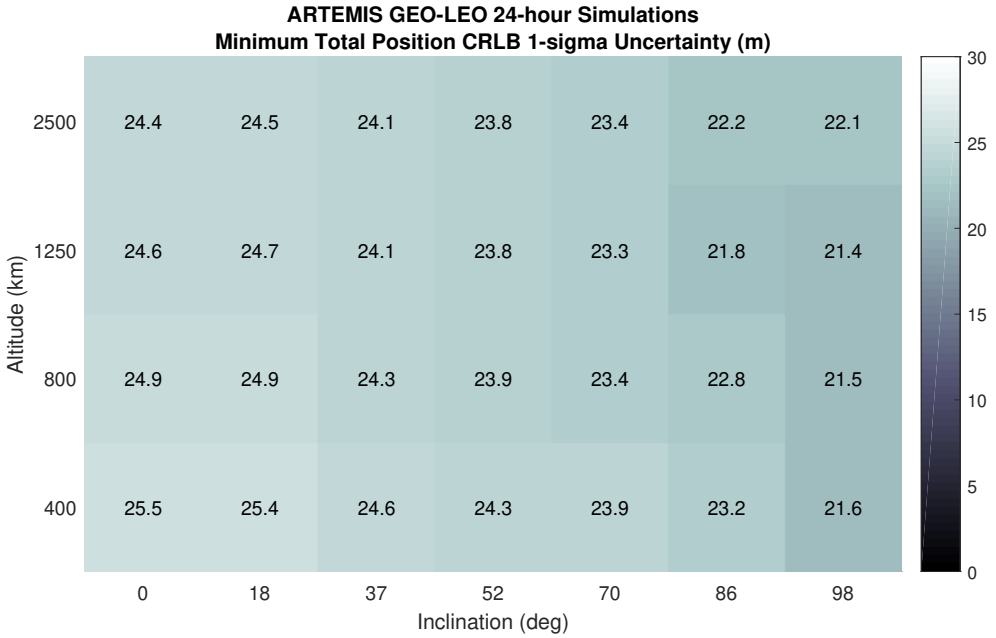


Figure 4-4: Summary of minimum achieved CRLB over 24-hour simulation for GEO satellite (ARTEMIS) with LEO satellites in varying orbital altitudes and inclinations.

Figure 4-4 shows a colormap summary of the minimum position uncertainty achieved over the 24-hour simulation period for the GEO relay satellite, focusing only on the cases with a LEO user at varying altitudes and inclinations. The x-axis is in order of increasing inclination, and the y-axis is in order of increasing altitude. It is notable there is little variation in the colors/values across the parameter-sweep uncertainty results. This demonstrates that the altitude and inclination of a LEO user is nearly inconsequential to the GEO relay satellite's uncertainty in position estimation, though a slight edge can be awarded to highly-inclined (polar) orbits.

Now we flip perspective to the LEO user satellite. Figure 4-5 shows the variability of navigation performance with respect to varying orbital altitude and inclination, from the perspective of the user satellite, which changes in each simulation. This plot shows different values of results from that of the GEO relay satellite in Figure 4-3, but the trend of higher uncertainty from users at higher altitude is consistent. Notice that from the user perspective, there is now some difference in accuracy between LEO

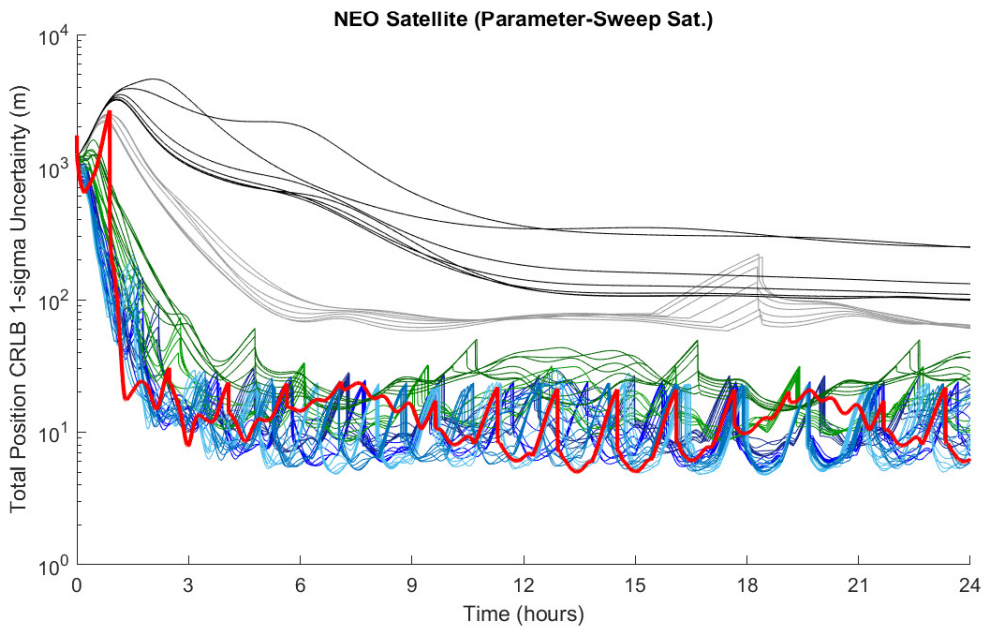


Figure 4-5: CRLB analysis results for the parameter-sweep satellite (varying orbit) in simulations of GEO satellite with other Earth-orbiting satellites in varying orbital altitudes and inclinations. LEO altitudes are shown in blue, lower MEO altitudes in green, GPS altitude in gray, and GEO altitude in black. Baseline GEO-LEO example of ARTEMIS & OICETS is shown in red. Lower altitude is better.

and lower MEO altitudes, which was not seen in Figure 4-3. Also, satellites at the same altitude seem to have variability in performance, which may imply correlation with the other varying parameter, orbital inclination, as investigated next.

Figure 4-6 shows a colormap summary of the minimum position uncertainty achieved over the 24-hour simulation period focusing only on user satellites in LEO altitudes and inclinations. As before, the x-axis is in order of increasing inclination, and the y-axis is in order of increasing altitude, though we changed the colors of the colormap in order to differentiate the new range of values. These results demonstrate clear improvement at lower altitudes and higher inclinations from the perspective of the user satellite. The improvement at lower altitudes is understandable, given that lower altitudes better sample the dominant forces of the dynamics model, and thus see gains in estimation observability and reductions in uncertainty. However, in order to understand the improvement at higher inclinations, we must look at a different

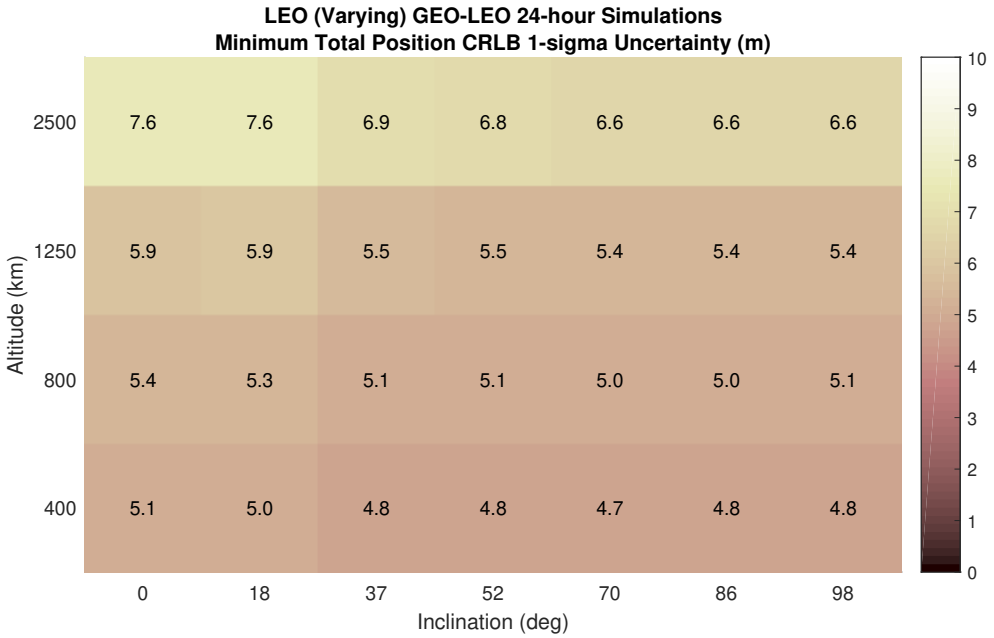


Figure 4-6: Summary of minimum achieved CRLB over 24-hour simulation for the parameter-sweep satellite (varying orbit) in parameter-sweep simulations of GEO satellite with LEO satellites in varying orbital altitudes and inclinations.

metric that varies as a result of the changes in intersatellite geometry.

Figure 4-7 shows a colormap summary of the total access times between the relay and user satellites over the 24-hour simulation period. We use yet another colormap in order to differentiate these results from the previous two figures, due to the new range of values and, more importantly, the new units. The total access time increases for users at higher altitudes and higher inclinations. This helps to explains why errors tend to be lower for users at higher inclinations, since more access time involves more measurements for use in the estimation algorithm, and thus reducing uncertainty. However, the gains in access time generated by the higher altitude orbits do not result in better uncertainty performance compared with lower orbits. This means that the altitude of the user satellite is of greater importance in reducing uncertainty than maximizing total access time between the orbits.

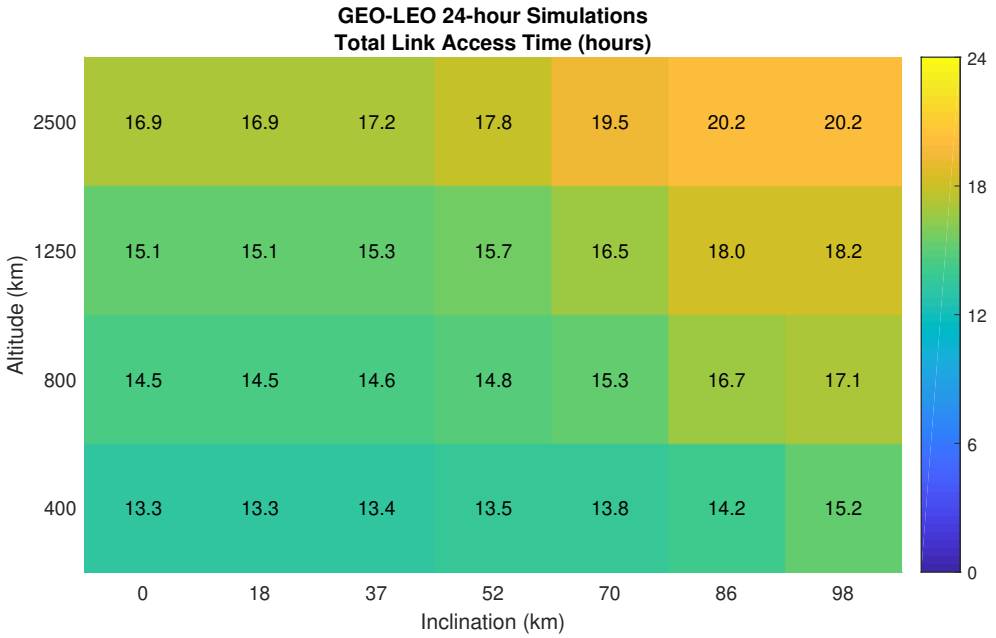


Figure 4-7: Summary of total link-access times over 24-hour simulations of GEO "relay" satellite with LEO "user" satellites in varying orbital altitudes and inclinations. More access time is better.

Example User in Highly-Elliptical Orbit (HEO)

In this scenario, we employ the same analysis as we did earlier with OICETS as an example user in LEO, but this time for a potential user in a high-altitude or highly-elliptical orbit. For this study, we use the NASA Magnetospheric MultiScale (MMS) mission as a model for HEO spacecraft.

The MMS mission is made up of four formation-flying satellites launched in 2015 that have completed operations in two science mission phases, and have now embarked on an extended mission phase after its designed mission completed earlier in 2019. The first science mission phase (from 2015 to 2017) had the four spacecraft in a 1.2 $R_E \times 12$ R_E orbit, with a mean-motion of nearly one revolution-per-day. The second science phase (from 2017 to 2019) had a raised apogee of roughly 25 R_E, orbiting with a mean-motion of nearly 1/3 of a revolution-per-day. In order to view both apogee and perigee in a single 24-hour simulation, we chose to model our example HEO user based on the MMS mission from its initial science phase. Figure 4-8 shows the results

of an EKF error analysis for this GEO-HEO scenario.

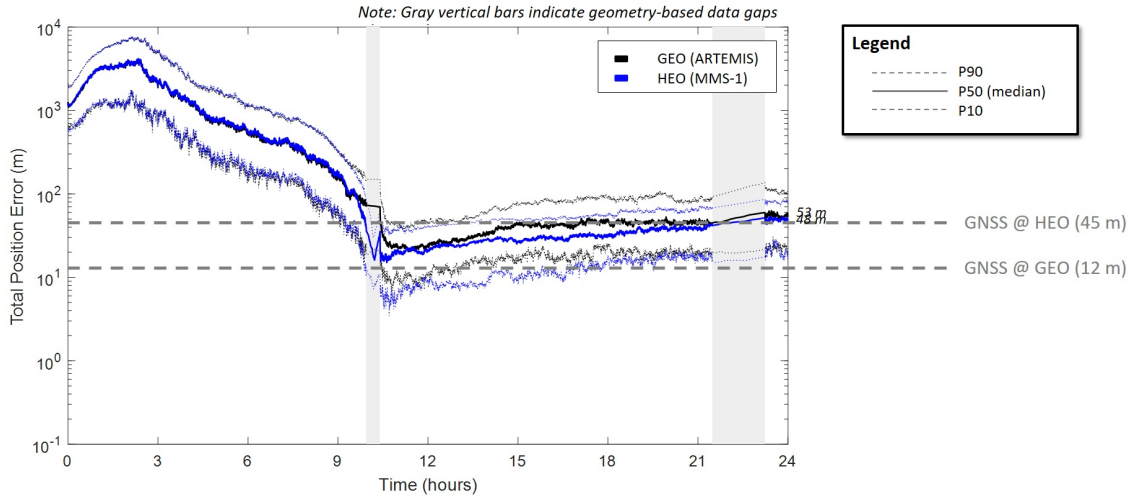


Figure 4-8: EKF results (10/50/90 percentiles) from Monte Carlo simulations (N=100) of a single GEO-relay satellite, ARTEMIS (black), with a single HEO user, MMS-1 (blue).

As before, gray vertical bars are used to indicate periods of geometry-based outages in the link between the two spacecraft. When the link is available, measurements are collected at a time-step of $T = 10$ seconds (a sample frequency of 0.1 Hz). The HEO user is at apogee at the start and end of the simulation period, and reaches perigee just before $t = 12$ hours. Since both satellites operate in relatively similar inclinations, this leads to two geometry-based data gaps (due to Earth occlusion) about every 12 hours.

The results in Figure 4-8 demonstrate the variability in navigation performance due to the change in intersatellite geometry. In the baseline GEO-LEO example earlier, simulations showed navigation errors on the order of 3-5 meters for the LEO user satellite, and about 15 meters for the GEO relay satellite. In this scenario, steady navigation errors are on the order of 30-50 meters, and are only achieved after about 12 hours of simulation, after the HEO satellite travels through perigee. This reiterates the finding from the orbtial parameter-sweep analysis that better performance is expected from scenarios involving satellites in lower altitudes. Interestingly, both the GEO relay and HEO user satellites have similar performance throughout, likely due to

the reduced impact of J2 perturbing dynamics beyond a certain altitude. This result would likely be different if third-body perturbations is included in the filter dynamics model, as higher altitude satellites are more affected by the moon and sun, and this should be investigated for its impact on performance for high-altitude satellites in future work.

4.1.3 Multiple Relays and/or Multiple Users

CRLB Analysis of Different Scenarios

Multiple GEO-relay satellites can be deployed in order to maintain custody of user satellites on either side of the Earth, such that there are no geometry-based data gaps from the perspective of the user. We analyzed this concept by performing a CRLB uncertainty analysis for different scenarios of two or three GEO-relay satellites equally-spaced in longitude around the Earth, and either a single user in LEO or HEO (modeled by OICETS and MMS-1, respectively), or a heterogeneous set of both LEO and HEO users.

In the case of three GEO relay satellites (at 0°, 120°, and 240° longitude), all three relay hubs maintain constant crosslinks with one another, emulating a concept where user satellite data can be relayed to the optimal GEO spacecraft for downlink. Note that this is not geometrically possible with only two GEO relay satellites as their relative position vector is intersected by the Earth, being directly opposite one another in longitude at 0° and 180°. In all cases with multiple relay satellites, the user satellite(s) can have 1 or 2 GEO satellites available for a crosslink at any given time, therefore the link-selection framework developed in this thesis is activated in this scenario, with evaluation cycles every 15 minutes for LEO users and every 180 minutes for HEO users.

Figures 4-9, 4-10, and 4-11 show the predicted uncertainty results of these simulations from the perspective of the LEO user, the HEO user, and the GEO relay satellites, respectively. Dotted lines are used to denote scenarios with a single LEO user, dashed lines are used to denote scenarios with a single HEO user, and solid lines

are used to denote scenarios with the set of both users. Colors are used to denote the number of GEO relay satellites used in each scenario: blue for 1, orange for 2, and gold for 3.

Note that there are no gray vertical bars in any of the three figures. Although data gaps do exist for the 1x GEO relay cases (in blue), we decided to not display any data-gap indicators in order to avoid any confusion since the 2x and 3x GEO relay cases, which do not experience data gaps, are shown in the same figure. Still, periods of data gaps in the blue lines are still easily distinguishable spikes in uncertainty that occur nearly every 90 minutes for the LEO user, and the periods of steadily increasing uncertainty every 12 hours for the HEO user.

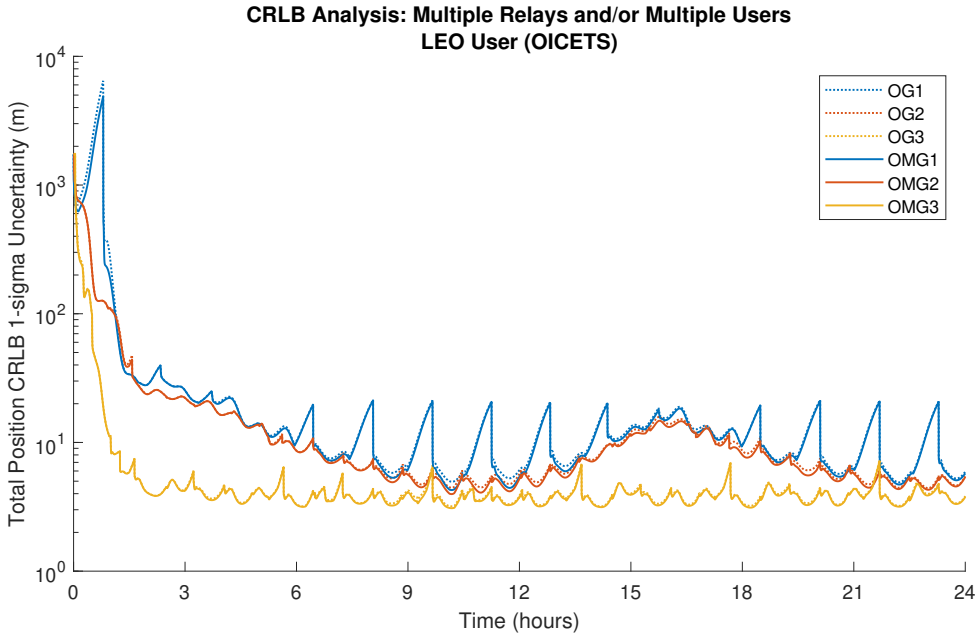


Figure 4-9: CRLB analysis results for LEO satellite (OICETS) in different multiple-relay/user configuration: 1x GEO-relay in blue, 2x GEO-relays in orange, and 3x interconnected GEO-relays in yellow. Dotted curves are used for single LEO user scenarios, and solid curves for multiple users (LEO+HEO). In the legend, 'O' denotes those scenarios that include the LEO user OICETS, 'M' denotes those scenarios that include the HEO user MMS-1, and 'G#' denotes the number of GEO relay satellites used in those scenarios.

Figure 4-9 shows the uncertainty results from the perspective of the LEO user satellite. The first observation we can make from these results is that there is little

difference made from including the HEO user. As shown in all three colored cases, the dotted lines and the solid lines are mostly indistinguishable. However, the number of GEO relay satellites has a perceivable impact in the stability and minimum value of achieved uncertainty. The blue and orange lines achieve similar minimum values, but the orange lines are clearly more stable due to the distribited relays to maintain custody of the user satellite and mitigate data gaps due to geometric link availability. However, both of these scenarios still experience observability effects, as seen by the long-term periodic behavior with a period of 12 hours. This effect is removed by the third scenario with 3x GEO relay satellites, when the interconnectivity of the constellation network is augmented by links between the relay satellites. This represents a closure of the constellation network, which allows gains in the estimation observability.

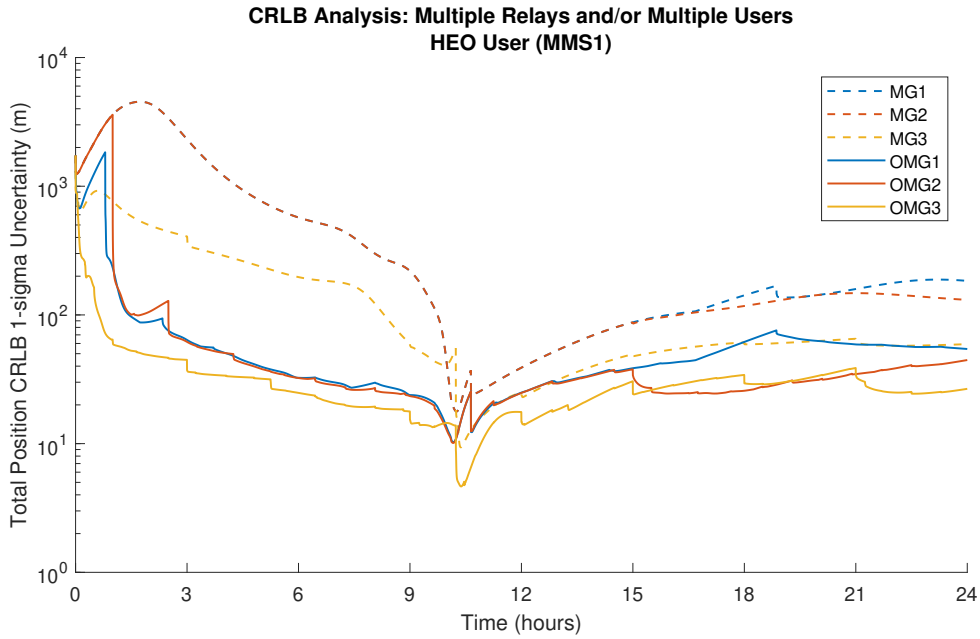


Figure 4-10: CRLB analysis results for HEO satellite (MMS-1) in different multiple-relay/user configuration: 1x GEO-relay in blue, 2x GEO-relays in orange, and 3x interconnected GEO-relays in yellow. Dashed curves are used for single HEO user scenarios, and solid curves for multiple users (LEO+HEO). In the legend, 'O' denotes those scenarios that include the LEO user OICETS, 'M' denotes those scenarios that include the HEO user MMS-1, and 'G#' denotes the number of GEO relay satellites used in those scenarios.

Figure 4-10 shows the uncertainty results from the perspective of the HEO user satellite. Note that the blue dashed line is hidden behind the orange dashed line for most of the simulation, as the same link between the HEO user and the 0° longitude GEO relay is activated by the link-selection algorithm up until $t = 15$ hours, when a different link is selected and the results begin to diverge.

From the perspective of the HEO user, there is a significant difference made from the inclusion of a user satellite in LEO. As seen in all three colors, the solid lines clearly outperform the dashed lines in achieving smaller uncertainty values throughout the period of the simulation. This again is due to the gains in filter observability provided by including a satellite at a lower-altitude, as shown in the parameter-sweep analyses earlier. The number of GEO relay satellites again shows a significant impact in the stability and minimum value of achieved uncertainty. Lower uncertainty results are achieved by those cases with more relay satellites, and thus more geometric distribution and network interconnections.

Finally, Figure 4-11 shows the uncertainty results from the perspective of the GEO relay satellite at 0° latitude. The relay satellites located at other latitudes (in the cases with multiple relays) are not shown, both because they are not used in every scenario, and they tend to achieve very similar results to those of the 0° satellite. Note that the blue dashed line is again hidden behind the orange dashed line for most of the simulation, due to these cases having equivalent link-selections up until $t = 15$ hours.

These results reinforce all of the previous observations, as the solid lines (representing the cases with both heterogeneous orbits, LEO & HEO) clearly outperform the dashed lines (representing the HEO only cases) in achieving smaller uncertainty values throughout the period of the simulation, but the dotted lines (the LEO only cases) are nearly equal to the solid lines. Also, lower uncertainty results are achieved by those cases with more distributed and interconnected relay satellites, as the gold cases (3x GEO) outperform the orange (2x GEO) and blue (1x GEO) cases.

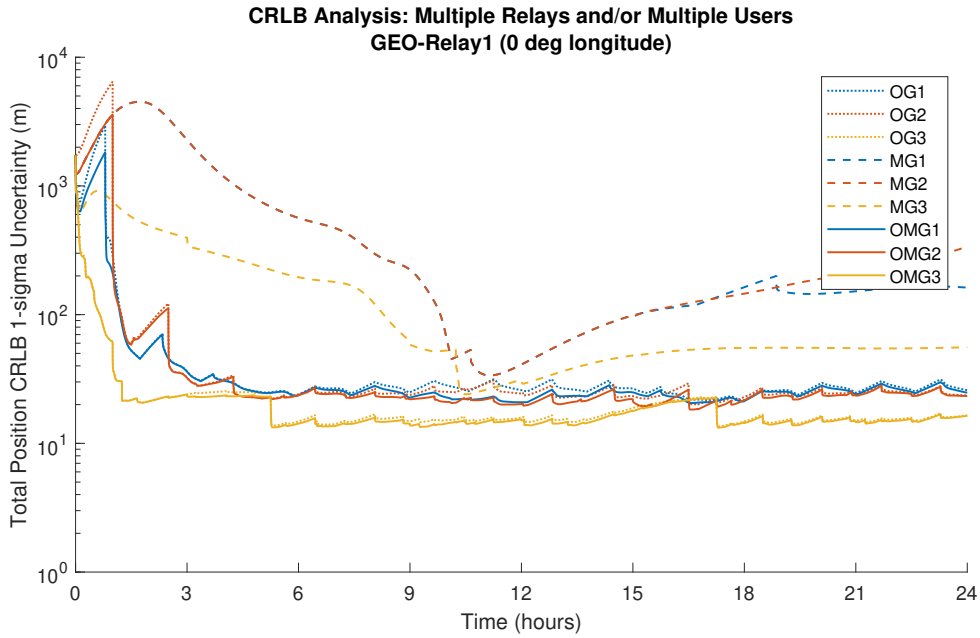


Figure 4-11: CRLB analysis results for GEO-Relay satellite #1 (at 0 deg longitude) in different multiple-relay/user configurations: 1x GEO-relay in blue, 2x GEO-relays in orange, and 3x interconnected GEO-relays in yellow. Dotted curves are used for single LEO user scenarios, dashed curves for single HEO user, and solid curves for multiple users (LEO+HEO). In the legend, 'O' denotes those scenarios that include the LEO user OICETS, 'M' denotes those scenarios that include the HEO user MMS-1, and 'G#' denotes the number of GEO relay satellites used in those scenarios.

EKF Error Analysis of 3x GEO Relay Scenarios

In order to gauge the navigation performance of a multiple relay/user scenario, and how it compares to our earlier performance results of a single relay and a single user, we simulated the best-performing scenario from the previous CRLB analysis (3x GEO) in a full error analysis using 100 Monte Carlo simulations (sampling noise/uncertainty in the measurements and estimation) with the EKF estimator. Figures 4-12 shows the result of a 3x GEO relay satellite scenario with a LEO user.

At the end of the 24-hour simulation period, the median total position errors are about 8 meters for the GEO satellites, and 1.4 meters for the LEO satellite. These results demonstrate improved performance (50% reductions in total position error) over the single GEO-relay case shown in Figure 4-2. This is expected due to the continuous availability of intersatellite measurements.

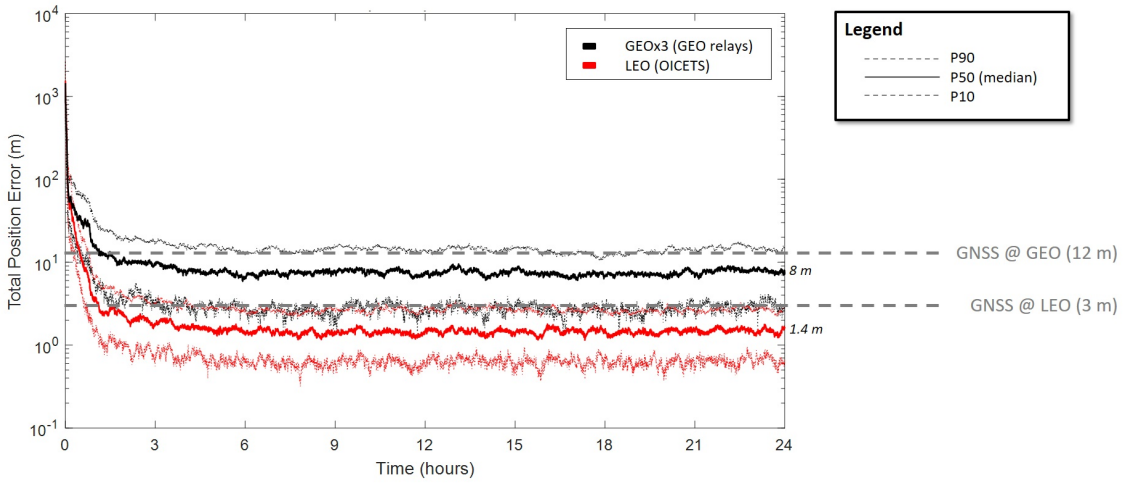


Figure 4-12: EKF results (10/50/90 percentiles) from Monte Carlo simulations (N=100) of three GEO-relay satellites equally separated in longitude (black) with a single LEO user, OICETS (red).

However, in addition to mitigating geometry-based link outages and their effect on short-term stability, the distributed GEO-relay satellites also help to mitigate the long-term geometry-based effects caused by estimation observability, since the link-selection framework selects the optimal GEO satellite that minimizes uncertainty on a 15-minute cycle, leading to a more stable estimation result.

Figure 4-13 shows the impact of multiple GEO relay satellites communicating with a single HEO user. These results again demonstrate improved performance over the previous single GEO relay case, due to continuous availability of links, and improved observability from the inclusion of more distributed and interconnected satellites.

4.2 Case Study E2: LEO Constellations

In this second case study of Earth-orbiting applications, we evaluate the navigation performance between satellites operating in low Earth orbit. With the cost of access to space (and LEO in particular) declining each year with new primary and secondary launch capabilities, many organizations have proposed using LEO satellites to serve a variety of missions, such as communications, remote sensing, secure data operations,

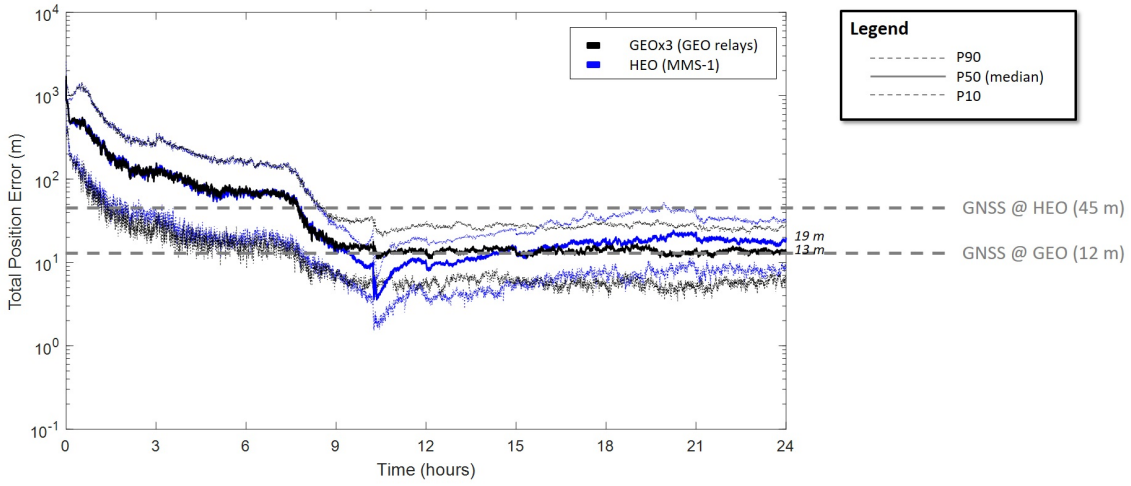


Figure 4-13: EKF results (10/50/90 percentiles) from Monte Carlo simulations (N=100) of three GEO-relay satellites equally separated in longitude (black) with a single HEO user, MMS-1 (blue).

and navigation. Low-latency communications in particular has recently become a commercial space-race, with quite a few companies proposing different designs for mega-constellation configurations, of which several expect to use optical intersatellite communications. With these types of missions in mind, both large and small, we simulate a few potential classes of constellation size/configuration in this study, starting simple with a two-satellite navigation scenarios, and increasing complexity with a few example constellations of varying size and distribution.

4.2.1 Setup of Scenarios

All two-satellite scenarios are simulated over the 24-hour time period between $t_0 =$ 16 Mar 2015 16:00:00 UTC and $t_N =$ 17 Mar 2015 16:00:00 UTC, with a time-step of $T = 10$ seconds. All constellation scenarios are simulated over a 6-hour time period with $t_N =$ 16 Mar 2015 22:00:00 UTC, with a time-step of $T = 10$ seconds.

Different scenarios are defined using different LEO-LEO configurations. Initial states for spacecraft modeled off of existing satellites are provided in the form of TLEs from the online database Space-Track.org [79]. The downloaded TLEs provided

Keplerian element states at for a given time of epoch, $t_e^{(i)}$ for the satellites TerraSAR-X and NFIRE. Table 4.4 shows the orbital elements and epoch-time for these spacecraft. We also the LEO subset of satellite orbits generated for parameter-sweep studies, shown in Table 4.2, set for the epoch-time $t_e = t_0$.

Table 4.4: Orbital parameters for LEO-LEO scenario.

Orbital Element	**TerraSAR-X (LEO)**	**NFIRE (LEO)**
n (rev/day)	15.1915	15.4337
e	1.50e-4	1.24e-3
i (deg)	97.45	48.16
ω (deg)	73.89	119.25
Ω (deg)	83.71	221.59
M (deg)	52.10	287.99
t_e	2015/03/16	2015/03/16
(UTC)	07:26:00.23	10:24:09.44

For all cases, the network topology is defined as $topo_{\text{ALL}}$. As we mentioned earlier, this type of network topology becomes increasingly more difficult as the quantity of satellites grows, as additional links available at the same time would require as many lasercom payloads as the maximum amount of simultaneous links for each satellite. Despite these challenges, we have chosen to simulate the "all-to-all" network architecture for all cases in order to isolate the impact of constellation size and distribution on navigation results. The impact of varying network architectures is a focus of the case studies in Chapter 5.

The minimum link altitude is defined for Earth at 100 km and kept constant for all cases. In order to show "best-case" performance results, all satellites are assumed to have no "off" mode durations ($T_{\text{off}} = 0$), and thus link-availability is not affected by duty-cycles. However, T_{on} is defined with non-zero values in order to maintain evaluation periods for the link-selection framework. For LEO satellites, link-selection is evaluated every 15 minutes.

4.2.2 Two-satellite Navigation

Baseline Example: TerraSAR-X & NFIRE (2008)

The baseline two-satellite case is modeled off of the LEO satellites TerraSAR-X and NFIRE, which were used for the first bidirectional lasercom crosslink between two LEO satellites in 2008 [31]. Figure 4-14 shows the 10th, 50th, and 90th percentile results from a 100-run Monte Carlo simulation of continuous lasercom crosslink measurements between the two spacecraft over a longer 72-hour period. A black vertical line is drawn at the end of 24 hours of simulation for easy comparison with other 24-hour studies.

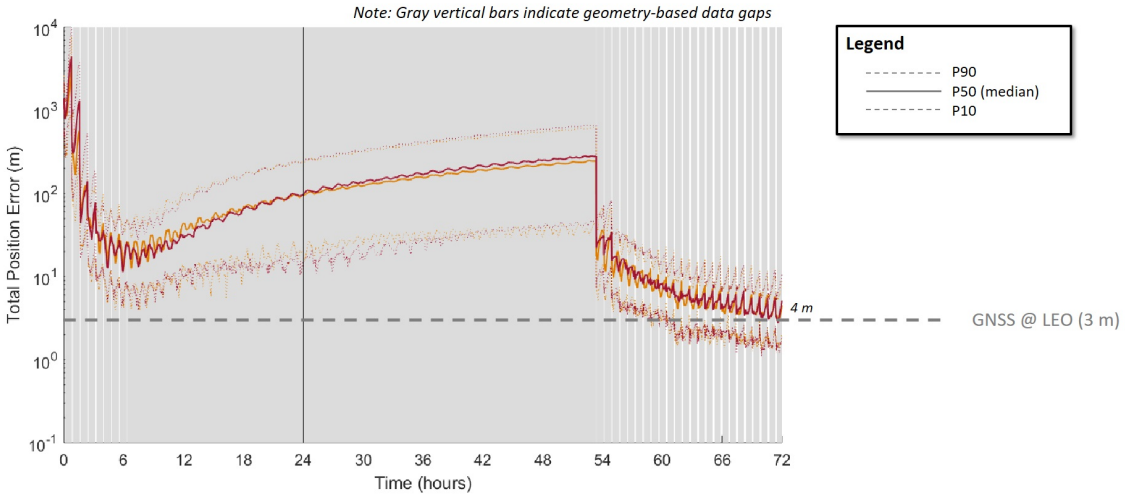


Figure 4-14: EKF results (10/50/90 percentiles) from Monte Carlo simulations (N=100) of baseline LEO two-satellite example: TerraSAR-X (orange) with NFIRE (red).

Gray vertical bars are used to indicate periods of geometry-based outages in the link between the two spacecraft. When the link is available, measurements are collected for the full availability period at a time-step of $T = 10$ seconds (a sample frequency of 0.1 Hz). Notice that most of the 72-hour simulation period is gray, with only short-duration links occurring in the time period from hours 0 to 6, and hours 54 to 72, and no links available for the full period between. At the end of the 72 hours of observation, the navigation filter is able to achieve median total position errors

of less than 3 meters for both satellites (meeting our performance objective for LEO satellites, based on the average performance of GNSS), but near 100 meters at the 24-hour mark due to the lack of measurement updates.

We use this result to serve as a baseline for navigation performance between two LEO satellites, to which we will compare the results of other combinations of orbits that generate different intersatellite geometries.

Parameter-sweep Analysis of Other Low Earth Orbits

We now perform a parameter-sweep analysis for other LEO-LEO configurations, using CRLB predicted uncertainty as a metric for comparing results.

The parameters that we alter to generate different orbit configurations are the altitude and inclination of the second LEO satellite, while keeping the orbit of the first LEO satellite consistent with that from the baseline scenario (TerraSAR-X). We use a subset of the altitudes used earlier in Section 4.1.2, keeping only those representative of LEO altitudes. We use the same set of inclinations as before. See Table 4.2 for the full sets of altitude and inclinations used in this study.

Figure 4-15 demonstrates the variability of navigation performance from different orbit configurations, from the state estimation perspective of the fixed LEO satellite, TerraSAR-X. From these results, it is unclear if there is a trend between navigation uncertainty and the altitude of the second LEO spacecraft, since the four different altitude scenarios achieve similar minimum uncertainty values, only at different times based on when link-access is geometrically available. However, the result of the baseline example (in red) relative to the other curves seems to indicate that certain inclinations may perform better or worse than others, as it is outperformed by the other curves with respect to minimum achieved uncertainty.

Figures 4-16 and 4-17 show colormap summaries of the minimum position uncertainty achieved over the 24-hour simulation period from the perspectives of the fixed LEO satellite (TerraSAR-X) and the partner LEO satellite at varying altitudes and inclinations, respectively. The x-axes are in order of increasing inclination, and the y-axes are in order of increasing altitude. Notice there is little variation both qualita-

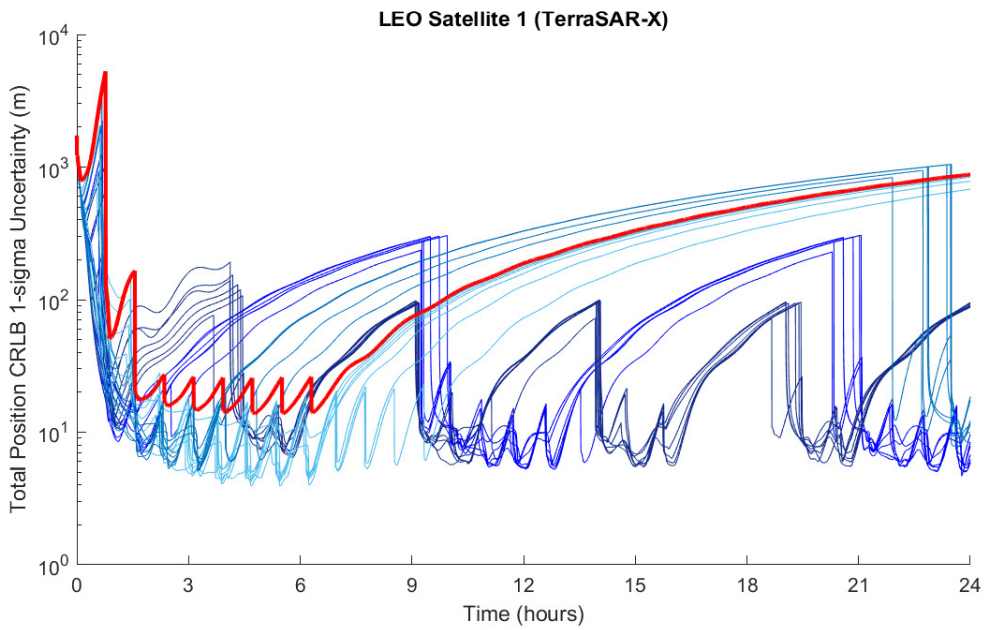


Figure 4-15: CRLB analysis results for TerraSAR-X LEO satellite with other LEO satellites in varying orbital altitudes and inclinations. Different altitudes are depicted by different shades of blue: lower altitudes are lighter shades, and higher altitudes are darker shades. Baseline example of TerraSAR-X & NFIRE is shown in red.

tively and quantitatively between the two sets of results (with a maximum difference of 0.8 meters). This is thought to be due to both satellites being in a relatively similar regime of orbital altitudes, and thus having relatively equal observability within the intersatellite navigation filter.

It is clear from these results that better minimum uncertainty values are achieved when the second LEO satellite is at a lower altitude of 400 km. However, that class of orbits also contains the highest minimum uncertainty value (9.1 meters) where $i = 98°$. This outlier is again due to the single unobservable case of this navigation filter (when the two spacecraft are in coplanar orbits at the same altitude and phasing). TerraSAR-X is in a 500-km altitude at $i = 98°$, which explains why the case at 400-km altitude and $i = 98°$ is the worst-performing case in this study, as it is close to the unobservable case.

In order to make any further observations, we also need to consider the total access times achieved by these sets of orbital configurations. Figure 4-18 shows a

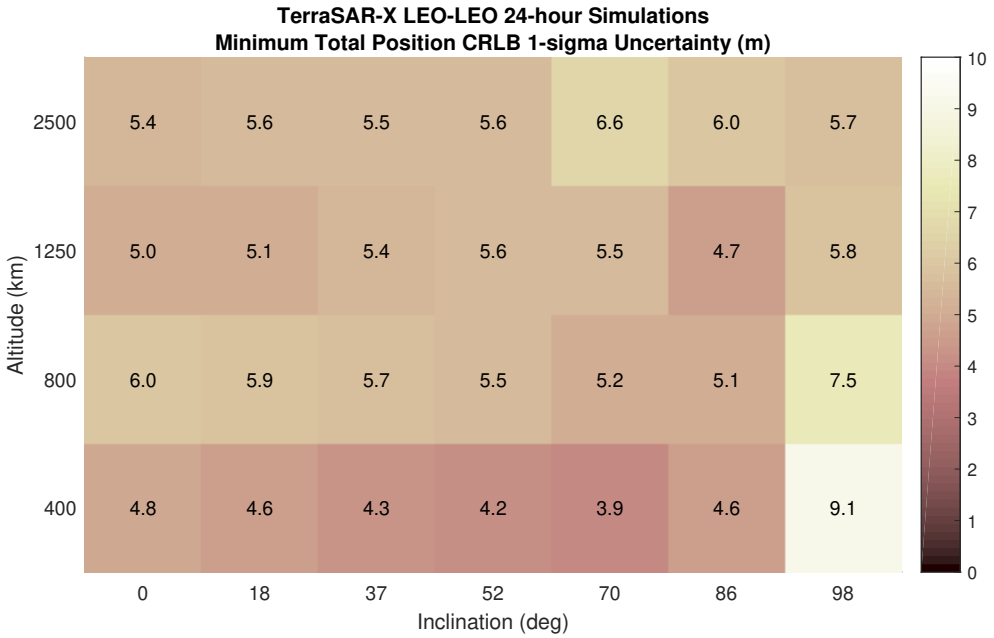


Figure 4-16: Summary of minimum achieved CRLB over 24-hour simulation for TerraSAR-X LEO satellite with other LEO satellites in varying orbital altitudes and inclinations.

colormap summary of the total access times between the LEO satellites over the 24-hour simulation period. The first observation we can make from these results is that the regime of 800-km altitude orbits suffers the most (relative to the other altitudes analyzed) in terms of total access times due to Earth-occlusion. This explains why orbits at 800 km tend to be outperformed by those at 1250 km, but not by those at 2500 km. Hence we conclude that lower altitudes produce better results from gaining observability in the navigation filter, but only benefit when there are adequate amounts of link-access times. At each altitude, the total access time also increases as the inclination increases, which should result in lesser uncertainty simply due to a higher quantity of measurements. However, we previously determined that higher inclinations also increase observability errors, since that increases proximity to the unobservable case. This seems to explain why uncertainty tends to be lowest for some middle inclination in each altitude regime, as there is a trade-off between maximizing total access time while minimizing observability effects.

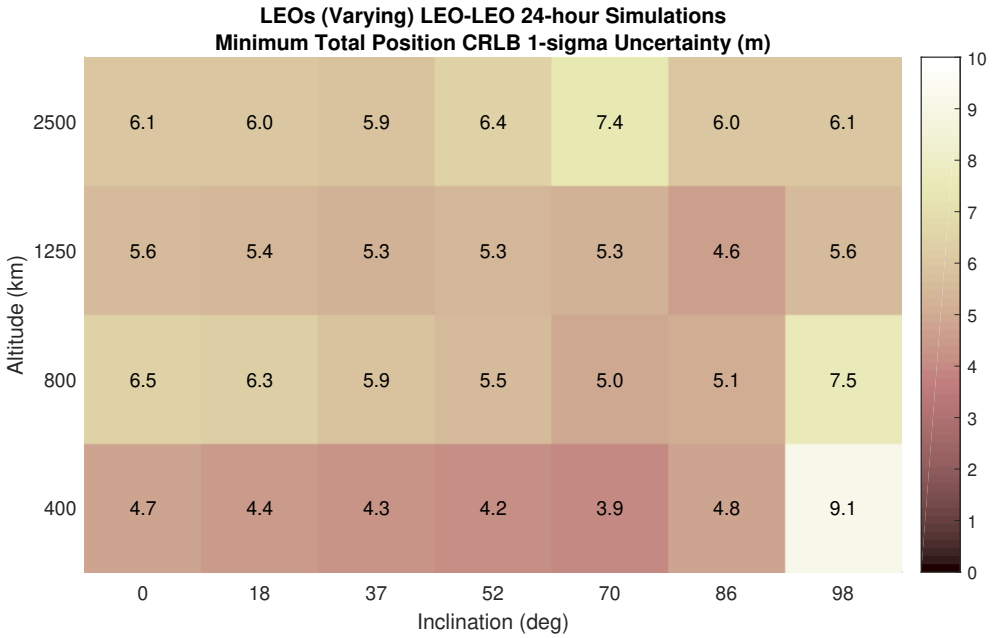


Figure 4-17: Summary of minimum achieved CRLB over 24-hour simulation for the parameter-sweep satellite (varying orbit) in parameter-sweep simulations of TerraSAR-X LEO satellite with other LEO satellites in varying orbital altitudes and inclinations.

4.2.3 Walker Constellation Navigation

Next, we evaluate intersatellite navigation using lasercom measurements for a few example LEO constellation cases. Figure 4-19 depicts the orbital scenario of the smallest and largest constellation configurations considered in this study.

It is expected that increasing the number of satellites that can serve as crosslink partners should improve navigation performance due to the higher quantity and geometric distribution of measurements, however larger constellations become more complex from a feasibility perspective, depending on the network connectivity. In most of the following cases, we assume that all satellites are able to link with all other satellites ("all-to-all" architecture), and also evaluate a best-case performance assuming the highest level of interconnectivity.

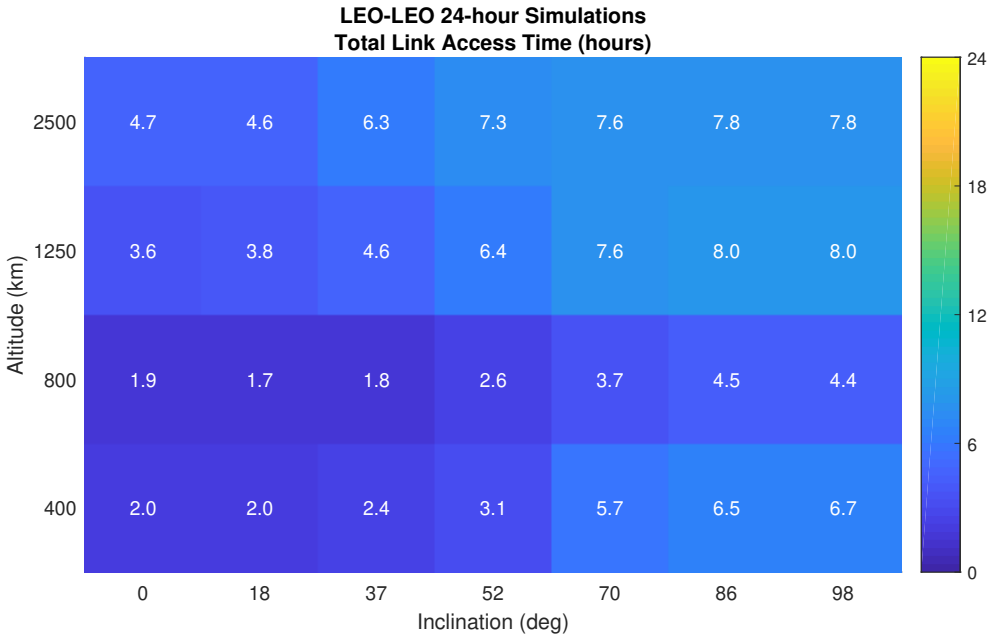


Figure 4-18: Summary of total link-access times over 24-hour simulations of TerraSAR-X LEO satellite with other LEO satellites in varying orbital altitudes and inclinations.

Baseline Example: Simple Configuration with Zero Simultaneous Links

The first notional LEO constellation we analyze is one that we designed, in which all satellites nearly continuously possess one, and only one, link access partner at a time. This was achieved with a Walker Delta 6/2/1 constellation (6 total satellites, in 2 planes, with automatic inter-plane spacing) at an altitude of 1,067 km and an inclination of 60 degrees. Figure 4-20 shows the 10th, 50th, and 90th percentile results from a 100-run Monte Carlo simulation for one of the satellites in this constellation over a 24-hour period. The bottom plot in the figure shows which link is available at each point in time for each satellite in the constellation. As intended, there are no overlaps between links, and gap-time is minimal.

Results are only shown for one of the satellites (Sat11) because all of the satellites exhibit near-equal results. This is expected due to the symmetry of the constellation. By the end of the simulation period, the median total position error is reduced from the $>$ 1-km initial error to about 0.9 meters. This result demonstrates one example for

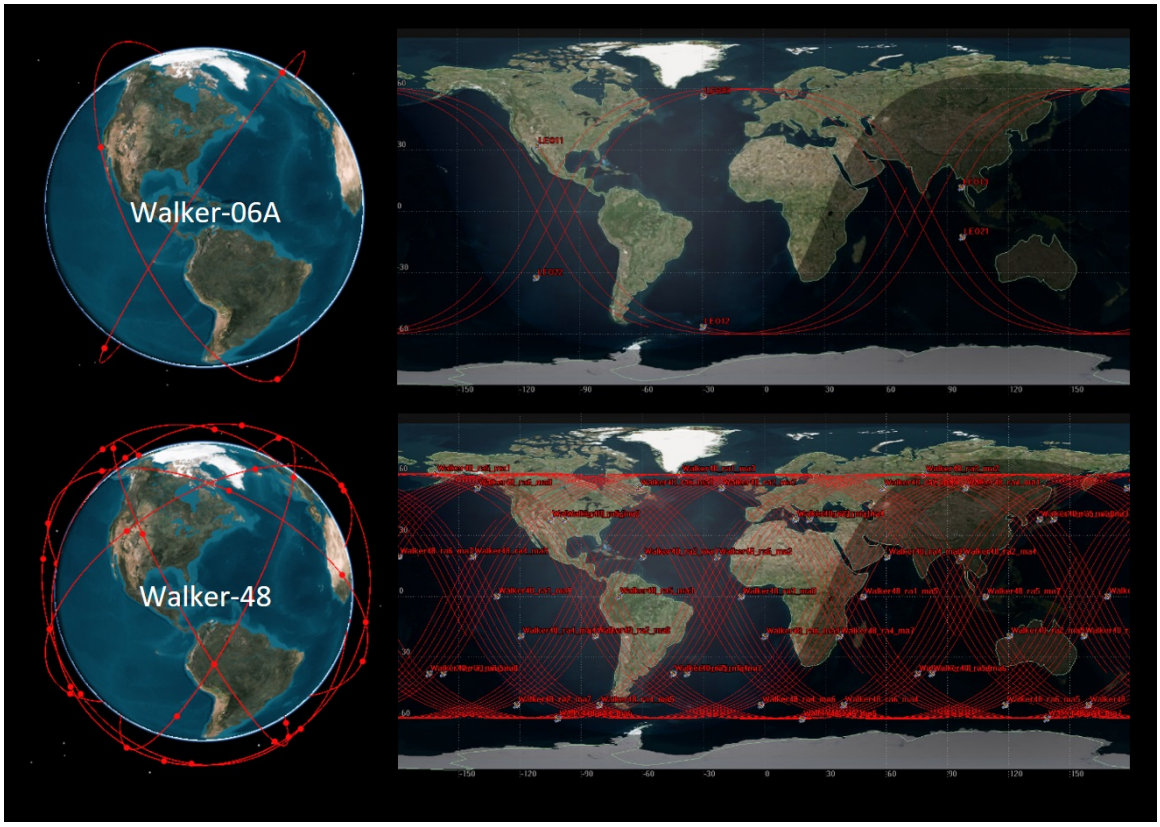


Figure 4-19: Diagrams of orbital scenario and ground-tracks for the smallest and largest constellations considered for a LEO Walker constellation case study, generated using AGI STK. Walker-06A denotes the Walker Delta 6/2/1 constellation, and Walker-48 denotes the Walker Delta 48/6/1 constellation.

potential sub-meter navigation performance by employing a lasercom-networked constellation architecture. In the next section, we evaluate the impact of constellations of greater quantity and/or distribution of satellites.

Uncertainty Analysis of More Complex Configurations

For this analysis, we maintain the same constellation altitude and inclination as used in the baseline example, but we steadily increase the size of the Walker Delta constellation from 6 satellites up to 48 satellites, with consideration to how the satellites are distributed within the constellation. See Table 4.5 for an overview of the Walker constellations used for this study.

The inter-plane spacing parameter, IPS, is a Boolean variable for offsetting the

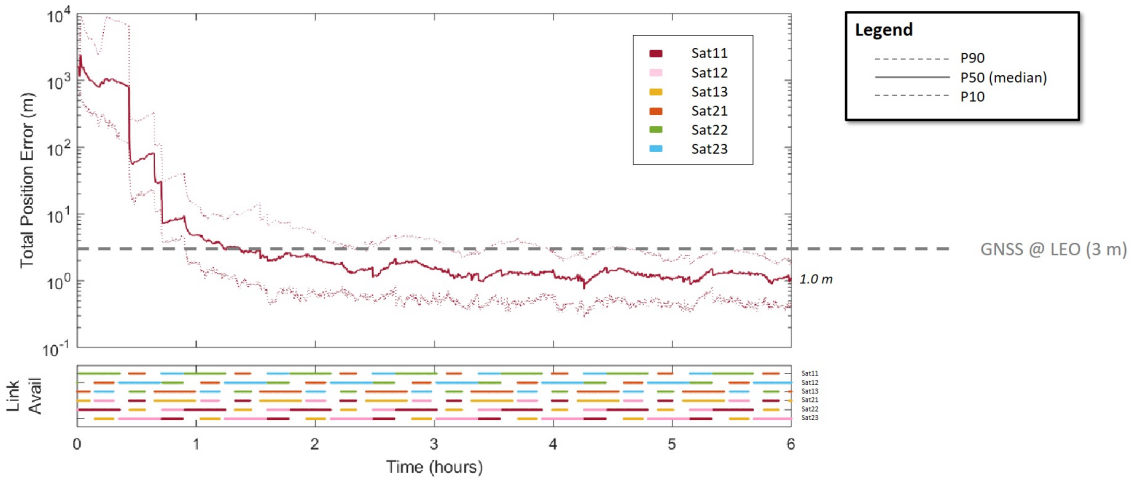


Figure 4-20: Top: EKF results (10/50/90 percentiles) from Monte Carlo simulations (N=100) of baseline LEO constellation example: a simple six-satellite configuration without any simultaneous links. Bottom: Depiction of which crosslink partner is available for each satellite at any given time over the course of the simulation period.

Table 4.5: Walker constellation configurations at a=7445.83 km, e=0, and i=60°.

Constellation	n_{op}	n_{sp}	IPS
Walker Delta 6/2/1	2	3	1
Walker Delta 6/2/0	2	3	0
Walker Delta 12/2/1	2	6	1
Walker Delta 12/4/0	4	3	0
Walker Delta 24/4/1	4	6	1
Walker Delta 48/6/1	6	8	1

mean anomaly of the satellites in each plane by an amount equal to:

$$M_{op} = \frac{(op-1) \times 360^{\circ}}{n_{op} \times n_{sp}} \tag{4.1}$$

where $op \in \{1 \cdots n_{op}\}$ is the orbital plane index, n_{op} is the total number of orbital planes, and n_{sp} is the total number of satellites in each plane.

Although the baseline example is successful in its design goal of ensuring that each satellite has near-continuous link-access with the other spacecraft without simultaneous links, it does not result in an ideal distribution of its satellites, and carries some

concern for orbital collisions (also known as conjunctions) where the planes intersect. This is partially due to the inter-plane spacing being set to 1, such that the mean anomaly offset is in phase with the natural spacing of the satellites in each plane. In a real-life utilization of this constellation design, conjunctions can still be mitigated by inserting an additional small offset in mean anomaly, or offsetting any of the other orbital parameters. In this case, the greatest distribution of the satellites is achieved by setting the inter-plane spacing to 0, and thus not using any offsets in mean anomaly between planes.

The first case we simulate is this six-satellite, two-plane configuration with the inter-plane spacing set to 0. Compared to the baseline case, this change effectively eliminates one of the three possible crosslink partners of each satellite, and thus reduces constellation connectivity. The second and third cases double the size of the constellation to 12 satellites, the prior by doubling the number of satellites in each plane n_{sp}, and the latter by doubling the number of planes n_{op}. The fourth case doubles the constellation size again to 24 satellites by combining the two previous 12-satellite designs. The final case is a 48-satellite constellation which augments both quantities n_{sp} and n_{op}, instead of doubling either one individually. For each, the inter-plane spacing parameter is set for greatest satellite distribution (and thus no conjunctions).

As mentioned earlier, an "all-to-all" network topology architecture is assumed for all cases. Though the complexity of implementing this increases exponentially with respect to constellation size, this assumption allows us to examine the effect of the quantity and distribution of satellites without additional modifications to the network connectivity, and also results in a best-case scenario. Figure 4-21 shows the resulting CRLB position uncertainty for these constellation cases.

As expected, the reduced connectivity of the Walker Delta 6/2/0 constellation results in higher uncertainty than the baseline example of a Walker Delta 6/2/1 configuration. When the constellation size is doubled to 12 satellites, the uncertainty drops by nearly 50%. This trend continues as the constellation size increases, as doing so opens new network links for each satellite and augments the total number of

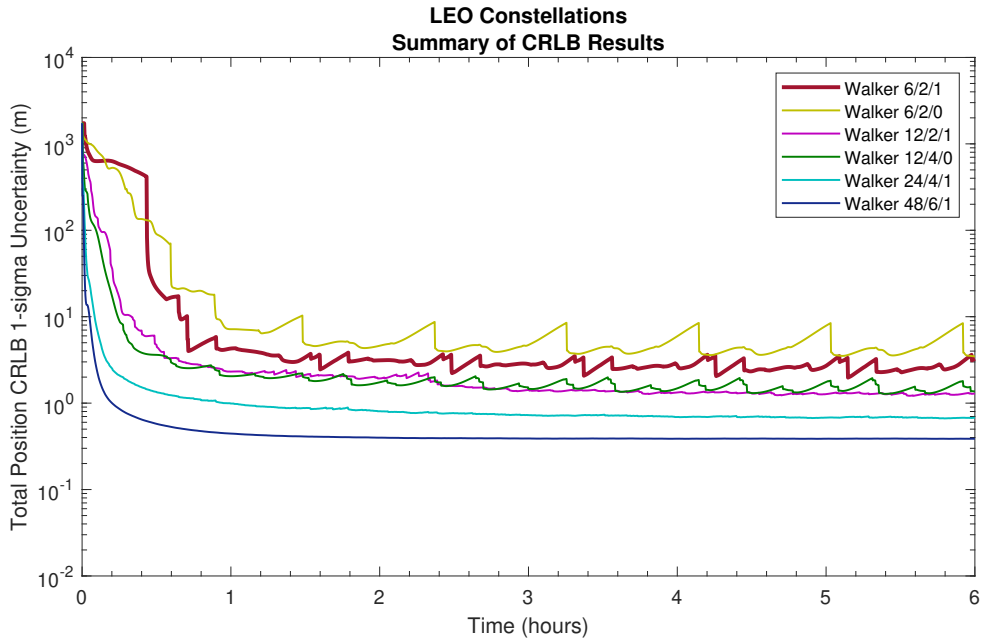


Figure 4-21: CRLB analysis results for LEO Walker Delta constellations of varying size and distribution. The baseline Walker Delta 6/2/1 example is denoted by the thick red curve.

measurements available to the filter to reduce estimation uncertainty. However, the improvement seems to be diminishing as the constellation is continually doubled in size.

At smaller constellation sizes, the only type of links accessible by each satellite are those from "crossing" satellites, those in other orbital planes that come into view near the intersections with those planes. Based on the altitude and inclination of the constellation, at a certain value of n_{sp}, the constellation now has enough spacecraft evenly distributed in mean-anomaly in order for each satellite to maintain constant links with "leading/trailing" satellites (those nearest in the same orbital plane). This is similarly true beyond a certain value of n_{op}, where the constellation now has enough spacecraft evenly distributed in RAAN in order for satellites to maintain constant links with "neighboring" satellites (those in the nearest orbital planes). Values beyond these critical points will only increase the number of satellites in the same or neighboring planes that can be accessed by a given satellite in the constellation. Though more connections leads to additional measurements to reduce estimation uncertainty, this

is expected to have diminishing returns, as greater quantities of satellites in an evenly distributed Walker Delta constellation also reduces the intersatellite range, which is an important observable in the intersatellite navigation filter.

Based on the altitude and inclination of the constellation configuration used in this study, and the minimum link altitude of 100 km, the constellations of 6, 12 and 24 satellites can only conduct links between "crossing" satellites. However, the 48-satellite constellation is large enough to add additional links with one "leading" satellite, one "trailing" satellite, and one "neighboring" satellite. We expect that constellations of larger size and distribution would likely see smaller uncertainty values, though with diminishing returns.

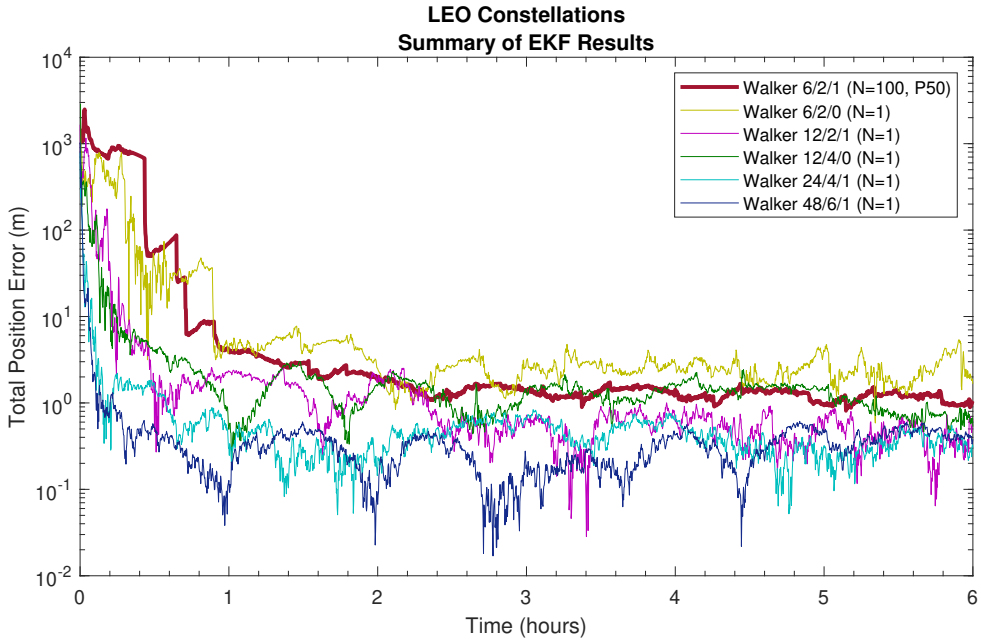


Figure 4-22: EKF results for LEO Walker Delta constellations of varying size and distribution. 50th percentile results (N=100) of the baseline Walker Delta 6/2/1 example is denoted by the thick red curve.

In order to gauge how these uncertainty analysis results translate to navigation errors, we ran a few sample EKF simulations for each of the constellations used in this study. The navigation errors from these results are shown in Figure 4-22. Note that other than the baseline example, which is shown as the 50th percentile of 100 Monte Carlo simulations, all other cases were only run once (i.e., N=1), and thus do

not have statistical significance for sampling noise distributions. This is because of the large run-time for each of these cases, due to full structure of the "all-to-all" link topology being evaluated at each time-step. However, these sample cases still serve to show the potential sub-meter performance of LEO constellations larger than six satellites, assuming a large degree of network connectivity between the satellites.

4.3 Summary of Results

This section summarizes the estimated performance of using an intersatellite navigation filter with measurements from lasercom crosslinks in the three orbital domains for Earth-orbiting applications: LEO, GEO, and high-altitude or HEO. Performance values are compared with the threshold goals as set from the literature review.

4.3.1 LEO Satellites

A summary of results from both case studies (GEO Relay and LEO Constellation) are assembled in Figure 4-23. From two-satellite navigation uncertainty analyses, we concluded that satellites at lower altitudes are preferred for leveraging the dynamics model of the intersatellite navigation filter in order to gain observability and reduce estimation error and uncertainty. The performance goal of 3 meters for LEO satellites is achieved by both scenarios: with 3x GEO relay satellites (for full orbit coverage and interconnectivity between relay hubs), and LEO Walker constellations (that leverage satellite distribution for more measurements and greater connectivity).

4.3.2 GEO Satellites

A summary of the results for GEO satellites from the GEO Relay case study are assembled in Figure 4-24. In this "hub-node" topology model, we concluded that the GEO satellites achieve best performance when linked with satellites at lower altitudes (to leverage estimation observability in the dynamics model), and with other GEO hubs (to gain interconnectivity effects in reducing error covariance). As such, the

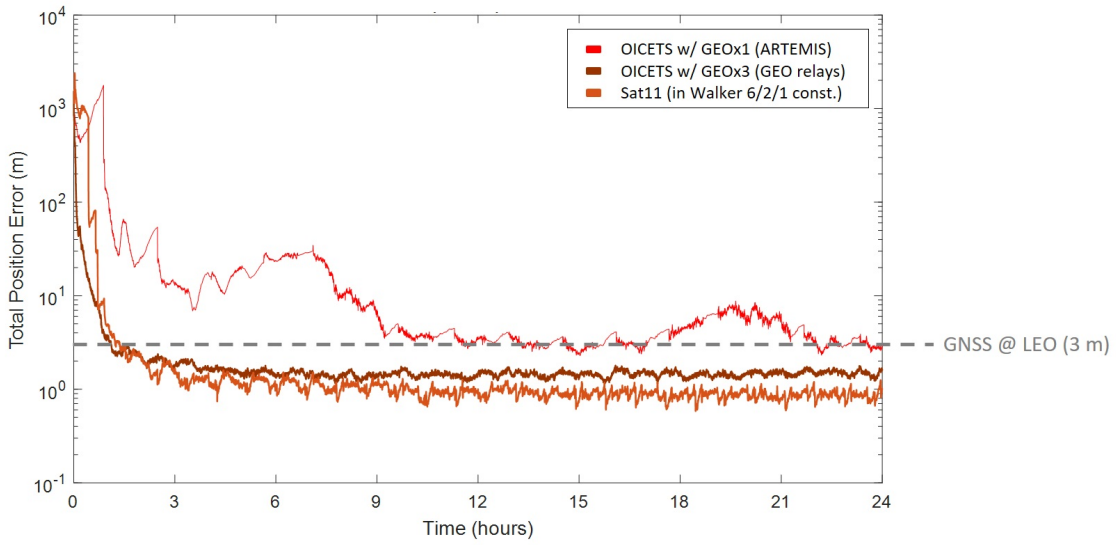


Figure 4-23: Summary of EKF results (50th percentile over 100 Monte Carlo simulations) for LEO satellites from both the GEO-Relay and LEO Constellation case studies.

performance goal of 12 meters for GEO satellites is achieved in the scenario with LEO users and multiple GEO hubs.

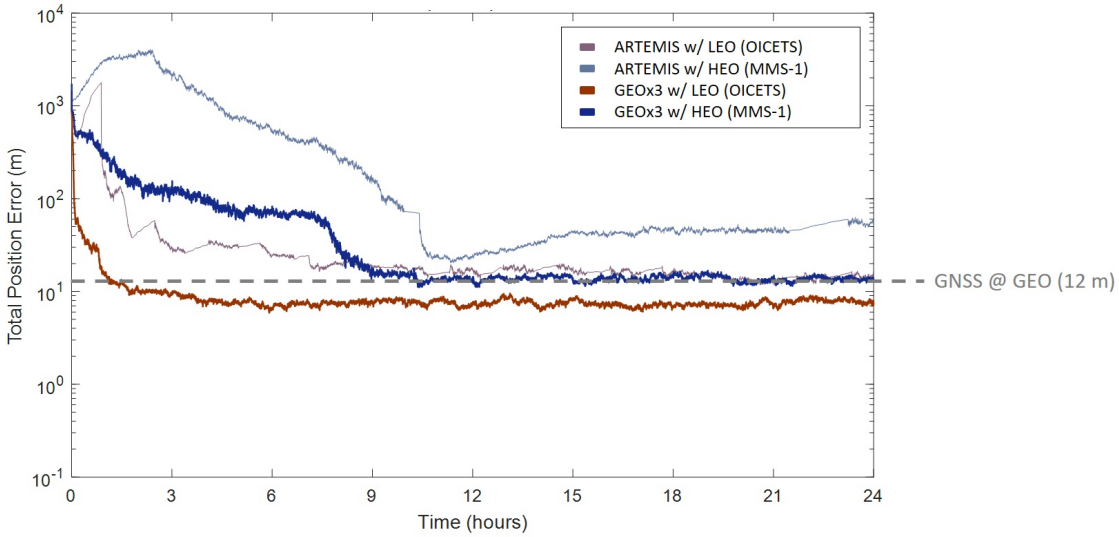


Figure 4-24: Summary of EKF results (50th percentile over 100 Monte Carlo simulations) for GEO satellites from the GEO-Relay case study.

4.3.3 HEO Satellites

A summary of the results for HEO satellites from the GEO Relay case study are assembled in Figure 4-25. Serving as a node in the GEO-relay "hub-node" topology model, the HEO satellites achieve best performance when multiple GEO relay satellites are available to maintain link availability throughout all portions of the elliptical orbit, and gain geometric diversity to reduce uncertainty. As such, the performance goal of 45 meters for HEO satellites is achieved in the scenario with multiple GEO hubs.

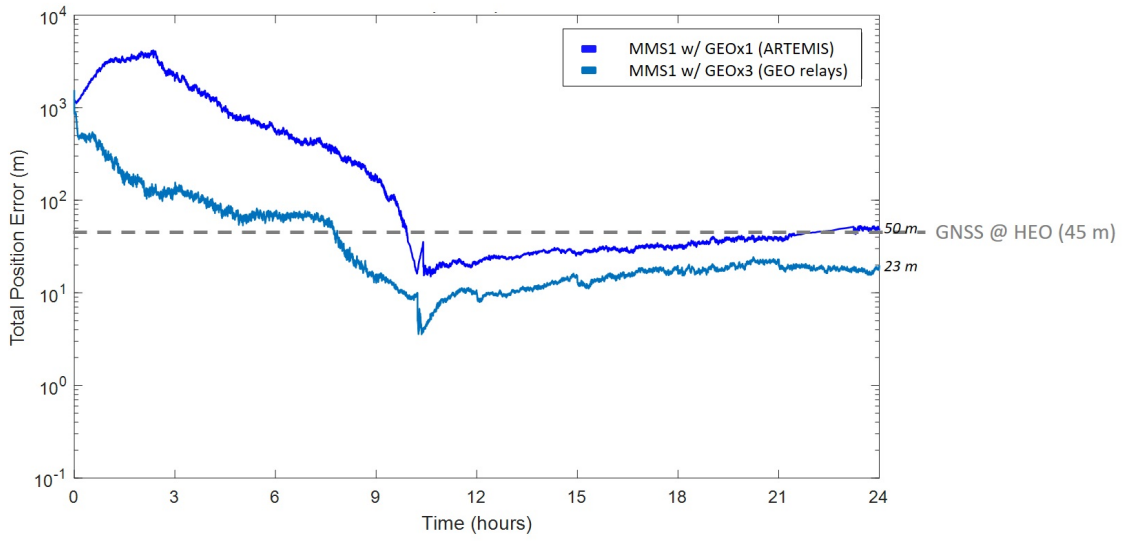


Figure 4-25: Summary of EKF results (50th percentile over 100 Monte Carlo simulations) for HEO satellites from the GEO-Relay case study.

Chapter 5

Deep-Space Orbital Applications

This chapter describes the results and analysis from evaluating the performance of the proposed navigation method using intersatellite lasercom measurements for deep-space orbital applications. We chose to focus on applications for Mars, as it is the location of many past, present, and future exploration missions that can be used as case studies. However, the lasercom navigation method can also be applied to other deep-space bodies, assuming that its mass and gravity characteristics are known well enough to model its primary gravitational forces. Thanks to a long history of Mars orbiter missions, Mars is well-characterized to over 100 terms in degree & order in the latest gravity models produced by NASA JPL [80]. In our first Mars case study, we simulate the performance of intersatellite navigation using lasercom for future science/exploration Mars-orbiters using the current missions as proxies. In the second case study, we simulate lasercom navigation performance for a possible six-satellite Mars communications constellation as proposed by Castellini et al. (2010) [14].

5.1 Case Study M1: Existing Mars Mission Orbits

In this first case study, we evaluate the navigation performance of different configurations of Mars exploration orbiters based on the orbital parameters of current Mars-orbiting missions. The purpose of using the current mission orbits is that they serve as ready-made examples for a set of potential future missions that could have

optical terminals for intersatellite lasercom, and provide more realistic examples of intersatellite geometry than randomly generating orbits with arbitrary state values.

Six satellites are currently operating in Mars orbit: 2001 Mars Odyssey (MO), Mars Express (MEX), Mars Reconnaissance Orbiter (MRO), Mars Orbiter Mission, "Mangalyaan" (MOM), Mars Atmosphere and Volatile Evolution (MAVEN), and ExoMars Trace Gas Orbiter (TGO). In this study, we focus on the results of a single spacecraft in the constellation, Mars Odyssey, which both launched and began Mars operations in 2001. Figure 5-1 provides a diagram of the orbital geometry of the six satellites considered in this study.

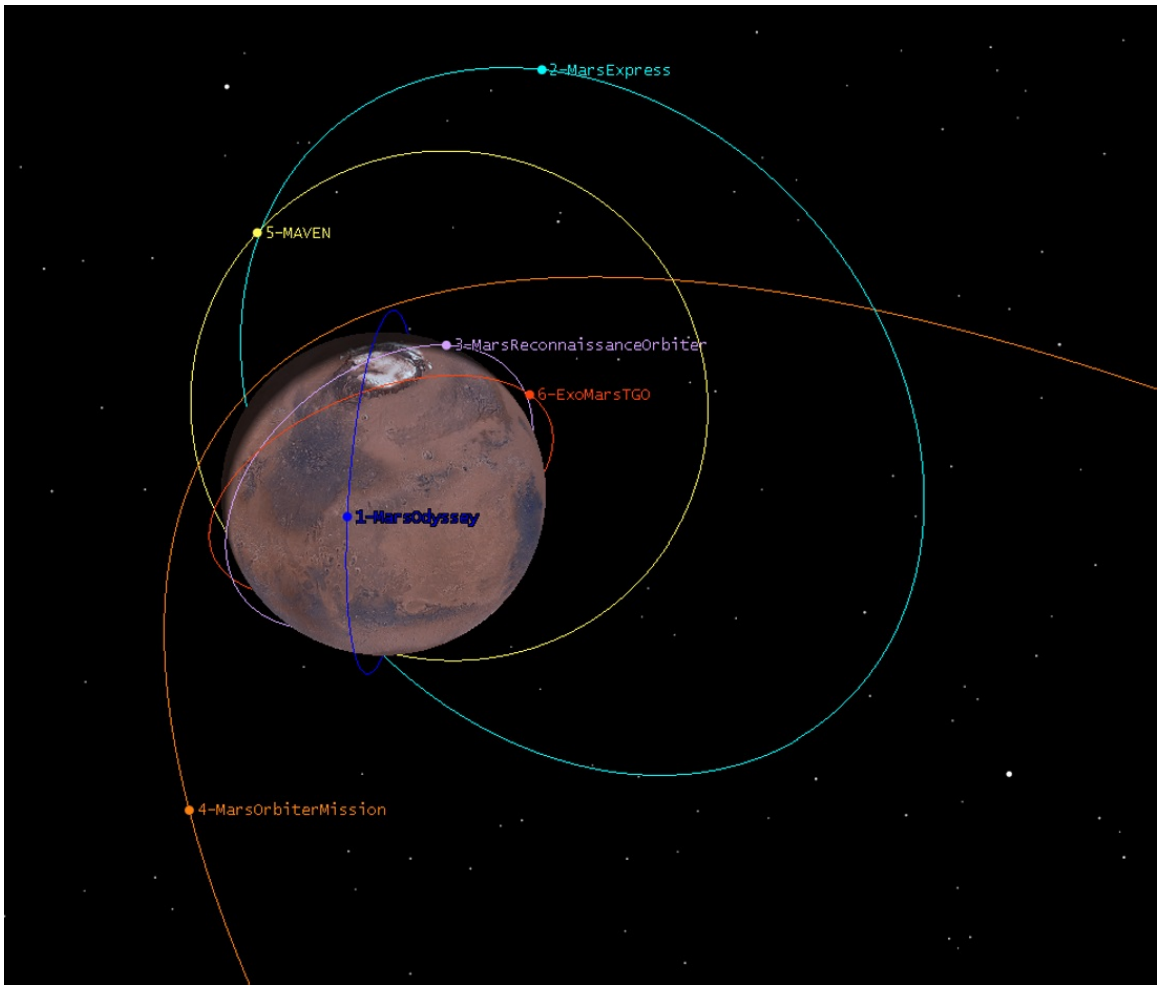


Figure 5-1: Diagram of orbital scenario for the Mars-orbiters case study, generated using AGI STK.

5.1.1 Setup of Scenarios

All scenarios are simulated over the 12-hour time period between t_0 = 02 Jun 2019 16:00:00 UTC and t_N = 03 Jun 2019 04:00:00 UTC, with a time-step of $T = 10$ seconds.

Different scenarios are defined using different constellation configurations (either two-satellite, or ad-hoc constellations of more than two satellites), with Mars Odyssey as the common spacecraft existing in all scenarios. The initial states of the six Mars-orbiter satellites were provided using JPL's HORIZONS online database of spacecraft ephemeris [81]. Table 5.1 shows the orbital elements of each spacecraft at the epoch $t_e = t_0$.

Table 5.1: Orbital parameters for considered Mars-orbiter missions. Note that R = 3,396 km for Mars.

Orbital Element	**MO (NASA)**	**MEX (ESA)**	**MRO (NASA)**	**MOM (ISRO)**	**MAVEN (NASA)**	**TGO (ESA)**
a (km)	3788.93	8818.71	3658.46	39634.8	5742.84	3785.09
e	6.68e-3	5.78e-1	9.03e-3	9.08e-1	3.84e-1	6.09e-3
i (deg)	93.04	86.99	92.67	155.34	74.78	73.55
ω (deg)	252.71	220.50	278.28	49.58	240.83	267.39
Ω (deg)	170.83	208.77	293.78	176.41	272.17	91.44
ν (deg)	41.08	146.85	303.85	197.80	310.97	125.08

For most of the scenarios simulated in this study, we assume an "all-to-all" type of network topology. For the two-satellite cases, this definition is obvious as only one pairing of satellites exist in the constellation, $topo_{2SAT}$. For the ad-hoc constellation cases, the "all-to-all" topology demonstrates the best possible performance, and helps to highlight the value of adding a new spacecraft to an asymmetric constellation.

If multiple payloads cannot be feasibly incorporated into the spacecraft design, a single lasercom system would only allow for point-to-point communications, where each satellite can only support one link at a time. Separate from the "all-to-all" and "some-to-some" architectures described earlier, this scenario requires a separate "one-to-one" network definition that cannot be fully described within the *topo* input

parameter. This is because the topology describes which satellites are available to a given satellite from a networking perspective, not how many links can be supported at a given time. For select scenarios, we also evaluate this "one-to-one" network topology in order to evaluate a more feasible scenario that can be implemented using current technologies and capabilities.

The minimum link altitude is defined for Mars at 80 km and kept constant for all cases. Since these are Mars exploration orbiters, we assume that 50% of the mission time will be spent performing tasks other than crosslink communications with a 30:30 minute duty-cycle (on:off).

5.1.2 Two-satellite Cases

We first evaluate any potential differences based on the orbit of the crosslinked partner satellite in two-satellite cases of Mars-orbiter navigation. Figure 5-2 shows the 50th percentile position navigation results for the Mars Odyssey spacecraft in different two-satellite configurations with the other current Mars-orbiters.

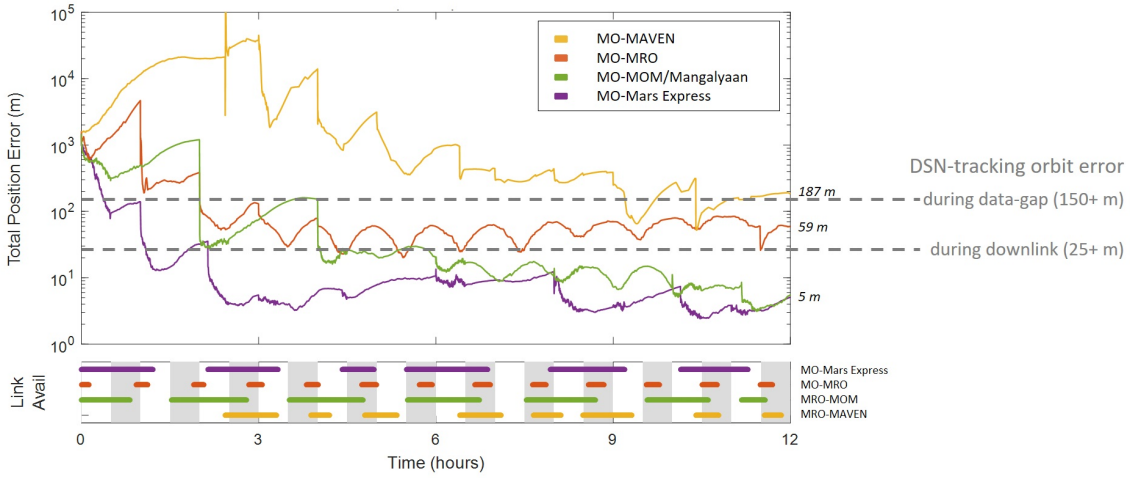


Figure 5-2: Top: Summary of EKF results (50th percentile only) from Monte Carlo simulations (N=100) of 2001 Mars Odyssey orbiter in separate two-satellite navigation scenarios with each of the other five current Mars-orbiters with a 30:30-minute communications duty cycle. Bottom: Depiction of when a link is available for each crosslink partner scenario at any given time over the course of the simulation period.

Note that ExoMarsTGO is not included in the results in Figure 5-2. This is because there are no links available between MO and TGO during the selected simulation period. Therefore, we decided to omit this case from this particular simulation.

The results show a wide spread of navigation errors at the end of the 12-hour simulation period, anywhere between 5 meters up to 190 meters. We believe that this is largely due to differences in access times over the course of the simulation. Looking at the link-availability plot in the bottom of the figure, the duty-cycles implemented in this study combined with the intersatellite geometry between satellite-pairings has a large impact in limiting the amount of link-access time available in each case.

The MO-MAVEN case results in poor performance in this study due to the long gap in link-access availability to start the simulation, and thus the uncertain initial state estimate is propagated without measurement updates causing even larger errors up to 21 km before the first measurements are available at $t \approx 2.5$ hours. The MO-MRO case also results in relatively high errors due to an inopportune overlap of duty-cycle "off" periods during times of link-access availability. The duty-cycled operations cuts link-access windows short towards the beginning and end of the simulation, and causes a long data-gap between $t = 5$ and $t \approx 10.5$ hours. These two cases demonstrate the impact of satellite orbits that do have irregular link-access opportunities or a poorly matched schedule of operations.

The other two cases demonstrate the power of leveraging satellite orbits with good intersatellite geometry in order to maximize potential link-access times. MarsExpress and MOM/Mangalyaan are both in higher altitude orbits which result in longer access times with the Mars Odyssey spacecraft. This helps to mitigate the impact of the duty-cycled operations such that enough measurements can still be collected to reduce uncertainty and estimation error in the navigation filter. MOM is at a much higher altitude during this simulation, as it operates in a highly-eccentric orbit around Mars, and therefore experiences less stability in navigation error than MarsExpress, though both crosslink partners allow Mars Odyssey to achieve errors less than the 25-meter performance target after a few hours of operation.

5.1.3 Ad-hoc Constellations

Uncertainty Analysis of 2+ Satellite Cases

In this next study, we evaluate the effect of having more than two satellites able to perform lasercom crosslinks in asymmetric, ad-hoc constellation configurations. The first scenario we investigate is the concept of an "additive" constellation, where satellites are gradually added to a constellation network in the order in which they are launched and begin operations, similar to how the current "constellation" of Mars-orbiters was created, starting with MarsOdyssey in 2001, then with MarsExpress in 2003, MRO in 2006, both MOM and MAVEN in 2014, and most recently ExoMarsTGO in 2016. Figure 5-3 shows the results from an uncertainty analysis of an additive, ad-hoc constellation.

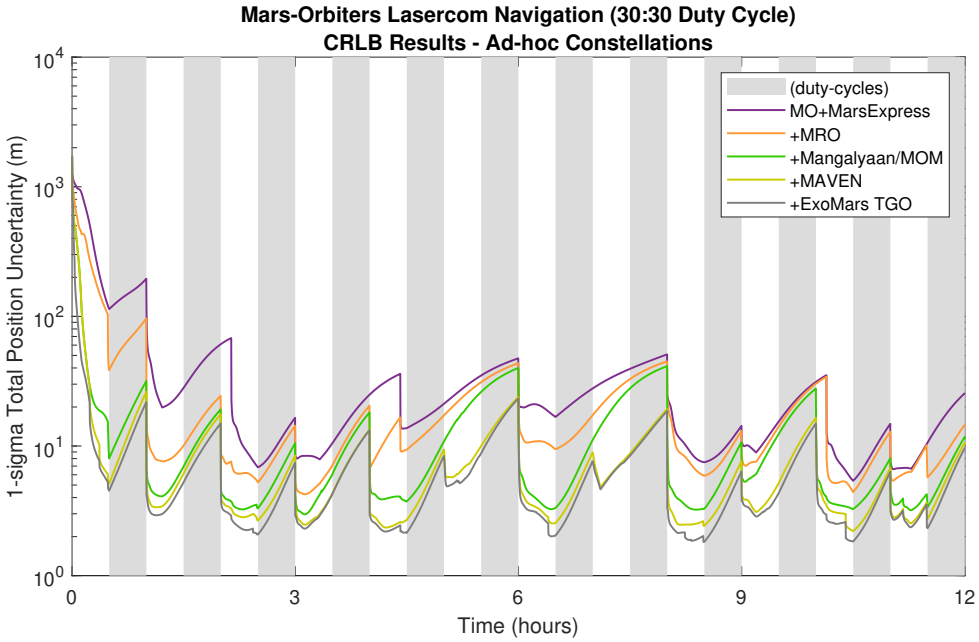


Figure 5-3: CRLB analysis results of 2001 Mars Odyssey orbiter in additive constellation navigation scenarios in order of launch of other five current Mars-orbiters with a 30:30-minute communications duty cycle.

As expected, the results show steady improvement from each addition to the overall Mars-orbiting constellation, reducing the minimum achievable uncertainty from 5.4 meters (in the MO-MEX two-satellite case) down to 1.7 meters (in the full six-satellite

case), which represents a 68% reduction in uncertainty. The results clearly show the added benefit of connecting more satellites into a crosslinked constellation. However, these results also assume that all satellites can communicate with each other, which becomes increasingly difficult with each satellite added to the constellation.

Baseline Example: NASA Spacecraft Constellation

In the second scenario of an ad-hoc constellation, we further investigate the impact of adding an additional satellite into an ad-hoc constellation. Of the six current Mars-orbiter missions, three were developed by NASA: Mars Odyssey, MRO, and MAVEN. As such, all three were designed with a common communications technology architecture in that frequencies, protocols, and even payload designs are consistent across all three satellites, simplifying ground communications and mission operations. If we were to assume that future NASA spacecraft could instead utilize common lasercom technology architectures, they could potentially communicate with one another.

In the previous study of two-satellite scenarios, the NASA satellites coincidentally represented the two worst-performing cases, MO-MRO and MO-MAVEN. In this study, we enable MRO and MAVEN to establish links with each other in addition to their links with MO, in order to close the network topology in a three-satellite fully connected configuration. We implement two network architectures, the "all-to-all" case and the "one-to-one" case. In the "all-to-all" case, each of the three spacecraft require the capability of maintaining two simultaneous crosslinks, most readily implemented by having two lasercom terminals, one that can track each of the other satellites. This mitigates the need for any link-selection, as each link can be made whenever it is available. In the "one-to-one" case, each spacecraft only possesses one lasercom terminal, and is thus only able to track and establish a crosslink with one other satellite at a given time. This represents a more readily achievable constellation case, as it simplifies the design of the spacecraft, though it does require a way to select between two simultaneously available links. This is implemented within the "network rules" portion of the link-selection framework developed in this thesis. Link-selection is executed from the perspective of the Mars Odyssey spacecraft, therefore

selecting between the MO-MRO link and the MO-MAVEN link. When both of these links are unavailable, but a link between MRO and MAVEN is available, that link is executed. Figure 5-4 summarizes the result of these scenarios for the three-satellite NASA spacecraft constellation orbiting Mars.

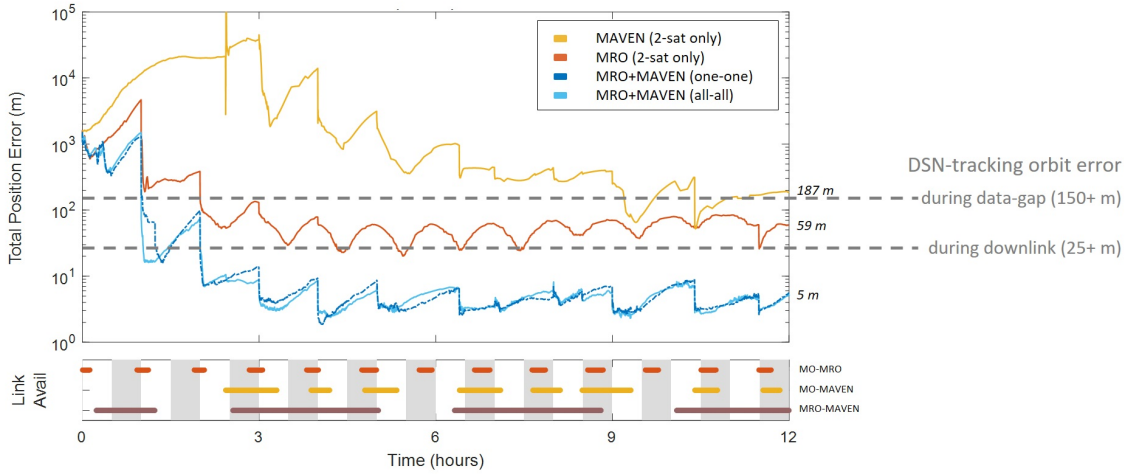


Figure 5-4: Top: Summary of EKF results (50th percentile only) from Monte Carlo simulations (N=100) of 2001 Mars Odyssey orbiter in different navigation scenarios with each of the other two current NASA Mars-orbiters with a 30:30-minute communications duty cycle. Bottom: Depiction of when a link is available for each crosslink partner scenario at any given time over the course of the simulation period.

Note that the two-satellite results from the previous section are also shown in this figure, in order to clearly show the improvement from connecting the three spacecraft into one network. Navigation errors are improved by an order of magnitude, from about 50 meters (in the case of MO-MRO) to about 5 meters (in the three-satellite cases). Whereas the two-satellite cases were heavily impacted by the duty-cycled operations, the three-satellite case completely mitigates those conditions by leveraging the additional link between MRO and MAVEN, in a a more connected constellation. Notably, both of the simulated network architectures are relatively consistent in their navigation error performance, meaning that the "one-to-one" case with a link-selection algorithm can perform just as well as the "all-to-all" with additional payloads on-board each of the spacecraft. Both cases produce more timely, more consistent, and more accurate error performance over the two-satellite cases shown earlier.

5.2 Case Study M2: Future Comms. Constellation

In the second Mars case study, we evaluate the navigation performance of a Mars-orbiting communications constellation for supporting future human and robotic operations, as proposed by Castellini et al. [14], consisting of six total satellites equipped with lasercom payloads for downlink communications with Earth. Although the authors assumed that intersatellite links would be conducted using radio transmissions, for the purposes of our study, we assume that they can be performed using lasercom payloads instead. Figure 5-5 provides a diagram of the orbital configuration of the considered constellation.

The constellation is configured as a Walker constellation with $n_{op} = 2$ orbital planes and $n_{sp} = 3$ satellites per plane. The satellites are described to be at an altitude of 17,030 km (therefore $a \approx 20,426$ km and $e = 0$) and an inclination of 37°. The phase shift (in mean-anomaly) between orbital planes is 5°, while the RAAN separation between orbital planes is 180°. We label satellites in the first plane as Sat11, Sat12, and Sat13, and satellites in the second plane are Sat21, Sat22, and Sat23.

5.2.1 Setup of Scenarios

All scenarios are simulated over a 24-hour time period, between t_0 = 02 Jun 2019 16:00:00 UTC and t_N = 03 Jun 2019 16:00:00 UTC, with a time-step of $T = 10$ seconds. All scenarios use all six satellites in the constellation. Table 5.2 shows the orbital elements of each spacecraft, based on the Walker constellation configuration described above, for the epoch $t_e = t_0$.

The minimum link altitude is defined for Mars at 80 km and kept constant for all cases. Similar to the previous case study, we model that 50% of the mission time will be spent performing tasks other than crosslink communications with a 30:30 minute duty-cycle (on:off).

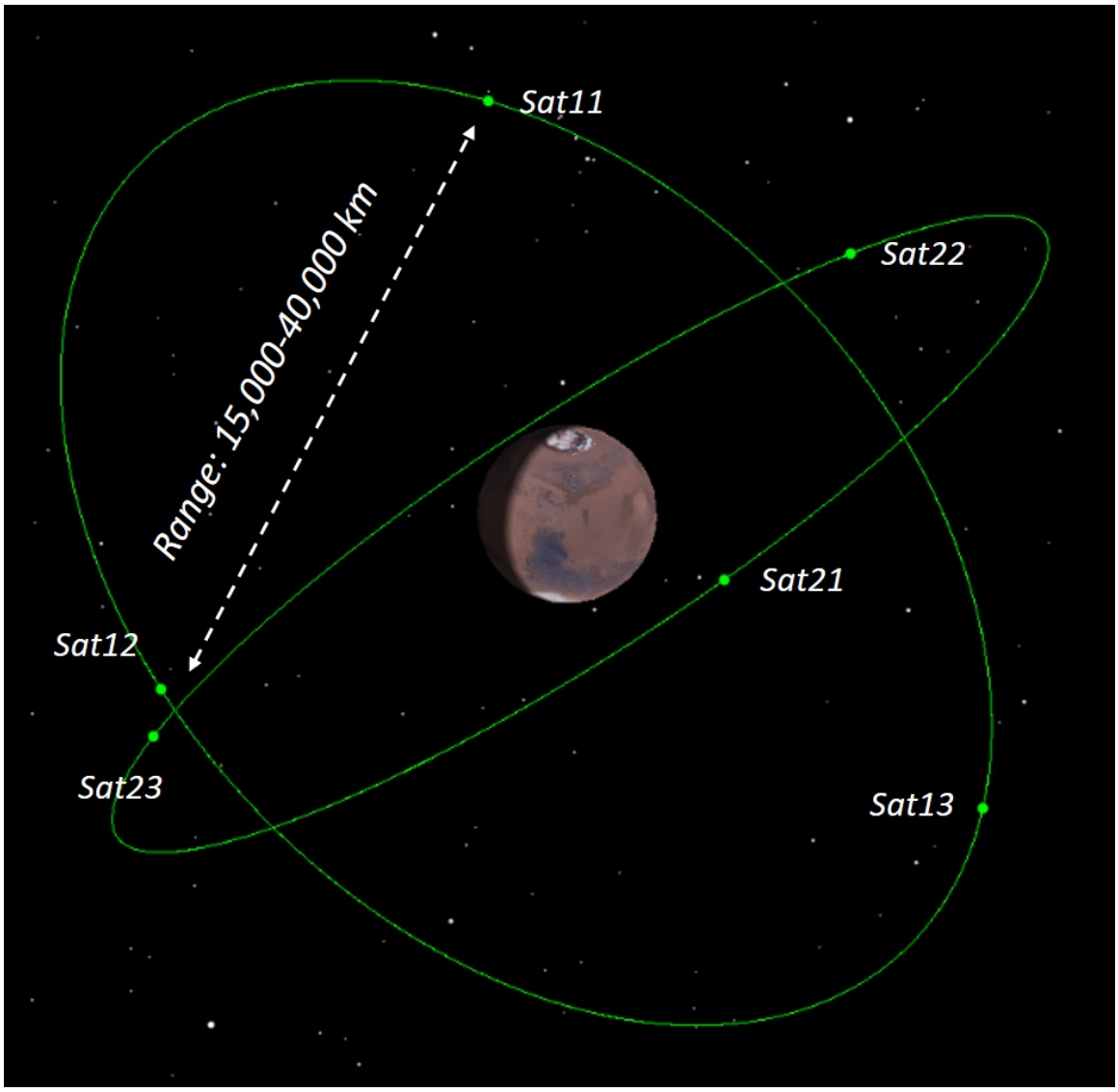


Figure 5-5: Diagram of orbital scenario for the Mars communications constellation case study, based on Castellini et al. (2010) [14], generated using AGI STK.

Network Architectures

Different scenarios are simulated based on the use of differing network architectures, which are enacted by altering the topology input parameter to the link-access computation (defining which satellite-pairings can be made) and the "network rules" used in the link-selection framework (constraining the number of simultaneous connections per satellite and/or simplifying the link-selection process based on the symmetric nature of the constellation). Although there is a nonsymmetric phase shift between

Table 5.2: Orbital parameters for Mars communications constellation, as proposed by Castellini et al. (2010).

Orbital Element	**Sat11**	**Sat12**	**Sat13**	**Sat21**	**Sat22**	**Sat23**
a (km)	20426.2	20426.2	20426.2	20426.2	20426.2	20426.2
e	0	0	0	0	0	0
i (deg)	37.01	37.01	37.01	37.01	37.01	37.01
ω (deg)	0	0	0	0	0	0
Ω (deg)	0	0	0	0	0	0
ν (deg)	0	120	240	5	125	245

orbital planes, we treat the constellation as symmetric since 5° is not a very significant phase shift. Table 5.3 and Figure 5-6 provide more details on the network architectures used in this study, and how each is implemented in the simulation approach.

The "all-to-all" scenario simulates the case where all spacecraft can establish crosslinks with all other spacecraft in the constellation, with no limitations on simultaneous link. This case is the easiest to model in the simulation framework, defining *topo* to be the set of all pairs between all satellites, and not requiring any link-selection (and thus zero "network rules"). However, this case is the most computation-heavy as well, as every possible satellite-pairing is represented in the measurement model $h(\mathbf{x}_k^{(i)}, \mathbf{x}_k^{(j)})$, leading to the largest number of terms being used in the navigation filter of all network architectures studied. This case is also the most difficult to technologically implement in the physical design of the spacecraft, as it would require either an optical transceiver payload capable of multiple two-way communications links [82, 83], or up to 5 lasercom payloads that can individually track and establish links with the five other spacecraft. Assuming these challenges can be overcome, the "all-to-all" case is expected to be the best-case scenario for reducing estimation uncertainty, as it exhibits the highest number of usable measurements for the navigation filter.

The "one-to-one" scenario simulates the case where all spacecraft can establish crosslinks with any of the other spacecraft in the constellation, but is limited to only one link at any given time. Thus, the *topo* variable is the same as the "all-to-all"

Table 5.3: Network architectures used in Mars communications constellation study.

Network Architecture	**Topology Definition**	**Simultaneous Links per Sat.**	**Fully Connected?**	**Closed Path?**
all-all	ALL → ALL	up to 5	Yes	Yes
one-one	ALL → ALL	1	Yes	Yes
some-some *(hub-node)*	Sat11 → ALL	Sat11: up to 5 all others: 1	Yes	No
some-some *(2 links)*	Sat11 → Sat22,Sat23 Sat12 → Sat21,Sat23 Sat13 → Sat21,Sat22	2	Yes	Yes
some-some *(1 link, same)*	Sat11 → Sat21 Sat12 → Sat22 Sat13 → Sat23	1	No	No
some-some *(1 link, diff)*	Sat11 → Sat22 Sat12 → Sat23 Sat13 → Sat21	1	No	No

case, the set of all pairs between all satellites. However, this case clearly requires the link-selection step in our simulation approach, as only one link can be made at any given time. Leveraging the symmetry of the constellation, we run the link-selection process for a single satellite in the constellation, Sat11, and implement "network rules" to constrain the number of simultaneous links and map the full constellation's link-schedule based on Sat11's link-selections. The mapping used for this scenario is shown in Table 5.4. Since all possible satellite-pairings are available in the network connectivity, the "one-to-one" case does not mitigate computational intensity, as it still requires as many terms in the navigation filter (though many are unused at a given time). However, this case does mitigate the design implementation challenges mentioned earlier, since each spacecraft only requires a single lasercom terminal, and

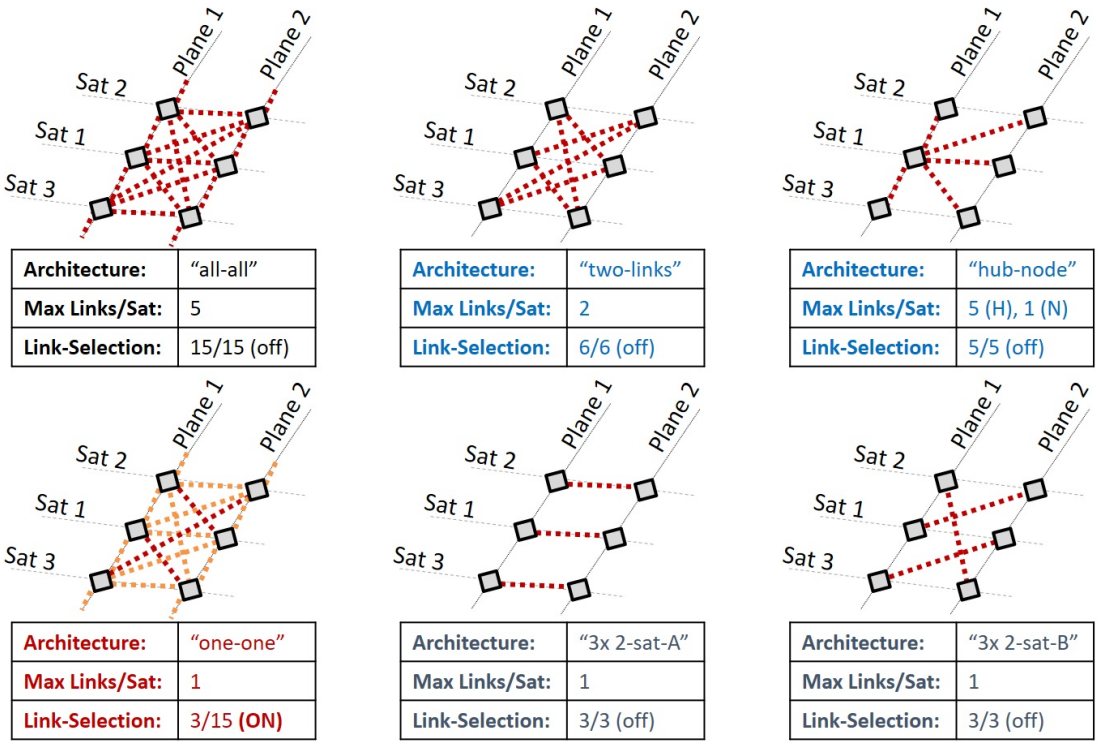


Figure 5-6: Node diagrams depicting the topologies considered in the Mars communications constellation study. "Max links/sat" describes the maximum number of simultaneous links each node can operate. "Link-Selection" shows how many of the total number of possible links in the total network can be used at one time, and if link-selection is active. Note that the font colors are chosen to be consistent with Figures 5-7 and 5-8.

only needs to track one other satellite at any given time.

Table 5.4: "One-to-one" case network map, implemented as "network rules" in Link-Selection framework.

Link-Selection Pairing	**Mapped Pair 1**	**Mapped Pair 2**
Sat11 → Sat12	Sat21 → Sat22	Sat13 → Sat23
Sat11 → Sat13	Sat21 → Sat23	Sat12 → Sat22
Sat11 → Sat21	Sat12 → Sat22	Sat13 → Sat23
Sat11 → Sat22	Sat12 → Sat23	Sat13 → Sat21
Sat11 → Sat23	Sat12 → Sat21	Sat13 → Sat22

Between these two network scenarios are the "some-to-some" cases where certain spacecraft may only be able to establish links with a limited subset of the other satellites in the constellation, and/or maintain a limited number of simultaneous links. There are many different possible enumerations of a "some-to-some" architecture. We limit this to a few cases that are easy to implement in our simulation approach and/or in a physical spacecraft design. The first is the "hub-node" model where certain spacecraft act as "hubs" capable of communicating with any and all of the other satellites, while the others act as "nodes" capable of communicating only with one hub at any time. Up to five satellites can act as hubs (not six, since that would be the same as the "all-to-all" case), but for this study, we implement the simplest case of a single hub, Sat11. This is an open network topology, since there is no path through the topology that returns to the originating satellite (in this case Sat11) through a separate link. However, it is fully connected since all spacecraft are contained within the same network, through the hub sat, Sat11.

A second "some-to-some" case is defined where every satellite can only communicate with a specific set of two other satellites in the constellation. That specific set is two satellites in the opposite orbital plane in the other two phase slots. For instance, Sat11 can link with Sat22 and Sat23, but not with Sat21 as it is in the same phase slot in the opposite plane. This is an example of a closed network topology, since a path can be made from Sat11 through the other satellites that returns to Sat11 from a separate link (Sat11 $\rightarrow$ Sat22 $\rightarrow$ Sat13 $\rightarrow$ Sat21 $\rightarrow$ Sat12 $\rightarrow$ Sat23 $\rightarrow$ Sat11). This is also clearly a fully connected network topology, as all satellites are represented in the closed network path.

The final network scenarios we evaluate are those where each satellite can only communicate one other satellite in the constellation, with no overlapping assignments. This is a special case of the "some-to-some" architecture definition, but essentially is the same as a two-satellite case, since it generates three unconnected networks with two satellites each. These are evaluated in this study to represent the lower order of network connectivity. For the constellation used in this study, there are two possible variations of this scenario that utilize both planes (to avoid coplanar unobservability

issues). The first is where satellites are paired with those from the same orbital phase slot (e.g., Sat11 ↔ Sat21). The second is where satellites are paired with those from a different orbital phase slot (e.g., Sat11 ↔ Sat22, or Sat11 ↔ Sat23).

5.2.2 CRLB Uncertainty Analysis

We first perform an uncertainty analysis using CRLB in order to predict differences in performance between the different network configurations, while keeping all other scenario configuration parameters fixed. Figure 5-7 shows the predicted uncertainty results from these CRLB computations.

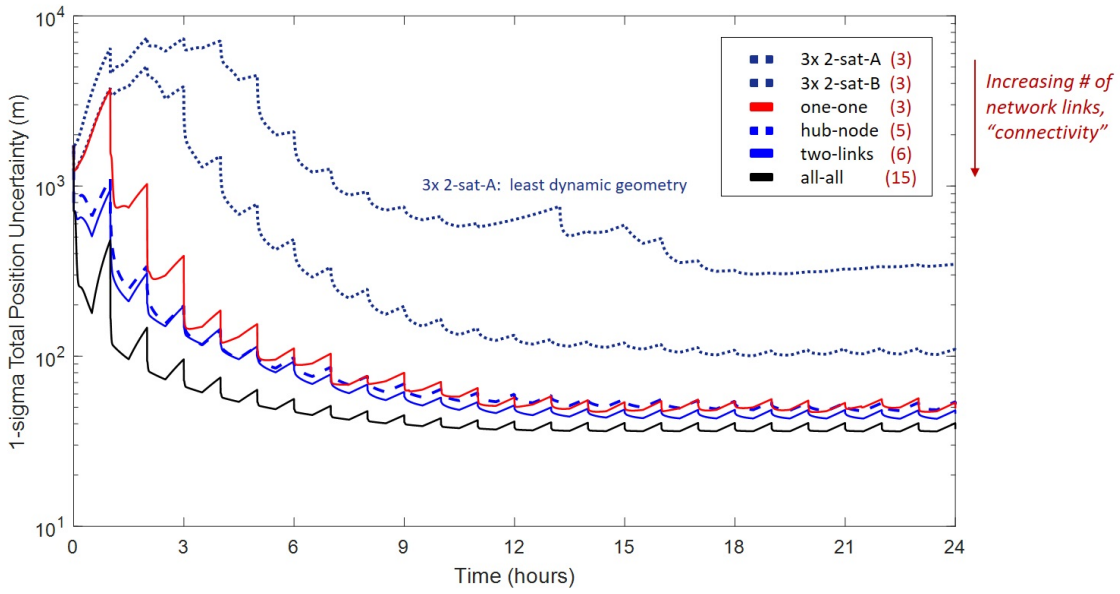


Figure 5-7: CRLB analysis results of a future 6-satellite Mars communications constellation [14] with a 30:30-minute communications duty cycle, and varying network topology architectures.

Note that only the results for Sat11 are shown. Due the near-symmetry of the constellation and network models used in this study, all satellites exhibit similar uncertainty results to one another in each of the scenarios simulated. Solid lines are used to depict network architectures that are fully connected and closed. The dashed line type is used to represent the one network scenario we simulated that is fully connected but open, the "hub-node" scenario with Sat11 as the sole hub. Finally,

dotted lines are used for the two single-link per satellite "some-to-some" scenarios that generate three unconnected 2-satellite networks within the constellation.

The black curve represents the "all-to-all" scenario, and as expected, outperforms all of the other network architectures simulated in this study, as the scenario with the highest degree of network connectivity. The "one-to-one" scenario is represented in red, and seems to achieve nearly equal uncertainty values as the fully connected "some-to-some" scenarios. This is notable since this scenario should be easier to technologically implement with respect to spacecraft design in the near-term, as it requires just one lasercom payload per spacecraft.

The different "some-to-some" scenarios are depicted in blue. As expected, estimation uncertainty is shown to be directly related to the degree of network connectivity within the constellation, as those least connected (comprised of three unconnected two-satellite networks) resulted in the highest uncertainty values and were steadily outperformed by networks of greater connectivity. The scenario where satellites in the same orbital phase slot are connected is the worst-performing scenario, due to the intersatellite geometry of the fixed link topology. Satellite-pairings between those in the same orbital phase slot are the only pairings that experience a link outage due to Mars occlusion. The result of such a data-gap can be seen in the top curve in Figure 5-7 near $t = 12$ hours where the uncertainty rises due to lack of measurements. The two fully connected network architectures perform better than the two unconnected scenarios, and the closed network performs better than the open "hub-node" network model.

5.2.3 EKF Performance Results

In order to gauge the prediction accuracy of the CRLB uncertainty analysis, and evaluate navigation errors for the proposed Mars communications constellation, we now conduct a Monte Carlo performance analysis using the EKF navigation filter on simulated measurements. Figure 5-8 shows the predicted uncertainty results from these simulations.

Note that the colors and line-types representing the different network scenarios

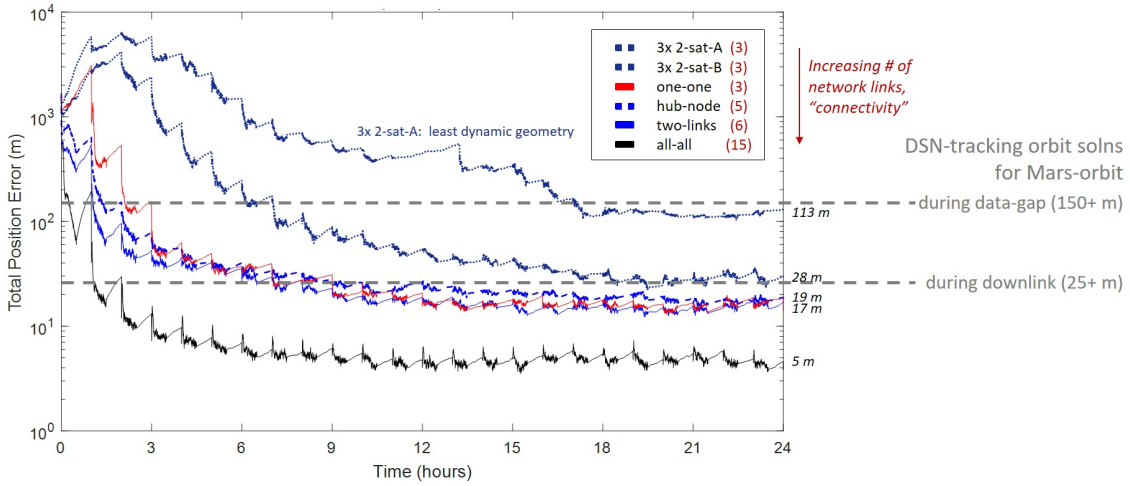


Figure 5-8: Summary of EKF results (50th percentile only) from Monte Carlo simulations (N=100) of a future 6-satellite Mars communications constellation [14] with a 30:30-minute communications duty cycle, and varying network topology architectures.

are consistent with those from the previous figure, however the y-axis range has been expanded to include results down to 1 meter. The 50th percentile navigation error results from the 100-sample Monte Carlo analysis are qualitatively consistent with the predicted results from the CRLB uncertainty analysis, which shows that CRLB can be a very useful tool for predicting performance due to network design decisions relative to a baseline scenario.

The best performing scenario is the "all-to-all" network configuration, as expected, achieving a 50th percentile error consistently below 5 meters after about 9 hours of 50% duty-cycled observation. In addition to the "all-to-all" case, three other network configurations are able to achieve position navigation errors better than the 25-meter performance target. The fully connected and open network scenario (with Sat11 as a hub in a "hub-node" model) achieves this result after about 12 hours of observation, while the fully connected and closed network scenario (with two constant links per satellite) achieves sub-25 meter errors after about 7 hours. Notably, the "one-to-one" network configuration performs consistently with the fully connected and closed network scenario throughout the simulation after about 5 hours of observation. This bodes well for being able to achieve this goal in a spacecraft design that is more

feasible in the short-term.

5.3 Summary of Results

A summary of the results for satellites from both the Mars-orbiting case studies are shown in Figure 5-9. Ad-hoc constellations for autonomous navigation can be gradually assembled by launching future Mars exploration orbiters equipped with lasercom terminals for crosslink communications. Our first study of a three-satellite case modeled off of the three current NASA spacecraft in Mars orbit (Mars Odyssey, MRO, and MAVEN) showed that single-terminal spacecraft can operate in a "one-to-one" network architecture (with a link-selection algorithm) and achieve similar results to a more connected yet harder to implement "all-to-all" network architecture. Our second study performed a deeper dive into the impact of differing network topology architectures of a larger constellation configuration based on a proposed Mars communications constellation design [14] at a higher altitude than typical science/exploration missions. The Mars-orbiter performance goal of 25 meters is achieved by both scenarios in their respective "one-to-one" network topology cases, which represents a more achievable spacecraft design architecture.

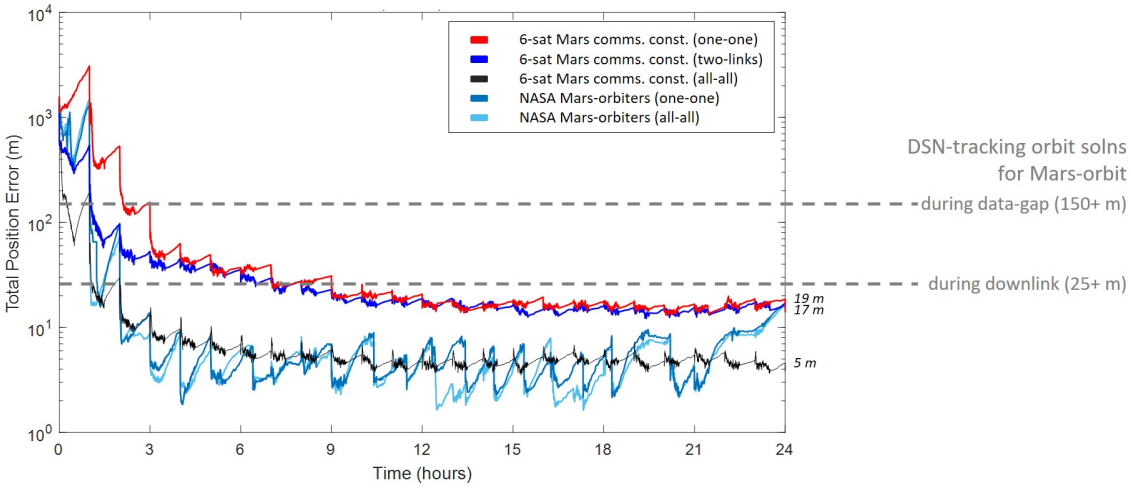


Figure 5-9: Summary of EKF results (50th percentile over 100 Monte Carlo simulations) for satellites from the Mars-orbiters and Mars communications constellation case studies.

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Chapter 6

Conclusion

This chapter concludes this thesis by summarizing all of the findings, contributions, and suggestions for future work.

6.1 Summary of Contributions

For this thesis, we have created a simulation environment to estimate the performance of autonomous spacecraft navigation using intersatellite measurements under varying configurations of orbital environments, constellation size and distribution, network architectures, and measurement models. This simulation environment includes a kinematic uncertainty approximation algorithm using Cramér Rao lower bound (CRLB) covariance estimates as a link-selection heuristic when multiple links are available. This link-selection framework is especially pertinent when a constellation network features a high degree of connectivity, but a limited number of simultaneous connections per spacecraft, which was a design tradeoff of particular interest in this thesis.

We first simulated a GEO-LEO example scenario in order to demonstrate the capability of the developed simulation environment, and illustrate any potential sensitivities to fixed input parameters. From this, we were able to confirm the expectation that an EKF estimator, which is used for this thesis, is sensitive to high uncertainty in the initial state estimate. We found that initial state uncertainty should be kept

below 10 km and 10 m/s for each Cartesian element of the position and velocity, respectively, in order to maintain capability of converging to a solution. We also learned that the filter is highly sensitive to both the time-step T, and the velocity component of process noise Q_v. Values for these input parameters were selected to minimize errors in our simulation approach. Finally, two additional classes of measurement noise were simulated, one that represents a less capable system than the one we assume, and one that represents a more capable system. Simulations of these cases showed an expected result in that navigation performance is proportional to the measurement noise, though not significantly so within the range of values we selected.

We then utilized our simulation environment to evaluate navigation performance in a few relevant past, present, and future satellite missions in order to demonstrate the applicability of navigation using lasercom measurements to both near-Earth and deep-space environments. For potential Earth-orbiting applications, we simulated a GEO relay satellite system, akin to the currently active EDRS operations developed by ESA as well as future missions developed by NASA (e.g., LCRD), and LEO-LEO systems, starting with two-satellite systems and followed by larger Walker constellations. In both case studies, we evaluated different scenarios where we varied the quantity and distribution of satellites in order to determine effects from different intersatellite geometries. The GEO relay case study utilized the link-selection framework we developed in order for the user satellites to predict which relay satellite would best reduce uncertainty, and thus generate a schedule for the day of observations. The best performing LEO-GEO case resulted in errors of about 1.5 meters for the LEO satellite and 8 meters for the GEO satellite(s), as shown in Figure 4-12.

We performed parameter-sweep analyses to evaluate the relationship of estimation uncertainty to differences in the user satellite's orbital altitude and inclination, and confirmed observability effects in the navigation filter. We learned that low-altitude spacecraft have the most value in reducing estimation errors and uncertainty, due to their ability to more dynamically sample the the space where the primary forces exist, in closer proximity to the central-body. Thus, it is recommended to include a low-altitude satellite in any navigation system design based on intersatellite laser-

com measurements, or to schedule crosslink operations during low-altitude portions of satellites in eccentric orbits. We also learned that the orbits of the satellites should be properly configured in order to maximize link-access time without sacrificing the navigation performance benefits of lower altitudes, and while avoiding proximity to the filter's unobservable scenario. Finally, we confirmed our hypothesis that constellations of higher connectivity can be beneficial for a few reasons. Higher quantities of satellites increase the number of available measurements in order to converge to a better solution. Greater distributions of satellites increases observability in the estimator in order to reduce uncertainties in the different state vector elements, leading to better navigation performance.

For potential deep-space applications, we chose to focus on Mars, since there are number of past, present, and future missions that can be used for case studies. In our first case study simulated ad-hoc constellations of future Mars exploration orbiters. For a second case study, we simulated a future Mars communications constellation, as proposed by Castellini et al. (2010) [14]. Through both case studies, we investigated the effects of different orbital geometries, duty-cycled operations, and network architectures on constellation navigation. These studies reiterated the importance of considering intersatellite orbital geometry in order to maximize potential link-access times, especially considering any duty-cycled operations due to competing priorities of other mission tasks that must be scheduled and performed. In investigating different network architectures, we modeled network constraints that relate to the technological feasibility of designing satellites for crosslink operations. At the highest connectivity, the "all-to-all" architecture enables each satellite to perform multiple links at the same time, but would require a complex set of on-board systems to achieve this. At the lowest connectivity, the spacecraft are configured to only establish links with one fixed partnering satellite, but this would require the least implementation effort in the design and operations of the spacecraft. We also modeled a few other network architectures between the two extremes that represent more feasible designs for immediate demonstration potential. Most notable of these architectures is the "one-to-one" case, where each satellite can only perform one link at any given time,

but makes use of the the link-selection algorithm developed in this thesis in order to fully connect the constellation in a closed network design. From these simulations, we learned that fully-connected and closed network designs are preferred in order to gain observability in the state estimation filter, and that different network configurations can be employed to best utilize the capabilities afforded by different system design choices, such as the spacecraft design (which constrains the number of simultaneous links achievable per satellite), and the constellation configuration (which dictates the link-accesses available per satellite).

Finally, through the course of performing these case studies in different operating environments, we have demonstrated the potential for improved navigation performance over the current state-of-the-art spacecraft navigation techniques. In the Earth-orbiting domain, comparing against typical GNSS-based navigation performance, GEO relay satellite systems were shown to achieve median total position errors better than the 12-meter target for GEO satellites, and 45-meter target for HEO satellites. A dedicated 6-satellite LEO constellation was shown to achieve navigation error performance better than the 3-meter target for LEO satellites, while larger Walker constellations showed promise for achieving sub-meter navigation performance, assuming a feasible design can be made balancing constellation size/distribution with network connectivity.

Lastly, both Mars-orbiting case studies demonstrated achievement of median total position errors below the 25-meter target set by typical performance by orbit estimation based on DSN radiometric tracking data. Both an ad-hoc constellation comprised of three or more exploration orbiters at low altitudes, and a dedicated high-altitude Mars communications constellation were able to maintain navigation errors on the order of 5-10 meters by using a reasonably feasible "one-to-one" network architecture with a link-selection framework.

6.2 Future Work

This thesis paves the way for a few areas for future work, mostly with respect to increasing the fidelity of individual components in the simulation environment in order to more accurately model operations of a real spacecraft system.

- **Dynamics & Estimation**:
  - Incorporate smoothing to reduce noise in state estimates
  - Convert satellite state definition to Keplerian elements, which can be beneficial with smoothing or narrow ranges of known possible values for potentially faster and more reliable convergence
  - Implement distributed filters instead of a centralized filter to emulate true-to-life navigation aboard individual satellites [65, 68, 84]
  - Increase fidelity of dynamics model – in particular, adding third-body perturbations, for additional domains of estimation observability
- **Systems Modeling**:
  - Simulate other additional orbital scenarios (e.g., lunar-orbit, cis-lunar spacecraft, halo orbits, heliocentric orbits, interplanetary relays)
  - Simulate other network topology options – in particular, expand all possible combinations for error/uncertainty comparison analysis
  - Increase fidelity of lasercom measurement models (e.g., dependencies to timing bias/uncertainty, orbital position, pointing vector, or relative position/velocity)
  - Increase fidelity of link-access model (e.g., additional constraints due to geometry, time, or network)
  - Model different classes of capability for different satellites (different or time-variant values for initial state knowledge or measurement uncertainty)

  - Include additional measurements – in particular, range-rate, to mitigate sensitivity to velocity process noise
  - Include additional sources of measurements – in particular, from down-links, to provide updates to state estimate or on-board clock

- **Autonomy & Decision-making**:
  - More efficient link-selection heuristic (e.g., filtered/smoothed dilution-of-precision)
  - More robust link-selection algorithm for handling large constellations, asymmetric constellations, and selection conflicts

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